

BTCVC 602 FOUNDATION ENGINEERING THEORY SYLLABUS

Teaching Scheme: (3 Lectures + 1 Tutorial) hours/week

Course Contents

Module 1: Introduction (6 Lectures)

General requirement to be satisfied for satisfactory performance of foundation, soil exploration: Necessity, planning, exploration methods, soil sampling disturbed and undisturbed, rock drilling and sampling, core barrels, core boxes, core recovery, field tests for bearing capacity evaluation, test procedures and limitations.

Module 2: Bearing capacity analysis (7 Lectures)

Failure modes, Terzaghi's Analysis, specialization of Terzaghi's equations, Skempton values for N_c , Meyerhof's analysis, I.S. code method of bearing capacity evaluation, effect of water table, eccentricity of load. Safe bearing capacity and allowable bearing pressure, settlement analysis: Immediate settlement – consolidation settlement, differential settlement, Tolerable settlement, angular distortion

Module 3: Foundation for difficult soils (6 Lectures)

Guidelines for weak and compressible soils, expansive soil, parameters of expansive soils, collapsible soils and corrosive soils, causes of moisture changes in soils, effects of swelling on buildings, preventive measures for expansive soils, modification of expansive soils, design of foundation on swelling soils, ground improvement methods, for general considerations for cohesive soils and for cohesion less soils

Module 4: Shallow foundations (5 Lectures)

Assumptions and limitation of rigid design analysis, safe bearing pressure, settlement of footings. Design of isolated, combined, strap footing (rigid analysis) Raft foundation (Elastic analysis), I.S. Code of practice for design of raft foundation

Module 5: Deep foundation (7 Lectures)

Pile foundation: Classification, pile driving, load carrying capacity of piles, single pile capacity, dynamic formulae, static formulae, pile load tests, penetration tests, negative skin friction, under reamed piles, group action of piles.

Caisson foundation: Box, pneumatic, open caissons, forces, grip length, well sinking, practical difficulties and remedial measures.

Sheet piles: Classification, design of cantilever sheet pile in cohesion less and cohesive soils. Design of anchored sheet pile by free earth support method, cellular cofferdam: Types, cell fill stability considerations.

Module 6: Slope stability (6 Lectures)

Different definitions of factor of safety, types of slope failures, stability of an infinite slope of cohesion less soils, stability of infinite slope in cohesive soils, stability of finite slopes – slip circle method, semi graphical and graphical methods, friction circle method, stability number concepts and its use

Text Books

- 1) Kasamalkar B. J., “Foundation Engineering”, Pune Vidyarthi Griha Prakashan Pune
- 2) Murthy V.N.S., “Soil Mechanics & Foundation Engineering” , U.B.S. Pub. And Distri. N. Delhi 32
- 3) Punmia B.S., “ Soil Mechanics & Foundation Engineering” , Laxmi Publications
- 4) Arora K. R., “ Soil Mechanics” Standard Publishers, N. Delhi
- 5) Gopal R Rao “Basic Soil Mechanics”
- 6) Nayak N.V., “Foundation design manual” Dhanpat Rai and Sons
- 7) Brahma S.P. Foundation Engineering. Tata Mcgraw hill
- 8) Braja Das Principle of Geotechnical Engineering
- 9) Bowels J.E. Foundation analysis and design McGraw hill

Reference Books

- 1) Teng W.C. “Foundation Design”
- 2) Tomlinson M J “Foundation design and construction”
- 3) Lee “Sheet pile” Concrete publication, 1961
- 4) Relevant IS codes (IS 6403, IS 1904, IS 4091)

Course Outcomes:

On completion of the course, the students will be able to:

CO1: To predict soil behaviour under the application of loads and come up with appropriate solution to foundation design queries

CO2: Analyse the stability of slope by graphical method

CO3: Analyze the results of in situ tests and transform measurements associated uncertainties into relevant design parameters.

CO4: Synthesize the concept of allowable stress design, appropriate factor of safety, margin of safety, and reliability

Module 1: Introduction (6 Lectures)

General requirement to be satisfied for satisfactory performance of foundation, soil exploration: Necessity, planning, exploration methods, soil sampling disturbed and undisturbed, rock drilling and sampling, core barrels, core boxes, core recovery, field tests for bearing capacity evaluation, test procedures and limitations.

1.1 Introduction.

Every building consists of two basic components which are the super-structure and substructure or foundations. The super-structure is usually that part of the building which is above ground, and which serves the purpose of its intended use. The sub structure or foundations is the lower portion of the building, usually located below the ground level, which transmits the load of the super-structure to the sub-soil. Foundation is part of the structure which is in direct contact with the ground to which the loads are transmitted. The soil which is located immediately below the base of the foundation is called the sub-soil or foundation soil, while the lowermost portion of the foundation which is in direct contact with the sub-soil is called the footing.

1.1.1 Functions of Foundation

- 1) Reduction of load intensity. Foundation distributes the loads of the super-structure, to a larger area so that the intensity of the load at its base does not exceed the safe bearing capacity of the sub-soil.
- 2) Even distribution of load: Foundations distribute the none uniform load of the super-structure evenly to be sub-soil. For example, two columns carrying unequal loads can have a combined footing which may transmit the load to sub soil evenly with uniform soil pressure. Due to this, unequal or differential settlements are minimized.
- 3) Provision of level surface: Foundation provide leveled and hard surface over which the super-structure can be built.
- 4) Lateral stability: The stability of the building, against sliding and overturning, due to horizontal forces (such as wind, earthquake etc.) is increased due to foundations.
- 5) Safety against undermining: It provide the structural safety against undermining or scouring due to burrowing animals and flood water.
- 6) Protection against soil movements: Special foundation measures prevents or minimizes the cracks in the super structure, due to expansion or contraction of the sub-soil because of moisture movement in some problematic soils.

1.1.2 Essential Requirements of a Good Foundation

Foundation should be constructed to satisfy the following requirements.

- 1) The foundations shall be constructed to sustain the dead load and imposed loads and to transmit these to the sub-soil in such a way that pressure on it will not cause settlement which would impair the stability of the building or adjoining structures.
- 2) Foundation base should be rigid so that differential settlements are minimized specially for the case when super imposed loads are not evenly distributed.
- 3) Foundation should be taken sufficiently deep to ground the building against damage or distress caused by swelling or shrinkage of the sub soil.
- 4) Foundation should be so located that its performance may not be affected due to any unexpected future influence.

1.2 Soil exploration

Earthwork forms the largest activity of a Civil Engineer. It is well understood that irrespective of the type of civil engineering structure on earth –

- It has to be rested either in soil (e.g., foundations)
- Rested on soil (e.g., pavements) or
- The structure is itself constructed making use of soil (e.g., Earthen dams).

This implies that a better knowledge of the spatial variation of the soils encountered is essential. Therefore, before construction of any civil engineering work a thorough investigation of the site is essential.

Soil exploration/Site investigation is the exercise of undertaking a planned sequence of exploratory holes, with associated field and laboratory testing, in order to bring our understanding of the ground at a site, to an acceptable level of confidence for a particular project.

Site investigation refers to the methodology of determining surface and subsurface features of the proposed area. Information on surface conditions (**surface exploration**) is necessary for planning the accessibility of site, for deciding the disposal of removed material (particularly in urban areas), for removal of surface water in water logged areas, for movement of construction equipment, and other factors that could affect construction procedures. This can be done by studying the various maps (topographical, contour, geological, aerial photographs etc.) and by reconnaissance of the area.

Information on subsurface conditions (**subsurface exploration**) is more critical requirement in planning and designing the foundations of structures, dewatering systems, shoring or bracing of excavations, the materials of construction and site improvement methods can be done by various methods of exploration which is discussed below.

1.2.1 Necessity of soil exploration:

Site investigations constitute an essential and important engineering program which, while guiding in assessing the general suitability of the site for the proposed

works, enables the engineer to prepare an adequate and economic design and to foresee and provide against difficulties that may arise during the construction phase. Site investigations are equally necessary in reporting upon the safety or causes of failures of existing works or in examining the suitability and availability of construction materials.

Objectives of Soil exploration: The information from soil investigations will enable a Civil engineer to plan, decide, design, and execute a construction project. Soil investigations are done to obtain the information that is useful for one or more of the following purposes

- 1.To know the geological condition of rock and soil formation.
- 2.To establish the groundwater levels and determine the properties of water.
- 3.To select the type and depth of foundation for proposed structure
- 4.To determine the bearing capacity of the site.
- 5.To estimate the probable maximum and differential settlements.
6. To predict the lateral earth pressure against retaining walls and abutments.
7. To select suitable construction techniques
- 8.To predict and to solve potential foundation problems
- 9.To ascertain the suitability of the soil as a construction material.
- 10.To determine soil properties required for design
- 11.Establish procedures for soil improvement to suit design purpose
- 12.To investigate the safety of existing structures and to suggest the remedial measures.
- 13.To observe the soil the soil performance after construction.
- 14.To locate suitable transportation routes
- 15.Selection of borrow areas for embankments.

1.2.2 Planning of exploration program: The actual planning of a subsurface exploration program includes some or all of the following steps:

1. Assembly of all available information on dimensions, column spacing, type and use of the structure, basement requirements, any special architectural considerations of the proposed building, and tentative location on the proposed site. Foundation regulations in the local building code should be consulted for any special requirements. For bridges the soil engineer should have access to type and span lengths as well as pier loadings and their tentative location. This information will indicate any settlement limitations and can be used to estimate foundation loads.

2. Reconnaissance of the area. This may be in the form of a field trip to the site, which can reveal information on the type and behavior of adjacent structures such as cracks, noticeable sags, and possibly sticking doors and windows. The type of local existing structures may influence to a considerable extent the exploration program and the best type of foundation for the proposed adjacent structure. Since nearby existing structures must be maintained in their "as is" condition, excavations or

construction vibrations will have to be carefully controlled, and this can have considerable influence on the "type" of foundation that can be used.

Erosion in existing cuts (or ditches) may also be observed, but this information may be of limited use in the foundation analysis of buildings. For highways, however, runoff patterns, as well as soil stratification to the depth of the erosion or cut, may be observed. Rock outcrops may give an indication of the presence or the depth of bedrock.

The reconnaissance may also be in the form of a study of the various sources of information available, some of which include the following (maps, literature, etc.)

3. A preliminary site investigation. In this phase a few borings (one to about four) are made or a test pit is opened to establish in a general manner the stratification, types of soil to be expected, and possibly the location of the groundwater table. If the initial borings indicate that the upper soil is loose or highly compressible, one or more borings should be taken to rock or competent strata.

It is common at this stage to limit the recovery of good-quality samples to only three or four for laboratory testing. These tests, together with strength and settlement correlations using index properties such as liquid limit, plasticity index, and penetration test data as well as unconfined compression tests on disturbed samples recovered during penetration testing, are usually adequate for determining if the site is suitable.

4. A detailed site investigation. Where the preliminary site investigation has established the feasibility and overall project economics, a more detailed exploration program is undertaken. The preliminary borings and data are used as a basis for locating additional borings, which should be confirmatory in nature, and determining the additional samples required. Note that if the soil is relatively uniformly stratified, a rather orderly spacing of borings at locations close to critical superstructure elements should be made.

On occasion additional borings will be required to delineate zones of poor soil, rock outcrops, fills, and other areas that can influence the design and construction of the foundation. In the detailed program phase it is generally considered good practice to extend at least one boring to competent rock if the overlying soil is soft to medium stiff. This is particularly true if the structure is multiple-storied or requires settlement control

The planning of an exploration program depends upon

1. The type and importance of the structure
2. The nature of the soil strata.
3. The depth, thickness, extent, and composition of each of the strata,
4. The depth of the rock,
5. The depth to the ground water table is important items sought to be determined by an exploration program.

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The two important aspects of a boring program are '**spacing of borings**' and '**depth of borings**'.

Exploration methods:

The methods available for soil exploration may be classified as follows:

- Direct methods ... Test pits, trial pits or trenches
- Semi-direct methods ... Borings
- Indirect methods ... Soundings or penetration tests and geophysical methods

Direct method: Test pit

In this method trench is excavated and the soil in the trench is visually examined and samples can be collected directly and can be used for determining strength and other properties of soil.



- Applicable to all types of soils
- Provide for visual examination in their natural condition

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- Disturbed and undisturbed samples can be conveniently obtained at different depths
- Depth of investigation: limited to 3 to 3.5 m.

Advantages

- Cost effective
- Provide detailed information of stratigraphy
- Large quantities of disturbed soils are available for testing
- Large blocks of undisturbed samples can be carved out from the pits
- Field tests can be conducted at the bottom of the pits

Disadvantages

- Depth limited to about 6m
- Deep pits uneconomical
- Excavation below groundwater and into rock difficult and costly
- Too many pits may scar site and require backfill soils.

Limitations

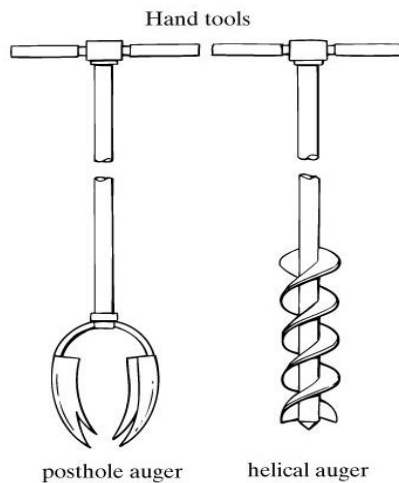
- Undisturbed sampling is difficult
- Collapse in granular soils or below ground water table

Semidirect method : In this boreholes are drilled in order to obtain the samples of soil or rock at desired depth (which is called boring).

The common method of advancing the boreholes are

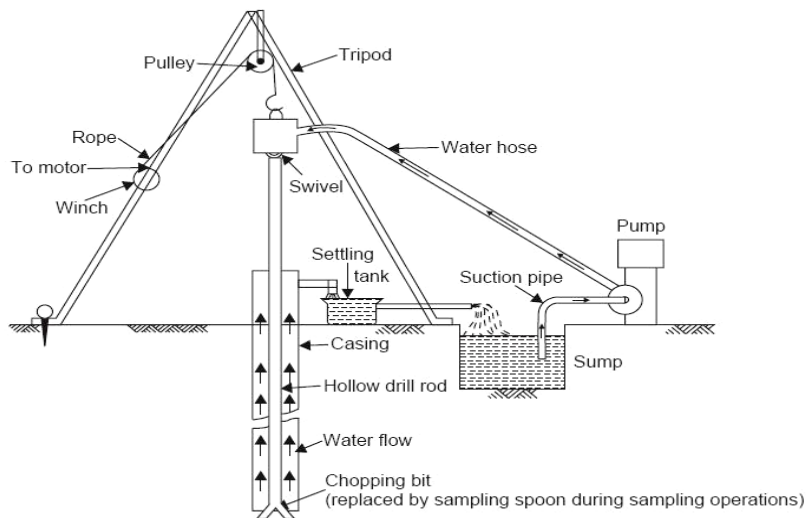
- a) Auger boring
- b) Wash boring
- c) Percussion drilling
- d) Rotary drilling

Auger boring : In this boreholes are drilled by using augers, which is a device that is useful for advancing a bore hole into the ground. Augers may be hand-operated or power-driven; the former are used for relatively small depths (less than 3 to 5 m), while the latter are used for greater depths. It is Suitable in all soils above GWT but only in cohesive soil below GWT.



Wash boring : The rotary wash boring method is generally the most appropriate method for use in soil formations below the groundwater level. In rotary wash borings, the sides of the borehole are supported either with casing or with the use of a drilling fluid

- Wash boring is commonly used for exploration below ground water table for which the auger method is unsuitable.
- This method may be used in all kinds of soils except those mixed with gravel and boulders.
- A casing pipe is pushed in and driven with a drop weight.
- A hollow drill bit is screwed to a hollow drill rod connected to a rope passing over a pulley and supported by a tripod.
- Water jet under pressure is forced through the rod and the bit into the hole. This loosens the soil at the lower end and forces the soil-water suspension upwards along the annular surface between the rod and the side of the hole.
- This suspension is led to a settling tank where the soil particles settle while the water overflows into a sump. The water collected in the sump is used for circulation again
- The soil particles collected represent a very disturbed sample and is not very useful for the evaluation of the engineering properties.
- Wash borings are primarily used for advancing bore holes; whenever a soil sample is required, the chopping bit is to be replaced by a sampler.
- The change of the rate of progress and change of colour of wash water indicate changes in soil strata



Set-up for Wash Boring

Where drill casing is used, the boring or is advanced sequentially by: (a) driving the casing to the desired sample depth, (b) cleaning out the hole to the bottom of the casing, and (c) inserting the sampling device and obtaining the sample from below the bottom of the casing.

Percussion Boring: Grinding the soil by repeated lifting and dropping of heavy chisels or drilling bits. Water is added to form slurry of cuttings. Slurry removed by bailers or pumps. In general, a machine used to drill holes is called a drill rig (generally power driven, but may be hand driven). A winch is provided to raise and lower the drilling tools into the hole. The slurry of pulverized material is bailed out at intervals. The method is suitable for advancing a hole in all types of soils, boulders and rock. The formation, however, get disturbed by the impact.

Rotary drilling: Rotary Boring or rotary drilling is a very fast method of advancing hole in both rocks and soils. A drill bit, fixed to the lower end of the drill rods, is rotated by a suitable chuck, and is always kept in firm contact with the bottom of the hole. A drilling mud, usually a water solution of bentonite, with or without other admixtures, is continuously forced down to the hollow drill rods. The mud returning upwards brings the cutting to the surface. The method is also known as mud rotary drilling and the hole usually requires no casing.



Drilling bit for percussion drilling

Indirect method : In these methods properties of soil are determined indirectly by measuring some other parameter of soil which is then correlated with the properties of soil. Following are the methods in this category

- A) Sounding method (SPT/SCPT/DCPT)
- B) Geo-physical methods (Electrical resistivity/seismic refraction)

Sounding method : In these methods probe is inserted in the ground (statically or dynamically) and resistance offered by the soil against its penetration is recorded. following are the various sounding methods

- a) Standard Penetration Test(SPT)
- b) Static Cone penetration test(SCPT)
- c) Dynamic Cone penetration test(DCPT)

Soil Sampling: Basic aim of soil exploration is to determine soil properties for which soil is collected from the site of exploration and is then tested in the laboratory. The process of collecting the soil from the field is called soil sampling, soil collected is called soil sample and the instrument used for collecting the soil sample is called soil sampler.

Need for sampling: Sampling is carried out in order that soil and rock description, and laboratory testing can be carried out.

Laboratory tests typically consist of:

- i). Index tests (for example, specific gravity, water content)
- ii). Classification tests (for example, Atterberg limit tests on clays); and

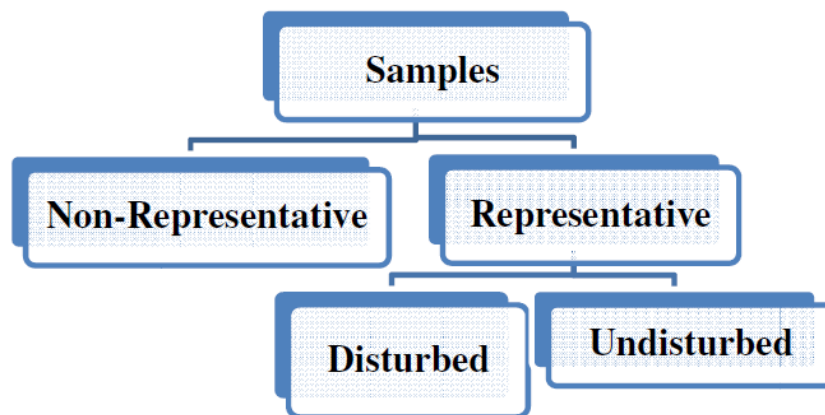
iii) Tests to determine engineering design parameters (for example Strength, compressibility, and permeability).

Factors to be considered while sampling soil

- i) Samples should be representative of the ground from which they are taken.
- ii) They should be large enough to contain representative particle sizes, fabric, and fissuring and fracturing.
- iii) They should be taken in such a way that they have not lost fractions of the *situ* soil
(for example, coarse or fine particles) and, where strength and compressibility tests are planned they should be subject to as little disturbance as possible.

Type of soil samples:

Soil samples are classified as below



Non-Representative samples: Non-Representative soil samples are those in which neither the in-situ soil structure, moisture content nor the soil particles are preserved.

- They are not representative
- They cannot be used for any tests as the soil particles either gets mixed up or some particles may be lost.

e.g., Samples that are obtained through wash boring or percussion drilling

Disturbed soil samples: Disturbed soil samples are those in which the in-situ soil structure and moisture content are lost, but the soil particles are intact.

- They are representative
- They can be used for grain size analysis, liquid and plastic limit, specific gravity, compaction tests, moisture content, organic content determination and soil classification test performed in the lab

e.g., obtained through cuttings while auguring, grab, split spoon (SPT), etc

Undisturbed soil samples: Undisturbed soil samples are those in which the in-situ soil structure and moisture content are preserved.

- They are representative and also intact
- These are used for **consolidation, permeability or shear strengths test**

(Engineering properties)

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- More complex jobs or where **clay** exist
- In **sand** is very difficult to obtain undisturbed sample
- Obtained by using Shelby tube (thin wall), piston sampler, surface (box), vacuum, freezing, etc.,

Causes of Soil disturbances

- _ Friction between the soil and the sampling tube
- _ The wall thickness of the sampling tube
- _ The sharpness of the cutting edge
- _ Care and handling during transportation of the sample tube

To minimize friction

The sampling tube should be pushed instead of driven into the ground

Sampling tube that are in common use have been designed to minimize sampling disturbances

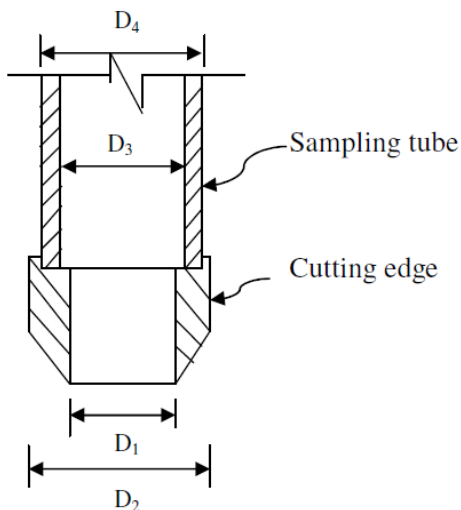
Design Features affecting the sample disturbance

- Area ratio
- Inside Clearance
- Outside Clearance
- Recovery Ratio
- Inside wall friction
- Design of non-return valve
- Method of applying force
- sizes of sampling tubes

Area ratio: It is defined as the ratio of c/s area of cutting edge of sampler to the c/s area of sample

$$A_r = \frac{D_2^2 - D_1^2}{D_1^2}$$

- For obtaining good quality undisturbed samples, the area ratio should be less than or equal to 10%.
- It may be high as 110% for thick wall sampler like split spoon sampler and may be as low as 6 to 9% for thin wall samples like Shelby tube.



Inside Clearance: It is defined as ratio of difference between inner diameter of cutting edge and sampling tube to that of inner diameter of sampling tube

$$C_I = \frac{D_3 - D_1}{D_1} \times 100$$

- The inside clearance allows **elastic expansion** of the sample when it enters the sampling tube.
- It helps in reducing the **frictional drag** on the sample, and also helps **to retain the core**.
- For an undisturbed sample, the inside clearance should be between **0.5 and 3%**.

Outside Clearance: It is defined as the ratio of difference of outer diameter of the cutting edge and sampling tube to that of outer diameter of the sampling tube.

$$C_O = \frac{D_2 - D_4}{D_4} \times 100$$

- Outside clearance facilitates the **withdrawal** of the sample from the ground.
- For **reducing the driving force**, the outside clearance should be as small as possible.
- Normally, it lies between zero and 2%.
- C_O Should not be more than C_I

Recovery Ratio: It is defined as the ratio of the actual length of sample recovered from the sampling tube to the depth of penetration of sampling tube in to the ground.

$$R_r = \frac{L}{H} \times 100$$

$R_r = 96 - 98 \%$ for getting a satisfactory undisturbed sample

Inside wall friction

- The friction on the inside wall of the sampling tube causes disturbances of the sample.
- Therefore, the inside surface of the sampler should be as smooth as possible.
- It is usually smeared with oil before use to reduce friction

Design of non-return valve

- The non – return valve provided on the sampler should be of proper design.
- It should have an orifice of large area to allow air, water or slurry to escape quickly when the sampler is driven.
- It should close when the sample is withdrawn

Method of applying force

- The degree of disturbance depends upon the method of applying force during sampling and depends upon the rate of penetration of the sample.
- For obtaining **undisturbed samples, the sampler should be pushed and not driven**

Rock Core Drilling and Sampling (Non-destructive Rock Core Drilling):

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Bedrock sampling is required for most subsurface explorations, particularly subsurface explorations for structures and roadway rock cuts and for project sites where rock is at a shallow depth. When bedrock sampling is required, rock core drilling must be performed. Rock core sampling is performed to:

- Determine the elevation of the top of bedrock
- Obtain rock core samples for visual inspection and laboratory testing
- Identify the bedrock type
- Describe bedrock color, hardness, weathering, bedding and discontinuity spacing, dip magnitude, and Rock Quality Designation (RQD).

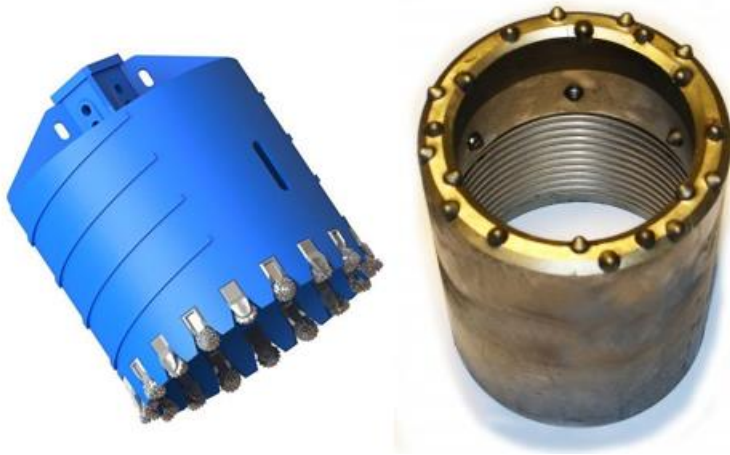
During rock core sampling, subsurface features encountered that could influence the determination of a proper foundation may include voids, mines, or other zones of weakness such as soil or clay infillings, discontinuities, and shear zones/faults. These features are often identified not only in the recovered rock core samples but also during changes in drilling such as:

- Rate of Drilling – Increased or decreased drilling rate, which include tool drops or rapid rates of advancement as these could be indicators of voids, soft zones, clay soil seams, or highly weathered rock. Decreased drilling rates or lack of advancement could indicate very hard rock conditions. Blocking off or plugging of the core barrel can indicate highly fractured rock conditions or presence of clay seams.
- Drill Water Conditions– Changes in the amount of drill water return to the surface or loss of drill water return entirely can indicate the presence of voids, open discontinuities, faults, weathered zones, and soil seams. Drill water return color should be documented for each core run as this can be an indicator to the degree of rock weathering.
- Rock Core Sample Observations and Handling– If rock core losses are observed documentation should identify and provide an interpretation for the core loss particularly if the losses are thought to represent conditions different from the core recovered. Measurements and observations such as core loss, core recovery, and RQD should be measured while the core remains in the split inner tube, and not after core segments are fitted together and placed in the core box. Rock core samples should be oriented so that bedding or foliation is observed at its maximum dip.

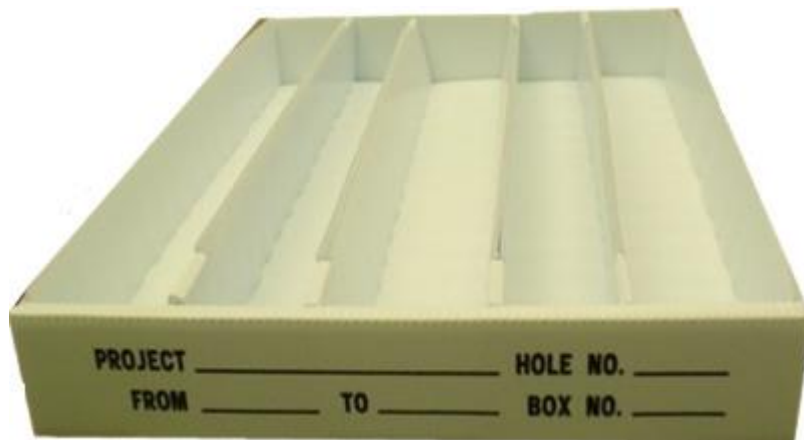
Rock core drilling is the process of recovering cylindrical samples (cores) of bedrock. A core barrel equipped with a drill bit is attached to the bottom of the drill rods, and as the rods and core barrel are rotated and advanced, the core is collected in the core barrel. Drilling fluid, which typically consists of water but can also be a variety of slurries, is pumped through the drilling rods and core barrel during advancement to flush cuttings from the boring and to lubricate and cool the drill bit.

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Core barrel: A hollow cylinder attached to a specially designed bit and which is used to obtain and to preserve a continuous section, or core, of the rocks penetrated in drilling.



Core boxes: These are the box which is used for placing and preserving samples after they were taken out from the core barrels and is further used for investigation to find its quality and strength.



Field tests for bearing capacity:

The test which is conducted at the site of construction before constructing the foundation is called field test

Major advantages of field tests are

- Sampling not required
- Soil disturbance minimum
- Large soil mass is tested which is not possible in the lab.

Major disadvantages of field tests are

- Laborious
- Time consuming
- Cannot control initial state of stress during test

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- Many times stress induced during the testing are horizontal while building loads are vertical
- Heavy equipment to be carried to field
- Short duration behavior
- Many times results are empirical

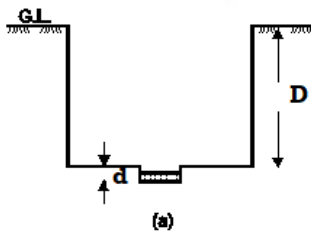
Bearing capacity of soil in the field can be evaluated by using following two commonly used methods.

- A) Plate load test and
- B) Standard penetration test

1. Plate load test:

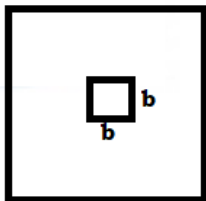
Object: Field test to determine ultimate bearing capacity of soil and the probable settlement under a given loading.

Specifications: [IS 1888-1962]



a) **Bearing plate:** Shape (square or circular), size (300mm to 750mm), thickness: decided based on bending stress consideration however it should not be less than 25mm.

b) **Pit:** Size – five times the size of plate, depth should satisfy $\frac{D}{B} = \frac{d}{b}$



Procedure: The experimental arrangement is as shown.

- 1) To start with seating load (7kPa) is applied (to Compensate unevenness of ground), which is released before starting actual test and dial gauges are set to read zero.
- 2) Loads are applied in cumulative load increments of not more than minimum of following values a) 20% of expected safe bearing capacity, b) 10% of ultimate bearing capacity c) twice the design load.
- 3) Each load is maintained constant until the settlement rate is less than 0.002 mm/hr.
- 4) Settlements are recorded at following time interval of 1, 2, 5, 10, 20, 40min, 1hr, 2 hr, 3 hr ... 24 hrs.
- 5) Apply the next increment and repeat the procedure.

- 6) Testing is continued until one of following stage is reached
- Applied pressure exceeds 3 times allowable pressure
 - Total settlement exceeds 10% width of plate

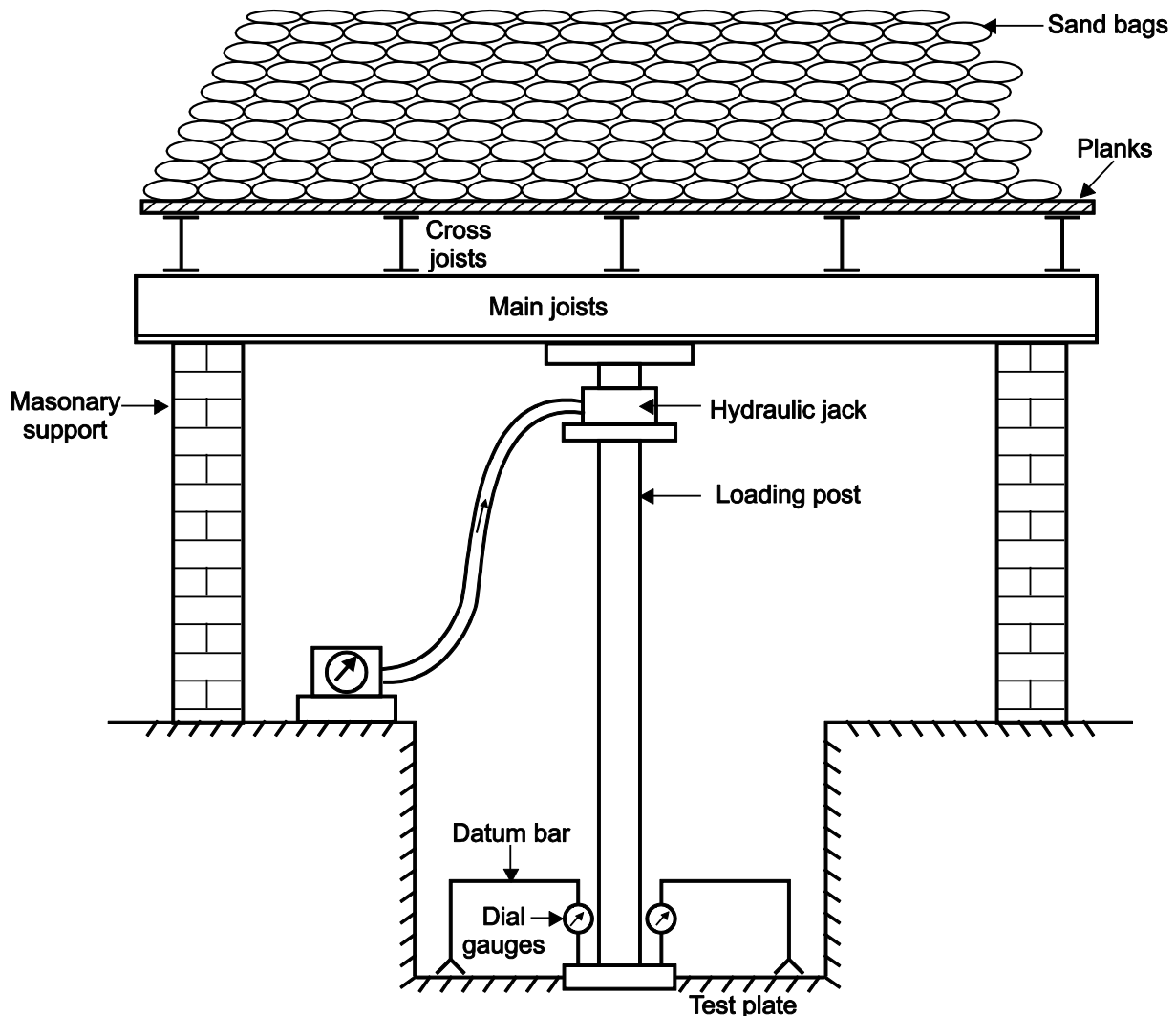


Fig Plate load test set up

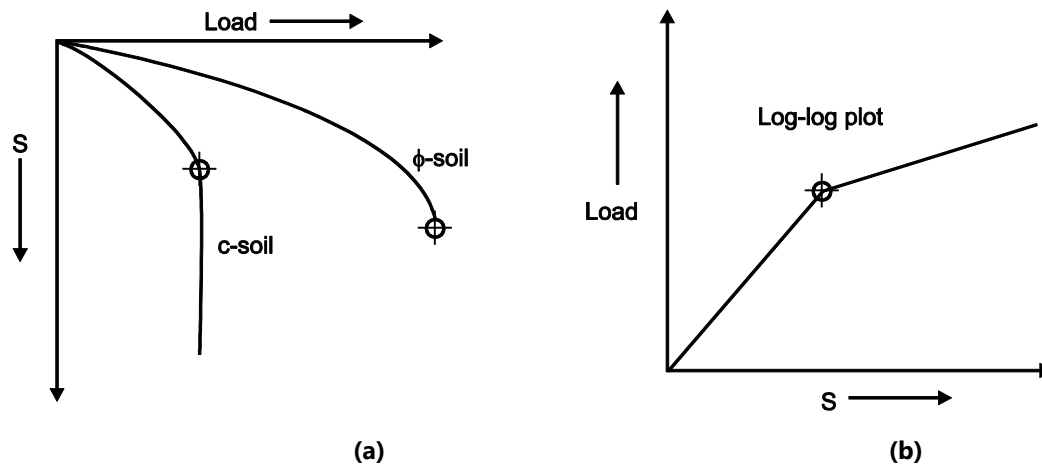
Interpretation:

A) Ultimate bearing capacity of plate (q_{up}): It is the ordinate corresponding to the point where graph changes its direction suddenly; intersection of tangent drawn to initial and final portion of the curve, load corresponding to settlement of 20% width of plate and if the graph consist of vertical q_u (p) is obtained from the graph.

B) Ultimate bearing capacity of footing (q_{uf}):

a) for clayey soil $q_{uf} = q_{up}$

b) for sandy soil $q_{uf} = q_{up} \frac{B}{b}$



C) Settlement of footing: If for a given pressure settlement of plate is S_p , then settlement of footing for same pressure is given by

a) Clayey soil

$$S_f = S_p \frac{B}{b}$$

b) Sandy soil

$$S_f = S_p \left[\frac{B(b+0.3)}{b(B+0.3)} \right]^2$$

In above equations b is width of plate and B is width of footing both need to be taken in meters.

D) Safe settlement pressure: (Taylors method)

$$\frac{q}{s} = C_1 \left[1 + 2 \frac{D}{B} \right] + \frac{C_2}{B}$$

Where, S – Permissible settlement, q – safe settlement pressure

B and D – Width and depth of footing

C_1 and C_2 - Soil parameters to be obtained from two plate load test

E) Effect of depth on settlement:

When depth of embedment of footing is increased, the settlement of footing will be decreased. If S_1 and S_2 are settlement of footing (size B) at depth D_1 and D_2 . These settlements are related by following relation

$$\frac{S_1}{S_2} = \sqrt{\frac{1+2 \left(\frac{D_2}{B} \right)}{1+2 \left(\frac{D_1}{B} \right)}}$$

The load settlement curve should pass through the origin and in case it does not, the "zero correction" is determined by extending the earliest straight line portion to intersect the vertical axis. The intercept on settlement axis at zero load is "**zero correction**". Corrected load settlement curve is obtained by substituting the correction from the recorded settlement.

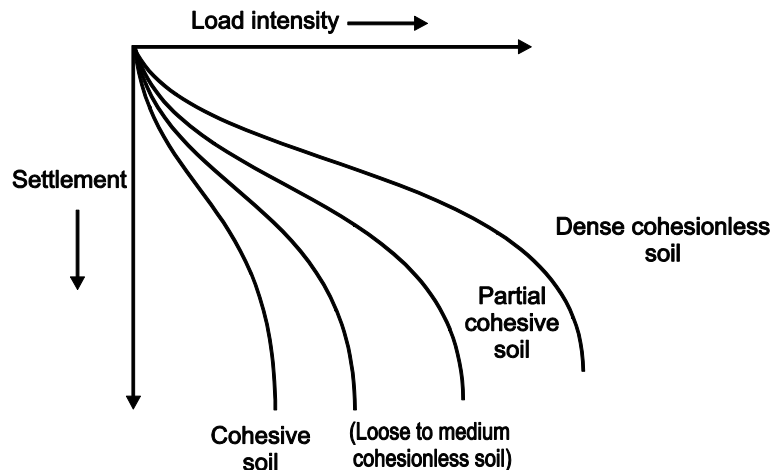


Fig. 2.11: Typical load settlement curve

The advantages of Plate Load Test are

1. It provides the allowable bearing pressure at the location considering both shear failure and settlement.
2. Being a field test, there is no requirement of extracting soil samples.
3. The loading techniques and other arrangements for field testing are identical to the actual conditions in the field.
4. It is a fast method of estimating ABP and $P - \Delta$ behavior of ground.

The disadvantages of Plate Load Test are

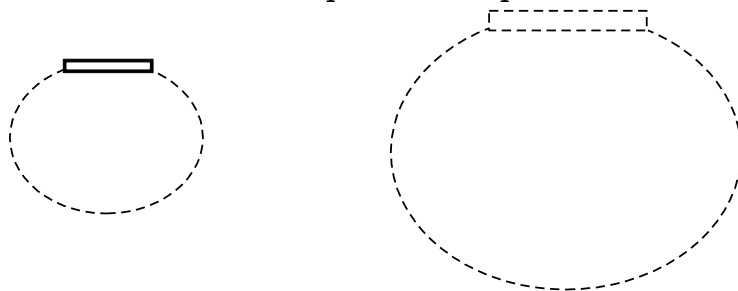
1. The test results reflect the behavior of soil below the plate (for a distance of $\sim 2B_p$), not that of actual footing which is generally very large.
2. It is essentially a short duration test. Hence, it does not reflect the long term consolidation settlement of clayey soil.
3. Size effect is pronounced in granular soil. Correction for size effect is essential in such soils.

It is a cumbersome procedure to carry equipment, apply huge load and carry out testing for several days in the tough field environment.

Limitations of Plate Load Test:

Results obtained by PLT may not be representative due to following factors

- **Size effect:** Results of the PLT test reflect the strength and settlement characteristics of soil within the pressure bulb of the test plate, which may not extend to the zone up to which pressure bulb of footing extends.



- **Scale effect:** Bearing capacity of cohesive soil is independent of size of footing, and that of cohesion-less soil depends on size of footing whereas settlement in both types of soil is function of size of footing. Thus bearing capacity and settlement of actual foundation will differ from those obtained due to small test plate
- **Time effect:** PLT is short duration test, thus for clayey soil it does not give ultimate settlement
- **Interpretation of failure load:** The failure load in PLT is not well defined except in case of general shear failure, so the load obtained in other types of failure will differ
- **Water table effect:** The position of water table lying below the test level has no influence on bearing capacity of plate however bearing capacity of actual footing get effected by water table position when it lies above the significant depth level.
- **Site and season:** It is difficult to obtain results that represent the whole site and the whole season. In summer it gives good results compared to that in wet season.
- **Shape of foundation:** Test ignores the effect of shape of footing on bearing capacity and settlement. As such results of PLT test using square plate cannot be applied to circular, rectangular or strip footing.

2. Standard Penetration test: (IS 2131 – 1981)

Procedure

- The borehole is advanced to the required depth and the bottom cleaned.
- The split-spoon sampler, attached to standard drill rods of required length is lowered into the borehole and rested at the bottom
- The split-spoon sampler is driven into the soil for a distance of 450mm by blows of a drop hammer (monkey) of 65 kg falling vertically and freely from a height of 750 mm.
- The number of blows required to penetrate every 150 mm is recorded while driving the sampler.
- The number of blows required for the last 300 mm of penetration is added together and recorded as the N value at that particular depth of the borehole.
- The number of blows required to effect the first 150mm of penetration, called the seating drive, is disregarded.

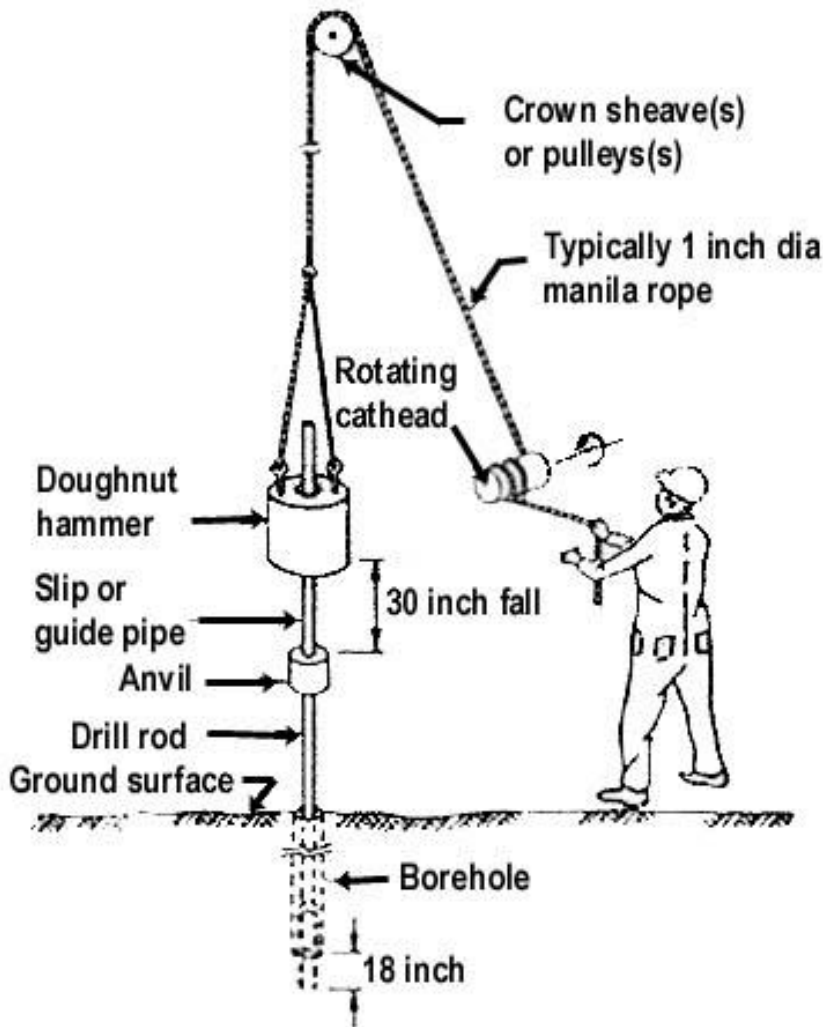
The split-spoon sampler is then withdrawn and is detached from the drill rods. The split-barrel is disconnected from the cutting shoe and the coupling. The soil sample collected inside the split barrel is carefully collected so as to preserve the natural moisture content and transported to the laboratory for tests. Sometimes, a thin liner is inserted within the split-barrel so that at the end of the SPT, the liner containing the soil sample is sealed with molten wax at both its ends before it is taken away to the laboratory.

The SPT is carried out at every 0.75 m vertical intervals in a borehole. This can be increased to 1.50 m if the depth of borehole is large. Due to the presence of boulders

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or rocks, it may not be possible to drive the sampler to a distance of 450 mm. In such a case, the N value can be recorded for the first 300 mm penetration.

The boring log shows refusal and the test is halted if 50 blows are required for any 150mm penetration 100 blows are required for 300m penetration 10 successive blows produce no advance.



Experimental arrangement for the test

Precautions

The drill rods should be of standard specification and should not be in bent condition.

The split spoon sampler must be in good condition and the cutting shoe must be free from wear and tear. The drop hammer must be of the right weight and the fall should be free, frictionless and vertical. The height of fall must be exactly 750 mm. Any change from this will seriously affect the N value.

The bottom of the borehole must be properly cleaned before the test is carried out. If this is not done, the test gets carried out in the loose, disturbed soil and not in the undisturbed soil. When a casing is used in borehole, it should be ensured that the

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casing is driven just short of the level at which the SPT is to be carried out. Otherwise, the test gets carried out in a soil plug enclosed at the bottom of the casing.

When the test is carried out in a sandy soil below the water table, it must be ensured that the water level in the borehole is always maintained slightly above the ground water level. If the water level in the borehole is lower than the ground water level, 'quick' condition may develop in the soil and very low N values may be recorded. In spite of all these imperfections, SPT is still extensively used because the test is simple and relatively economical. It is the only test that provides representative soil samples both for visual inspection in the field and for natural moisture content and classification tests in the laboratory. SPT values obtained in the field for sand have to be corrected before they are used in empirical correlations and design charts. IS: 2131-1981 recommends that the field value of N be corrected for two effects, namely, (a) effect of overburden pressure, and (b) effect of dilatancy.

(a) Correction for overburden pressure:

Several investigators have found that the penetration resistance or the N value in a granular soil is influenced by the overburden pressure. Of two granular soils possessing the same relative density but having different confining pressures, the one with a higher confining pressure gives a higher N value. Since the confining pressure (which is directly proportional to the overburden pressure) increases with depth, the N values at shallow depths are underestimated and the N values at larger depths are overestimated. To allow for this, N values recorded from field tests at different effective overburden pressures are corrected to a standard effective overburden pressure.

$$N' = C_N N_o$$

Where N_o – Observed value of blow count

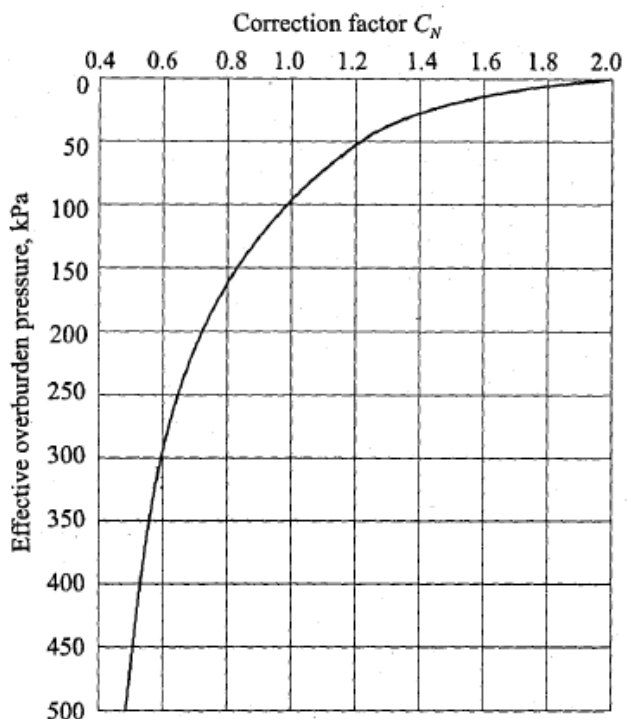
C_N – Normalized correction factor

N' – Blow count corrected for Overburden

the value of normalized correction factor C_N can be obtained by using chart or by using the formula as given below

$$C_N = 0.77 \log \frac{2000}{\sigma_o} \quad \text{for } \sigma_o \geq 25 \text{ kPa}$$

Where σ_o – effective Overburden pressure at the test level



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at $\sigma_o = 100kPa$ Correction factor = 1, at ground surface correction factor = 2

at $\sigma_o = 10kPa$ correction factor = 1.8, at $\sigma_o = 20kPa$ correction factor = 1.6

at $\sigma_o = 25kPa$ correction factor = 1.5, at $\sigma_o = 450kPa$ correction factor = 0.5

2) Dilatancy correction: The value of N_0 corrected for overburden need to be corrected for dilatancy if the subsoil consists of fine sand and silt below the water table. (Sand and silt below water table offer higher resistance to driving due to development of excess pore pressure, which could not be dissipated immediately) This correction is applied only when N_1 is more than 15

$$N_c = 15 + 0.5(N' - 15)$$

Correlations of blow count value with various soil properties are as below

for sandy soil				for clayey soil		
Nc	Relative density	ϕ		Nc	Consistency	UCS kPa
< 4	15%	28	Very loose	0-2	Very soft	< 25
4-10	15% - 35%	30	Loose	2-4	soft	25-50
10-20	35% - 50%	32	Moderately dense	4-8	Firm to medium	50-100
20-30	50% - 65%	36	Medium dense	8-16	stiff	100 -200
30-50	85% - 99%	42	Dense	16-32	Very stiff	200 - 400
>50	100%	45	Very dense	> 32	hard	> 400

Soil modulus: for sandy soil $E = 500(N+15)$ kPa and for clayey soil $E = 300(N+5)$ kPa

The values of C and ϕ were obtained from value of N_c and then by using Terzaghi equation bearing capacity can be calculated.

For strip footing: $q_{nu} = 0.471BN^2W_r + 0.785(100 + N^2)dW_q$ **kPa**

For square or circular footing $q_{nu} = 0.314BN^2W_r + 0.943(100 + N^2)dW_q$ **kPa**

Where B is width of footing, W_q and W_r are water table correction factors

Advantages of Standard Penetration Test are

1. Relatively quick & simple to perform
2. Equipment & expertise for test is widely available
3. Provides representative soil sample
4. Provides useful index for relative strength & compressibility of soil
5. Able to penetrate dense & stiff layers
6. Results reflect soil density, fabric, stress strain behavior
7. Numerous case histories available

Disadvantages of Standard Penetration Test are

1. Requires the preparation of bore hole.
2. Dynamic effort is related to mostly static performance
3. SPT is abused, standards regarding energy are not uniform
4. If hard stone is encountered, difficult to obtain reliable result.
5. Test procedure is tedious and requires heavy equipment.
6. Not possible to obtain properties continuously with depth.

Module 2: Bearing capacity analysis (7 Lectures)

Failure modes, Terzaghi's Analysis, specialization of Terzaghi's equations, Skempton values for N_c , Meyerhof's analysis, I.S. code method of bearing capacity evaluation, effect of water table, eccentricity of load. Safe bearing capacity and allowable bearing pressure, settlement analysis: Immediate settlement – consolidation settlement, differential settlement, Tolerable settlement, angular distortion

Introduction: Foundation is the lowermost part of structure which is in direct contact with the ground and transmits the load of structure to the subsoil. The subsoil should have sufficient strength to bear the load being transmitted by foundation without rupture or shear failure. This capacity of soil is called its 'bearing capacity'.

Basic terms

1. **Bearing Capacity:** Capacity of the subsoil to bear a load.
2. **Ultimate Bearing Capacity (q_u):** It is the gross pressure at which base of foundation at soil fails in shear.
3. **Net Ultimate Bearing Capacity (q_{nu}):** It is the minimum net pressure (Gross pressure – Pressure due to self-weight of soil) at which soil fails in shear.

$$q_{nu} = q_u - \gamma d$$

4. **Net Safe Bearing Capacity (q_{ns}):**

$$q_{ns} = \frac{q_{nu}}{F} \quad \text{where } F - \text{factor of safety } (= 3)$$

5. **Gross Safe Bearing Capacity or safe bearing capacity (q_s):** Maximum pressure which the soil can carry safely without risk of shear failure. $q_s = q_{ns} + \gamma d$
where, γ - unit weight of soil, d - depth of footing
6. **Net Safe Settlement Pressure (q_p):** Maximum net pressure intensity of the loading, the soil can carry without exceeding allowable settlement.
7. **Allowable Soil Pressure, ASP or Allowable bearing pressure(q_a) :** It is the net intensity of loading (maximum) at which the soil neither fails in shear nor there is excessive settlement.

MODES OF SHEAR FAILURE:

Bearing capacity failures of foundations can be grouped into three categories as indicated below:

- General shear failure
 - Local shear failure
 - Punching shear failure.
- a) **General Shear Failure:** This kind of failure occurs in soil possessing, brittle type stress-strain behaviour and is characterized by:
- Well defined failure pattern up to the ground surface.

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- Sudden and catastrophic failure accompanied by tilting of base.
 - Bulging of ground surface adjacent to foundation.
 - Load settlement curve shows pronounced peak.
 - Failure occur at very small vertical strain accompanied by large lateral strain.
 - For granular soil - This failure is observed if soil is dense.
 - For cohesive soil - This failure is observed if soil has stiff consistency.
- b) Local Shear Failure:** This kind of failure occurs in soils possessing elasto-plastic stress-strain behaviour and is characterized by:
- Well defined Wedge and slip surfaces only beneath the foundation.
 - There is neither visible collapse nor substantial tilt of foundation.
 - Slight bulging of ground surface adjacent to foundation.
 - It is difficult to locate failure point (load) on load settlement curve.
 - Failure occur at a very small lateral strain and large vertical strain.
 - For granular soil - This failure is observed if soil has medium density.
 - For cohesive soil - This failure is observed if soil has medium consistency.
- c) Punching Shear Failure:** This kind of failure occurs in soil possessing plastic type stress-strain behaviour and is characterized by:
- Poorly defined slip surfaces.
 - Footing sink into the soil.
 - Soil beyond the loaded area is little or not affected. So there is no bulging of soil.
 - Very difficult to locate the failure point on load settlement curve.
 - Failure occurs at negligible or zero lateral strain and substantial vertical strain.
 - For granular soil - Loose to very loose soil.
 - For cohesive soil - Very soft to soft consistency.

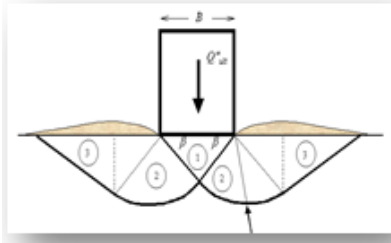
Comparison of General shear failure and local shear failure

	General Shear Failure	Local Shear Failure
1	Occurs in dense/stiff soil	Occurs in loose/soft soil
2	Soil parameters a) Angle of internal friction $\Phi > 36^\circ$ b) Blow count $N > 30$ c) Density index $ID > 70\%$ d) Undrained cohesion $C_u > 100\text{kPa}$ e) Void ratio of soil $e < 0.55$	Soil parameters a) Angle of internal friction $\Phi < 28^\circ$ b) Blow count $N < 5$ c) Density index $ID > 20\%$ d) Undrained cohesion $C_u < 50\text{kPa}$ e) Void ratio of soil $e > 0.75$
3	Results in small vertical strain ($< 5\%$)	Results in large vertical strain ($> 20\%$)
4	Failure pattern well defined & clear	Failure pattern not well defined
5	Well defined peak in P- Δ curve	No peak in P- Δ curve
6	Bulging formed in the neighborhood of footing at the surface	Slight Bulging observed in the neighborhood of footing
7	Extent of horizontal spread of	Extent of horizontal spread of disturbance

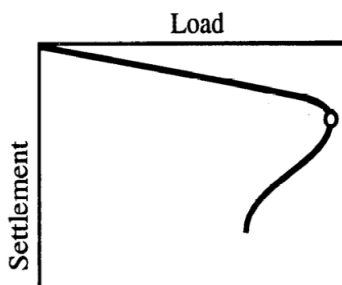
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	disturbance at the surface large	at the surface very small
8	Observed in shallow foundations	Observed in deep foundations
9	Failure is sudden & catastrophic	Failure is gradual (progressive)
10	Less settlement, but tilting failure observed	Considerable settlement of footing observed

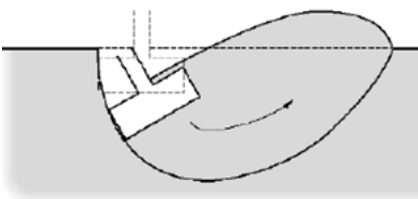
General Shear failure



Failure zones

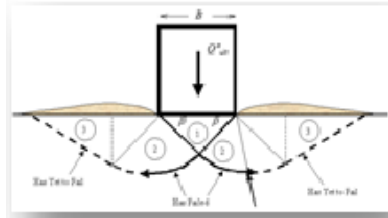


Load settlement curve

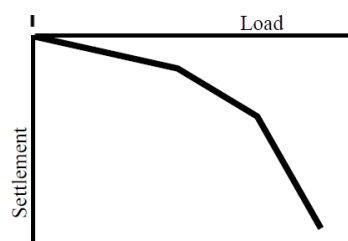


Footing movement

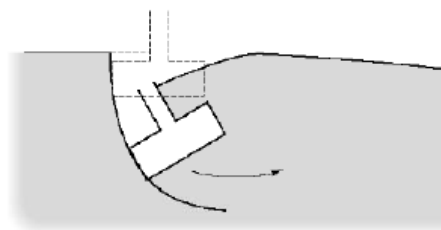
Local Shear failure



Failure zones

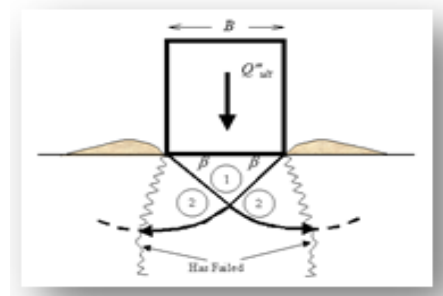


Load settlement curve

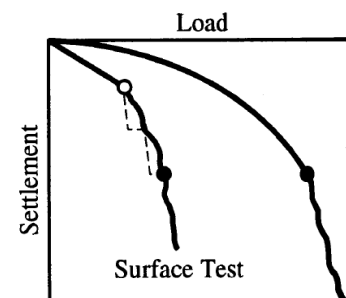


Footing movement

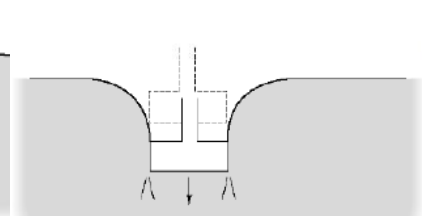
Punching Shear failure



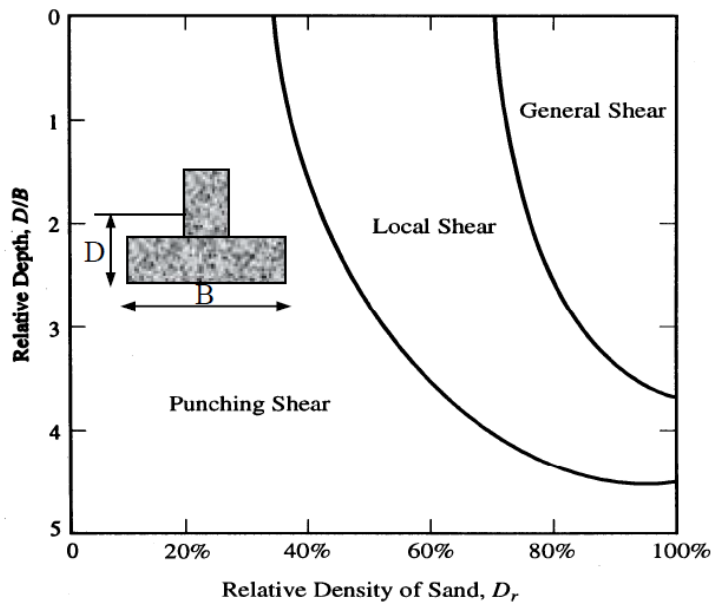
Failure zones



Load settlement curve



Footing movement

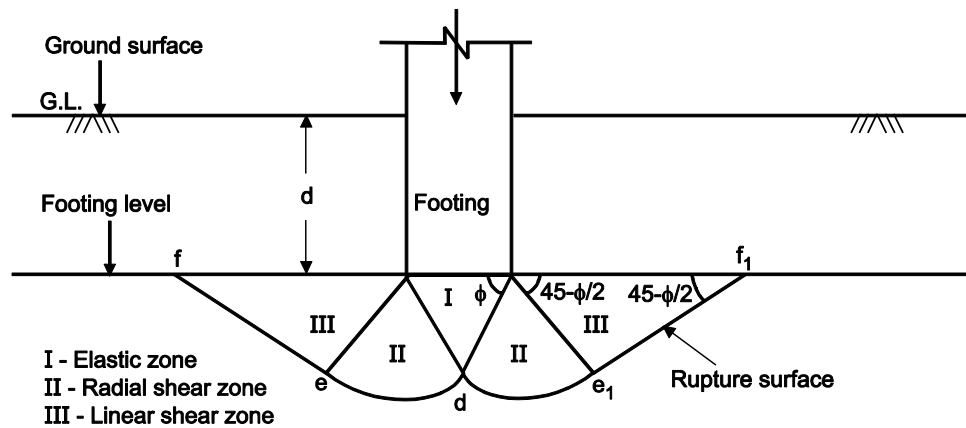


Types of Failure as a Function of Relative Density and Foundation Depth. (Vesic, 1963).

Terzaghi's bearing capacity equation:

Terzaghi gave a general theory of bearing capacity which is based on following assumptions:

- 1) Soil is homogeneous and Isotropic
- 2) Footing is laid at shallow depth
- 3) The shear strength of soil is represented by Mohr Coulombs Criteria.
- 4) The footing is of strip footing type with rough base. It is essentially a two dimensional plane strain problem.
- 5) Elastic zone has straight boundaries inclined at an angle equal to Φ to the horizontal.
- 6) Rupture surface extend only up to the level of base of footing.
- 7) Shear resistance of soil above the base of footing is neglected
- 8) Method of superposition is valid.
- 9) Passive pressure force has three components (PPC produced by cohesion, PPq produced by surcharge and PPy produced by weight of shear zone).
- 10) Effect of water table is neglected.
- 11) Base of footing is rough
- 12) Footing carries axial and vertical loads.
- 13) Footing and ground are horizontal
- 14) Limit equilibrium is reached simultaneously at all points. Complete shear failure Is mobilized at all points at the same time.
- 15) The properties of foundation soil do not change during the shear failure.



Terzaghi's analysis – Rupture surface

A strip footing of width B gradually compresses the foundation soil underneath due to the vertical load from superstructure. Let q_f be the final load at which the foundation soil experiences failure due to the mobilization of plastic equilibrium. The foundation soil fails along the composite failure surface and the region is divided into five zones, Zone 1 which is elastic, two numbers of Zone 2 which are the zones of radial shear and two zones of Zone 3 which are the zones of linear shear. Considering horizontal force equilibrium and incorporating empirical relation, the equation for ultimate bearing capacity is obtained as follows.

Let F = Factor of safety (usually 3),

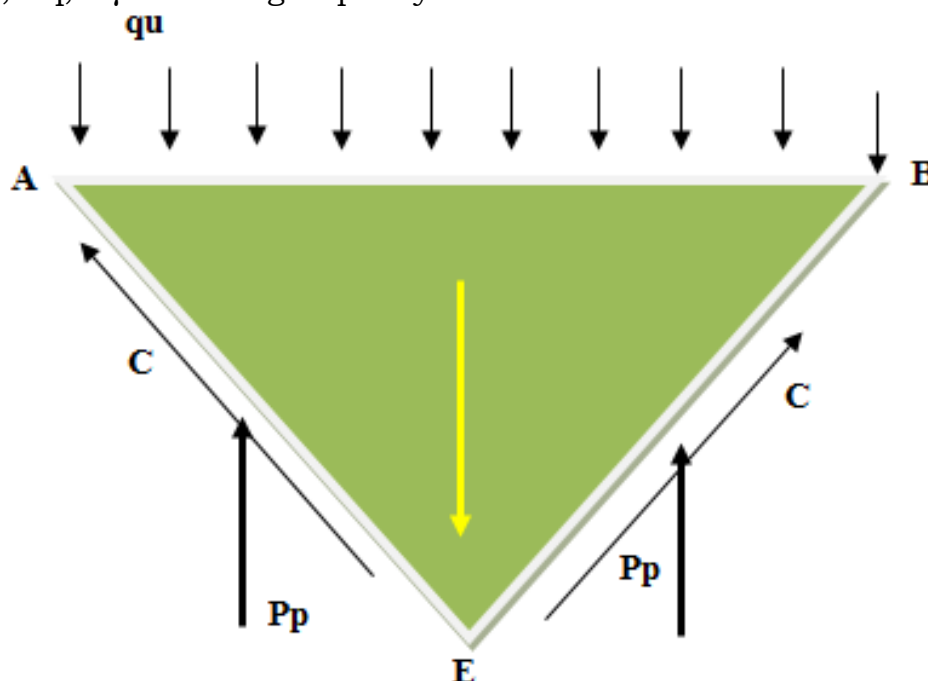
c = cohesion, γ = unit weight of soil and ϕ – angle of internal friction

D = Depth of foundation

q = Surcharge at the base of footing

B = Width of foundation

N_c, N_q, N_γ = Bearing Capacity factors



Free body diagram of triangular wedge

This triangular wedge is subjected to following forces

- 1) Contact pressure q_u acting downward
- 2) Self weight of wedge (W) acting downward
- 3) Cohesion acting on inclined surface of wedge
- 4) Passive pressure acting vertically on two inclined surface

Applying equilibrium equation to the wedge i.e. (sum of forces in vertical direction is zero)

$$q(BX1) + W - 2C\sin\phi - 2P_p = 0$$

$$W = \gamma \left[\frac{1}{2} B \frac{B}{2} \tan\phi \right] = \frac{1}{4} \gamma B^2 \tan\phi$$

C = Total cohesive force acting on each slant face of the wedge = $c \times (B/(2\cos\phi)) \times 1$

$$q_u(BX1) + \frac{1}{4} \gamma B^2 \tan\phi - 2\sin\phi \left(c \frac{B}{2\cos\phi} \right) - 2(P_{pc} + P_{pq} + P_{py}) = 0$$

Separating the terms containing Cohesion term, surcharge term and weight term i.e.

$$q_u B = [2P_{pc} + Bc \tan\phi] + [2P_{pq}] + \left[2P_{pr} - \frac{1}{4} \gamma B^2 \tan\phi \right]$$

$$q_u B = BcN_c + BqN_q + B(0.5rBN_r)$$

$$q_u = cN_c + qN_q + (0.5rBN_r)$$

This is the equation for calculating ultimate bearing capacity of strip footing

Values of Terzaghi's bearing capacity factors for various values of ϕ are as given below

ϕ	0	5	10	15	20	25	30	34	35	40	45	48	50
N_c	5.7	7.3	9.6	12.9	17.7	25.1	37.2	52.6	57.8	95.7	172.3	258.3	347.5
N_q	1.0	1.6	2.7	4.4	7.4	12.7	22.5	36.5	41.4	81.3	173.3	287.9	415.1
N_γ	0.0	0.5	1.2	2.5	5.0	9.7	19.7	36.0	42.4	100.4	297.5	780.1	1153.2

Specialization of Terzaghi's equations:

Terzaghi has derived an equation which is applicable for strip footing and for general shear failure of soil, later on this equation is modified to make it applicable for general case.

A) Modification of equation to account for Shape:

To account for the shape of footing each term of the equation is multiplied by the shape factor and modified equation will then be $q_u = cN_cS_c + qN_qS_q + 0.5rBN_rS_r$

The values of these shape factors are as given below

	Strip	circular	Square	Rectangular
S_c	1	1.3	1.3	$1 + 0.3(B/L)$
S_q	1	1.0	1.0	1
S_r	1	0.6	0.8	$1 - 0.2(B/L)$

B) Modification of equation to account for local shear failure:

To account for local shear type failure the equation is modified by modifying the value of shear parameters i.e. Replace c by c' ϕ by ϕ' and bearing capacity factors N by N' in the general equation given by Terzaghi where $c' = 0.67 c$ and $\tan \phi' = 0.67 \tan \phi$. The values of bearing capacity factors for modified values of ϕ need to be calculated. The values of bearing capacity factors for local shear failure are as given below

ϕ	0	5	10	15	20	25	30	35	40	45	50
N_c'	5.7	6.7	8.0	9.7	11.8	14.8	19.0	25.2	34.9	51.2	81.3
N_q'	1.0	1.4	1.9	2.7	3.9	5.6	8.3	12.6	20.5	34.1	65.6
N_γ'	0.0	0.2	0.5	0.9	1.7	3.2	5.7	10.1	18.8	37.7	87.1

Skempton's formula : Skempton proposed formula for purely cohesive soil

$$q_{nu} = c N_c \text{ where } c - \text{cohesion and } N_c - \text{Skempton's bearing capacity factor}$$

Case 1) for surface footing (D = 0)

$N_c = 5.14$ - for strip footing

$N_c = 6.20$ - for square or circular footing

$$N_c = 5.14 \left[1 + 0.2 \frac{B}{L} \right] \text{ for rectangular footing}$$

Case 2) for footing with [D/B < 2.5]

$$N_c = \left[1 + 0.2 \frac{D}{B} \right] \times [N_c]_{\text{Surface}}$$

Case 3) for footing with [D/B > 2.5]

$$N_c = 1.5 \times [N_c]_{\text{Surface}}$$

IS code method (IS 6403-1981) for bearing capacity:

IS code has given an equation to calculate the bearing capacity. Net ultimate bearing capacity is given by

$$q_{nu} = c N_c S_c d_c i_c + q (N_q - 1) S_q d_q i_q + 0.5 \gamma B N_\gamma S_\gamma d_\gamma i_\gamma W' - \text{For general shear failure}$$

$$q_{nu} = c' N_c' S_c d_c i_c + q (N_q' - 1) S_q d_q i_q + 0.5 \gamma B N_\gamma' S_\gamma d_\gamma i_\gamma W' - \text{For local shear failure}$$

Where

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N_c, N_q, N_γ – Bearing capacity factors for general shear

N'_c, N'_q, N'_γ – Bearing capacity factors for local shear

Bearing capacity factors for general shear failure can be obtained from the table given below

ϕ	0	5	10	15	20	25	30	35	40	45	50
N_c	5.14	6.49	8.35	10.98	14.38	20.72	30.14	46.12	75.31	138.88	266.89
N_q	1.0	1.57	2.47	3.94	6.40	10.66	18.40	33.30	64.20	134.88	319.07
N_γ	0.00	0.45	1.22	2.65	5.39	10.88	22.40	48.03	109.41	217.76	762.89

Note: for obtaining values of N'_c, N'_q, N'_γ can be calculated by calculating the value of ϕ' use this value to calculate the values of bearing capacity factors from above table instead of ϕ .

The values of **shape factor** for various shapes of footing is as given below

	Strip	circular	square	Rectangular
S_c	1.0	1.3	1.3	$1.0 + 0.2 (B / L)$
S_q	1.0	1.2	1.2	$1.0 + 0.2(B/L)$
S_γ	1.0	0.6	0.8	$1.0 - 0.4(B / L)$

The values of **depth factor** can be calculated by following equation

$$d_c = 1 + 0.2 \frac{D}{B} \sqrt{N_\phi}$$

$$d_q = d_\gamma = 1 \text{ for } \phi < 10^\circ$$

$$d_q = d_\gamma = 1 + 0.1 \frac{D}{B} \sqrt{N_\phi} \text{ for } \phi > 10^\circ$$

$$\text{Where } N_\phi = \tan^2 \left(45 + \frac{\phi}{2} \right)$$

Note: Depth correction is to be applied only when backfilling is done with proper compaction

The values of **inclination factor** can be calculated by following equation

$$i_c = i_q = \left(1 - \frac{\alpha}{90} \right)^2 \quad \text{and} \quad i_\gamma = \left(1 - \frac{\alpha}{\phi} \right)^2$$

Where α is inclination of load with vertical.

Meyerhoff's method:

Meyerhoff equation for ultimate bearing capacity of footing is as below

$$q_u = cN_c S_c d_c i_c + qN_q S_q d_q i_q + 0.5\gamma B N_\gamma S_\gamma d_\gamma i_\gamma$$

Where C – cohesion of soil (kPa), B – width of footing (m), D – depth of footing (m),

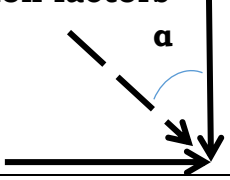
γ – Unit weight of soil (kN/m³), ϕ – Angle of internal friction for the soil

N – Meyerhoff's bearing capacity factors and are given by

$$N_q = e^{\pi \tan \phi} K_p, \quad N_c = (N_q - 1) \cot \phi \quad \text{and} \quad N_\gamma = (N_q - 1) \tan(1.4\phi)$$

Shape depth and inclination factors:

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Factors	Value	Applicable when ϕ is
Shape	$S_c = 1 + 0.2 K_p \frac{B}{L}$	Any value
	$S_q = S_r = 1 + 0.1 K_p \frac{B}{L}$	More than 10°
	$S_q = S_r = 1$	Equal to zero
Depth	$d_c = 1 + 0.2 \sqrt{K_p} \frac{D}{B}$	Any value
	$d_q = d_r = 1 + 0.1 \sqrt{K_p} \frac{D}{B}$	More than 10°
	$d_q = d_r = 1$	Equal to zero
Inclination factors 	$i_c = i_q = \left(1 - \frac{\alpha}{90}\right)^2$	Any value
	$i_\gamma = \left(1 - \frac{\alpha}{\phi}\right)^2$	More than 0°
	$i_\gamma = 0$	Equal to zero

Meyerhoffs bearing capacity factors:

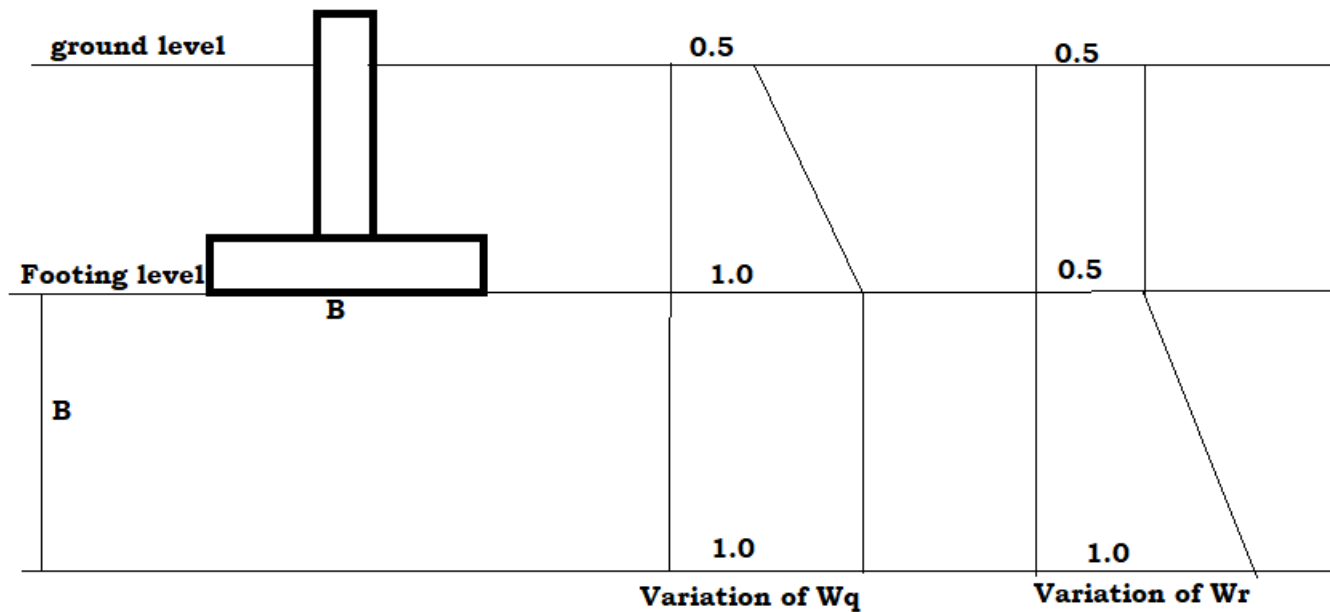
ϕ	0	5	10	15	20	25	30	35	40	45	50
N_c	5.14	6.5	8.3	11.0	14.8	20.7	30.1	46.1	75.3	133.9	266.9
N_q	1.0	1.6	2.5	3.9	6.4	10.7	18.4	33.3	64.2	134.9	319.0
N_γ	0	0.1	0.4	1.1	2.9	6.8	15.7	37.1	93.7	262.7	873.7

Effect of position of water table: The effective unit weight of the soil is used in the bearing-capacity equations for computing the ultimate capacity. This has already been defined for q in the qN_q term. Also wedge term $0.5\gamma BN_\gamma$ also uses the effective unit weight for the soil. Presence of water table in the vicinity of footing will decrease the bearing capacity of soil its effect is accounted either by introducing water table correction factors in the equation or by using equivalent unit weight of soil

Method 1) Using water table correction factors:

Multiply second term of the equation with water table correction factor [W_q or R_{w1}] and third term of the equation with water table correction factor [W_γ or R_{w2}]

i.e by $q_u = cN_cS_c + qN_qS_qW_q + 0.5\gamma BN_\gamma S_\gamma W_\gamma$



Method 2) Using equivalent unit weight for the soil: In this method weighted unit weight (equivalent unit weight) is determined. For soil above water table bulk unit weight is considered and for soil below water table submerged unit weight is considered.

When water table is in between ground level and footing level:

For surcharge term: $q = \gamma_e D = \gamma d + \gamma' (D - d)$

$$\gamma_e = \frac{\gamma d + \gamma' (D - d)}{D} = \gamma' + \frac{d}{D} (\gamma - \gamma')$$

For weight term : $\gamma B = \gamma_e B = \gamma' B$ i.e. $\gamma_e = \gamma'$

When water table is below footing

$q = \gamma_e D = \gamma d$ i.e. $\gamma_e = \gamma$

$\gamma B = \gamma_e B = \gamma b + \gamma' (B - b)$

i. e.

$$\gamma_e = \frac{\gamma b + \gamma' (B - b)}{B} = \gamma' + \frac{b}{B} (\gamma - \gamma')$$

Effect of eccentricity of load:

When the load acting on footing is not axial we need to modify the dimensions of footing in such a way that for this modified dimensions the load will be axial. These reduced dimensions need to be used in place of original dimensions. The modified dimensions are given by original dimension – twice the eccentricity in that direction.

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These dimensions need to be used while calculating various factors of the bearing capacity equation

$$L' = L - 2e_L, \quad B' = B - 2e_B \quad \text{and} \quad A' = L' \times B'$$

Settlement:

All structures that civil Engineers construct are founded on soil / rock. These structures apply load on the soil. When soil has to carry additional load, the soil particles have to adjust themselves and regroup themselves in a closer packing to withstand that load. This adjustment and regrouping results in causing the structures to settle.

The vertical downward movement of a loaded base is called Settlement. The external load acting on soil mass produces a time dependent accumulation of particle rolling, sliding, crushing and elastic distortion in a limited influence zone, beneath the loaded area. The statistical accumulation of movement in the direction of interest is called Settlement.

Causes of Settlement

The settlement (decrease in volume of a soil mass) is due to decrease in the volume of voids in the soil by expulsion of air, expulsion of water, compression of air, compression of soil solid and distortion of adsorbed water.

Table

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Internal Factor		External Factor
1. Compression of air	Initial	1. Static load
2. Expulsion of air		2. Dynamic load - (Machinery pile driving, explosion, earthquake etc.)
3. Rearrangement of particles (sliding, rolling)	Primary	3. W.T. fluctuation (stress variation and shrinkage)
4. Expulsion of water		4. Neighboring construction
5. Compression of soil particles		5. Other causes :
6. Reorientation of adsorbed water	Secondary	• Underground erosion • Thermal changes • Frost heave • Mining subsidence

Components of settlement: Total settlement of foundation is sum total of three components a) Initial settlement or elastic settlement, b) Primary consolidation settlement and c) secondary consolidation settlement

The settlement which will take place immediately after application of load is called immediate or elastic settlement

Mechanism of settlement of structures:

Mechanism	Nature	Duration
Elastic	Due to change in shape of stressed soil zone at constant volume.	Immediate
Compression of air	In partially saturated soil only, due to reduction of volume of air voids	Immediate
Consolidation Primary	Due to flow of water from voids of soil resulting in rearrangement of soil particles in a packing suitable to carry load imposed by the structure	Function of permeability and compressibility of soil
Secondary consolidation	Due to creep like behaviour of soil	Very slow

In general settlement depends on following factors

- Magnitude of the stresses applied.
- Geometry of plan area which is stressed.
- Depth at which stress is applied in relation to width of stressed area (D/B), and the thickness of compressible stratum.

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- Compressibility of stratum.
- Rigidity of the foundation element.

Types of Movement in different soils:

Principal soil types	Type of movement			
	Immediate	Consolidation	Creep	Swell
Rock	yes	No	No	some
Gravel	yes	No	No	No
Sand	yes	No	No	No
Silt	yes	Minor	No	yes
Clay	yes	Yes	No	yes
Organic	yes	Minor	No	yes

ELASTIC SETTLEMENT: The settlement which will take place immediately after applying load is called elastic settlement or immediate settlement. It occurs on account of elastic behaviour that produces distortion at constant volume. It is determined from elastic theory; it occurs in all types of soil due to elastic compression. It depends on the elastic properties of foundation soil, rigidity, size and shape of foundation.

It is computed by using

$$S_i = q B \left(\frac{1-\mu^2}{E} \right) I$$

Where q – contact pressure at base of footing, B – width of footing

I – influence coefficient, E – modulus of elasticity of soil

μ – poisons ratio for the soil

Typical Range of Poisson's Ratio		Soil modulus for various soil	
<i>Soil Type</i>	<i>Poissons ratio</i>	<i>Soil type</i>	<i>Soil modulus (kPa)</i>
Saturated clay	0.5	Very soft clay	400 – 3000
Sandy clay	0.3 – 0.4	Soft clay	1500 – 4000
Unsaturated clay	0.35 – 0.4	Medium clay	3000 – 8500
Loess	0.44	Hard clay	7000 – 17000
Silt	0.3 – 0.35	Sandy clay	28000 – 42000
Sand	0.15 – 0.30		
Rock	0.1 – 0.4		

The values of poisons ratio and modulus of elasticity values are only the representative values and can be used only in the absence of more accurate data. The influence factor I depends on the shape and flexibility of footing. Further, in flexible footing I is not constant. Table presents the different values of I for different soils.

Shape of footing	Flexible			Rigid
	Center	Corner	Average	
Circle	1	0.64	0.85	0.8
square	1.12	0.56	0.95	0.9
Rectangle L/B = 1.5	1.36	0.68	1.20	1.09
L/B = 2	1.52	0.77	1.31	1.22
L/B = 5	2.1	1.05	1.83	1.68
L/B = 10	2.52	1.26	2.25	2.02
L/B = 100	3.38	1.69	2.96	2.70

Consolidation settlement: The settlement of a soil which takes place due to expulsion of water from the soil and is mainly applicable to saturated or nearly saturated clay.

It occurs due to the process of consolidation, Clay and Organic soil are most prone to consolidation settlement, Consolidation is the process of reduction in volume due to expulsion of water under an increased load, it is a time related process occurring in saturated soil by draining water from void, it is often confused with Compaction. Consolidation theory is required to predict both rate and magnitude of settlement.

This settlement is given by

$$S = C_c \frac{H}{1+e} \log_{10} \frac{P+\Delta P}{\Delta P} \quad \text{or} \quad S = \frac{a_v}{1+e} \Delta P \times H = m_v \times \Delta P \times H$$

Where

S – Settlement

H – Thickness of compressible stratum

e - Initial void ratio

C_c – compression index

a_v - coefficient of compressibility

m_v – coefficient of volume compressibility

P – Initial pressure at center of layer (Geostatic stress)

ΔP – Increase in pressure at centre of layer due to load on footing (stress due to external load)

Secondary Compression: The settlement which occurs at very slow and negligible rate after completion of initial consolidation is called secondary consolidation or creep settlement

- This settlement starts after the primary consolidation is completely over.
- During this settlement, excess pore water pressure is zero.

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- The reasons for secondary settlement are not clear.
- This is creep settlement occurring due to the readjustment of particles to a stable equilibrium under sustained loading over a long time.
- This settlement is common in very sensitive clay, organic soils and loose sand with clay binders.

It is calculated by using

$$S_s = C_\alpha H \log \left(\frac{t_p + \Delta t}{t_p} \right)$$

Where C_α = Coefficient of secondary compression

H = Thickness of clay layer

Δt = Time increment producing S_s

t_p = Time taken for primary consolidation to complete

Typical values of C_α

Type of soil	C_α
Normally consolidated clays	0.005 – 0.02
Very plastic soils, organic soils	≥ 0.03
Pre compressed clays with OCR > 2	< 0.001

Types of Settlement

1. **Uniform Settlement:** If every part of structure settles by same amount.
2. **Non-Uniform Settlement:** If different parts of the structure settle by unequal amount then it is called Non-uniform settlement and the difference between settlement at any two points is called **Differential settlement**.

Differential settlement per unit length is called **Tilt or Angular distortion**.

Causes of Differential Settlement

1. Large loaded areas on flexible foundation.
2. Non-homogeneity in the soil.
3. Non-uniform loading.
4. Difference in time of construction of adjacent parts of structure.
5. Overlapping of stresses from adjoining structures.

Methods of Reducing Settlement

- (A) Total settlement or Uniform settlement
 - (a) Preloading
 - (b) By using floating foundation
- (B) Differential settlement
 - (a) By providing deep foundation
 - (b) By providing rigid material foundation
 - (c) Stabilization of soil
 - (d) By reinforcing the soil
 - (e) By providing additional loads on lightly loaded area
 - (f) By eccentrically loaded mass
 - (g) By increasing the rigidity of structure by providing column brackets.

Effect of Settlement

As long as settlement is uniform, no damage will be done to structure. The only effect it can have is on the service lines, such as water and sanitary pipe, telephone and electric cables etc. which can snap (Break suddenly) if settlement is considerable. Building in Mexico City have undergone such settlement in a large measure where the first floor, it seems has come down to ground floor level.

If settlement is non-uniform it will produce harmful effects:

1. Failure of pavement as a result of depression at corners and transverse joint.
2. Adverse effects on gradient of railways.
3. Cracking of plaster, displacement of door and window frames, loosening of brick (masonry).
4. Unequal settlement of pier of bridge causing traffic hazards.
5. In statically indeterminate structures (beams, frames, arches, vaults etc.). Settlement induces additional moments which will induce tension cracks in structure.

Tolerable settlement or allowable settlement: The settlement of foundation which will not cause any effect on the performance of structure is called allowable settlement or permissible settlement of foundation. Magnitude of this settlement depends on the type of foundation (Isolated / rigid) and type of soil (clay/sand). These values will be specified by IS code

Module 3: Foundation for difficult soils (6 Lectures)

Guidelines for weak and compressible soils, expansive soil, parameters of expansive soils, collapsible soils and corrosive soils, causes of moisture changes in soils, effects of swelling on buildings, preventive measures for expansive soils, modification of expansive soils, design of foundation on swelling soils, ground improvement methods, for general considerations for cohesive soils and for cohesion less soils

Introduction:

The soil which impose problems during construction of foundation or which impair

functioning of foundation and cause damage to the foundation and hence the structure

supported by foundation is called difficult soil.

Difficult soils are grouped into following categories

- Weak and compressible soil
- Expansive soil
- Collapsible soil
- Corrosive soil

WEAK AND COMPRESSIBLE SOIL:

The soil which is incapable of supporting the load imposed on it due to its compressibility and following are the soils which belong to this category

- a) Clays/Silts/Peats - These types of soil deposits are often found near the mouths of rivers, along the perimeters of bays, and beneath swamps or lagoons. Soil deposits with high organic content are often found in these low lying types of locations and can be especially troublesome. Since land features in which these troublesome soils are typically found are low lying, they are prone to flooding. Hence before buildings or roadways can be constructed on such soil deposits, the grade level must be raised by adding compacted fill. However, adding significant amounts of compacted fill puts significant loads on the soil which can cause significant settlements.
- b) Loose Saturated Sands - Loose saturated sand deposits that are located in seismically active regions are prone to liquefaction and settlements during strong ground motion.

Guidelines for construction over weak and compressible soils

Soil Improvement Among the various strategies used when encountering extremely weak or compressible soil layers are the following:

- *Removal and replacement.* This method can be employed when: the poor soil deposit is relatively small; the groundwater level is relatively deep; and good fill soil is readily available.
- *Temporary Surcharge Fills.* The idea here is to preload the weak/ compressible soil with a temporary surcharge. The underlying weak/ compressible soil is allowed to consolidate under the surcharge (again sand drains accelerate the process). The surcharge is removed before the proposed building construction occurs. Since the building is constructed on over-consolidated soil the displacements are considerably reduced.
- *Vibrocompaction.* This is particularly effective for loose sandy soils.
- *Chemical stabilization.* In the past, the weak clays and silts were often mixed with lime and the existing soil pore fluid to cement the soil grains together, making the soil stronger and less compressible. Presently, the trend in geotechnical engineering

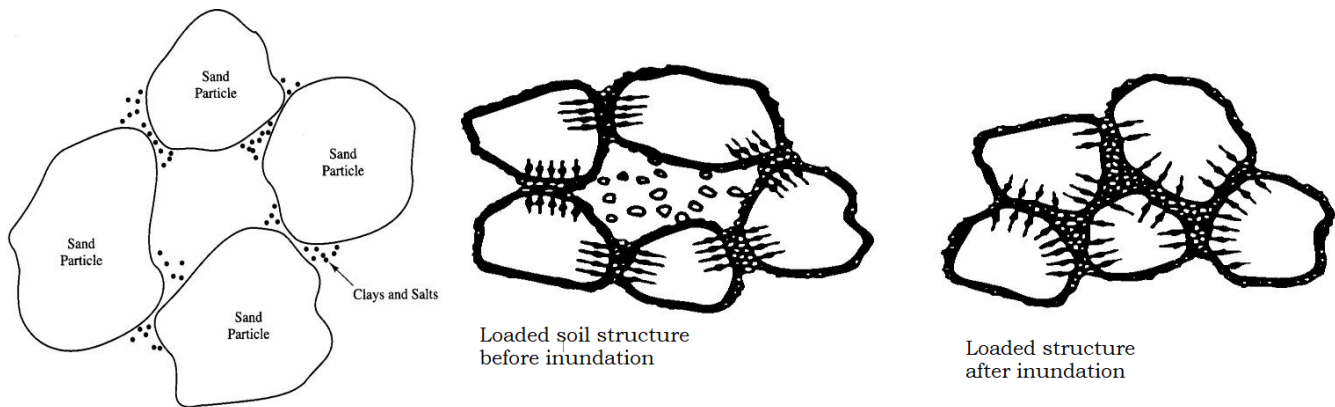
is away from using lime and toward using pulverized fly ash (pfa), which is a processed waste product from coal fired electric power generating plants. Again, the effect is to cement the soil grains together, increasing the soil strength and reducing both its compressibility potential expansivity.

COLLAPSIBLE SOILS:

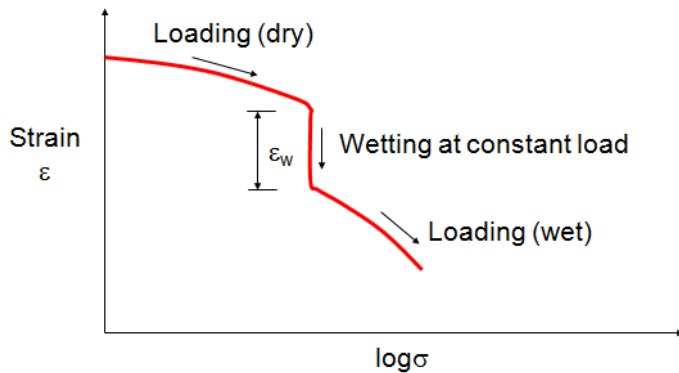
Collapsible soils are those that appear to be strong and stable in their natural (dry) state, but which rapidly consolidate under wetting, generating large and often unexpected settlements. This can yield disastrous consequences for structures unwittingly built on such deposits. Such soils are often termed “collapsible” or “metastable” and the process of their collapsing is often called any of “hydro-consolidation”, “hydro compression”, or “hydro-collapse.” “Loess” which is recognized as potentially collapsible. Collapsible soil deposits share two main features:

- (i) they are loose, cemented deposits; and
- (ii) they are naturally quite dry.

Loess soils consist primarily of silt sized particles loosely arranged in a cemented honeycombed structure. The loose structure is held together by small amounts of water softening or water soluble cementing agents such as clay minerals and CaCO_3 . The introduction of water dissolves or softens the bonds between the silt particles and allows them to take a denser packing under any type of compressive loading.



Single Oedometer Tests



Classification of Soil Collapsibility

Potential hydrocollapse strain, ϵ_w	Severity of Problem
0 – 0.01	No problem
0.01 – 0.05	Moderate trouble
0.05 – 0.10	Trouble
0.10 – 0.20	Severe trouble
> 0.20	Very severe trouble

Guidelines for construction in collapsible soil

Wetting depths of ≤ 2 meters

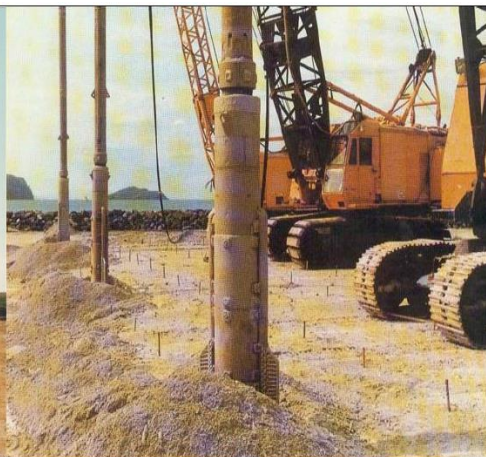
- Removal of the collapsible soil
- Avoidance or minimization of wetting
- Injection of chemical stabilizers or grout
- Prewetting
- Compaction with rollers or vehicles
- Compaction with displacement piles
- Controlled wetting
- Design structure to be tolerant of differential settlements

Wetting depth > 2 m

- Compaction by heavy tamping
- Vibroflotation
- Deep blasting combined with prewetting
- Transfer of load through the collapsible soils to the stable soils below



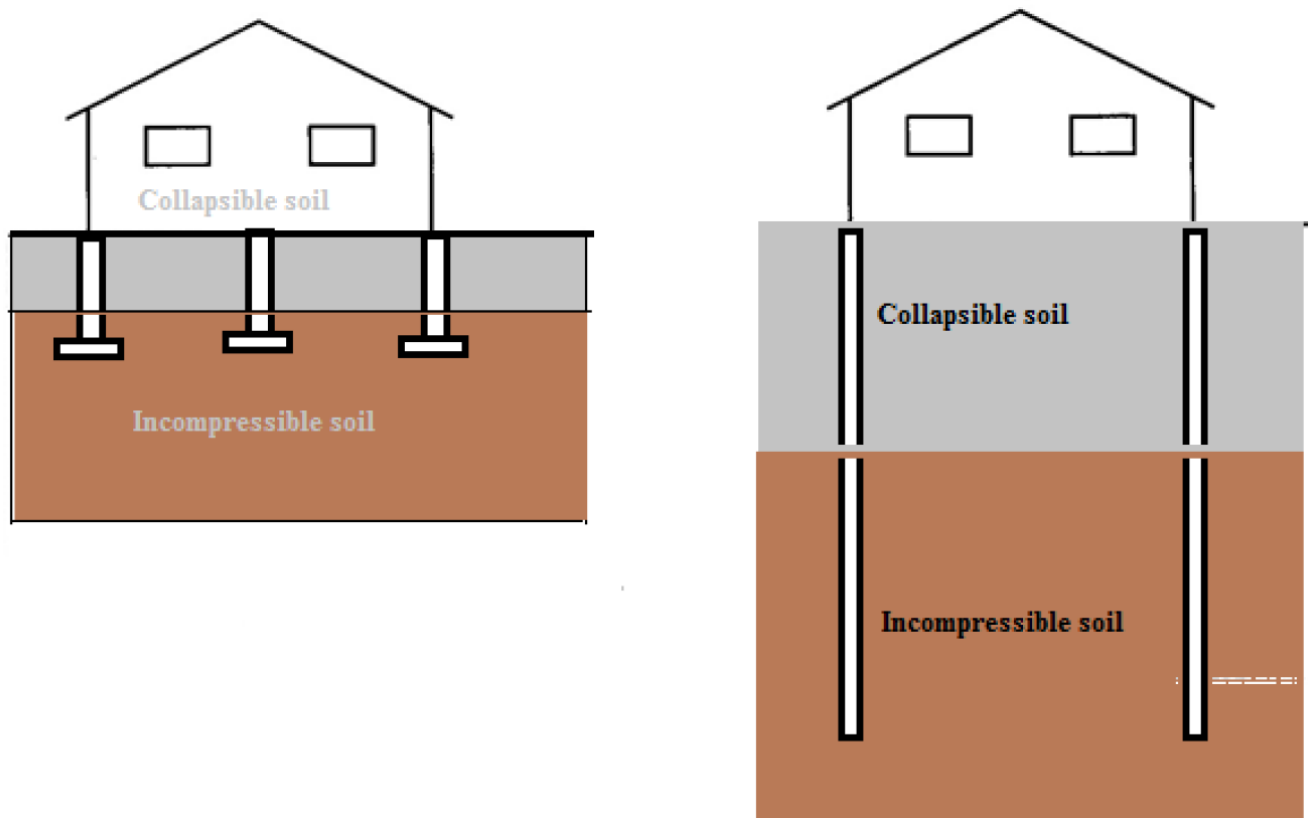
Deep compaction



Stone column



Grouting



EXPANSIVE SOIL:

Expansive soils are soils that expand when water is added, and shrink when they dry out. This continuous change in soil volume can cause homes built on this soil to move unevenly and crack. expansive soils cause damage to houses, other buildings, roads, pipelines, and other structures. This damage is more than twice the damage from floods, hurricanes, tornadoes, and earthquakes combined.

Expansive soil in India is commonly called as Black cotton soil These soils cover vast areas of Maharashtra Karnataka, Andhra, Madhya Pradesh and Gujarat

Causes of Moisture changes in Soils:

Moisture is the indication of amount of water present in a soil mass and is usually expressed in terms of percentage on mass basis. Expansive soils contain minerals (Montmorillonite) that are capable of absorbing water. When they absorb water, they increase in volume. Various causes of change in moisture (increase / decrease) of soil is as given below.

Natural causes to change moisture in the soil are rain, change in temperature, poor drainage, leakage through conveyance system etc. Artificial causes to change in moisture are construction activity can exacerbate the effect of expansive soils. For example, artificial irrigation causes more water to infiltrate the ground, while at the same time less water evaporates due to there being more roadways, parking lots, driveways, sidewalks, and buildings. This results in an increase in subsurface moisture.

Effects of Swelling on Buildings:

Due to increase in the moisture of soil volume of soil will increase. Expansions of ten percent or more are not uncommon. This change in volume can exert enough force on a building or other structure to cause upheaval of buildings and thus by damage it. Cracked foundations, floors, and basement walls are typical types of damage done by swelling soils. Damage to the upper floors of the building can occur when motion in the structure is significant. Expansive soils will also shrink when they dry out. This shrinkage can remove support from buildings or other structures and result in damaging subsidence. Fissures in the soil can also develop. These fissures can facilitate the deep penetration of water when moist conditions or runoff occurs. This cycle of shrinkage and swelling places repetitive stress on structures, and damage worsens over time.

Problems associated with expansive soil

- Low bearing capacity
- High compressibility (CH-MH)
- Swelling and swelling pressure
- Shrinkage

Common damage in building on expansive soil

- Separation of wall and roof slab
- Uplift of interior footing
- Diagonals crack near opening
- Separation of interior flooring from exterior walls
- Horizontal cracks along the longer span of slab
- Vertical crack between column and walls etc



Expansive soil

Guidelines for **Design** / construction in expansive soils

A) Remedial measures:

- Remove the expansive soil.
- Mix the soil with non-expansive material.
- Mix in chemicals to change the way the clay reacts with water.
- Keep the soil moisture constant.
- Use reinforced foundations that are designed to withstand soil volume changes.
- Ensure proper drainage away from the foundation.
- Provide under-reamed pile

B) Preventive measures

There are many ways to prevent damage to structures from expansive soils. These methods can be classified as follows:

- Controlling water's access to the soil;
- Altering the soil properties; and
- Altering the method of construction

a) **Controlling water's access to the soil**

The primary idea here is to keep all water away from the soil surrounding the foundations. This involves such common sense measures as:

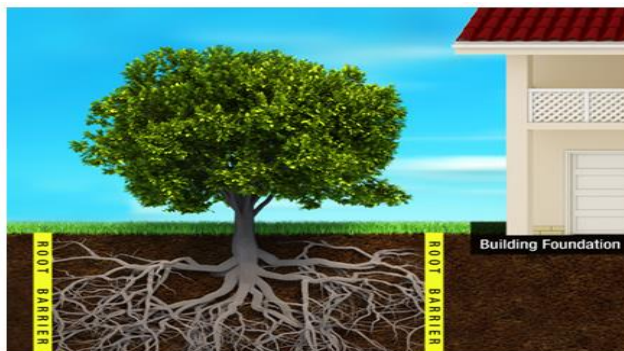
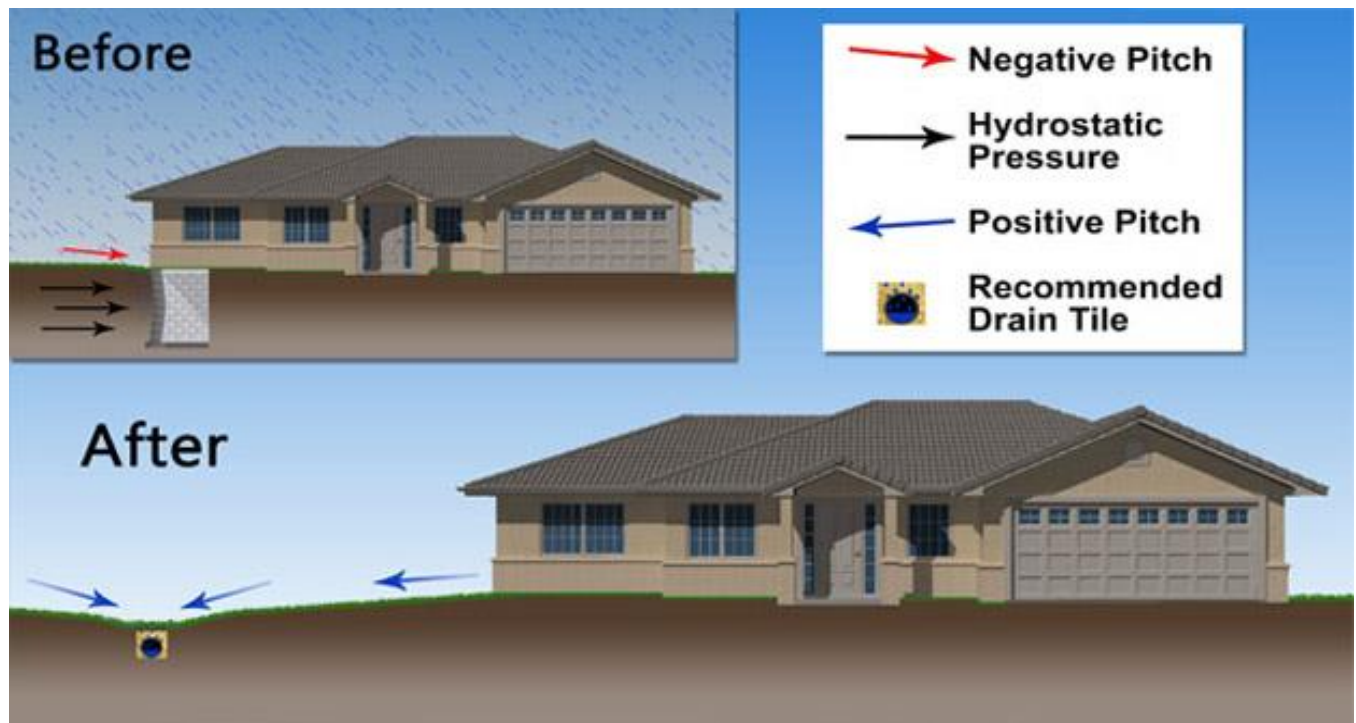
- 1) Controlling surface drainage by making sure that the ground surface slopes away from the structure and the foundation.
- 2) Draining rainfall sufficiently far away from the foundations.
- 3) Not placing landscaping that will require extensive watering near a structure.
- 4) Using impervious liners that serve as moisture barriers to isolate foundations from surface water

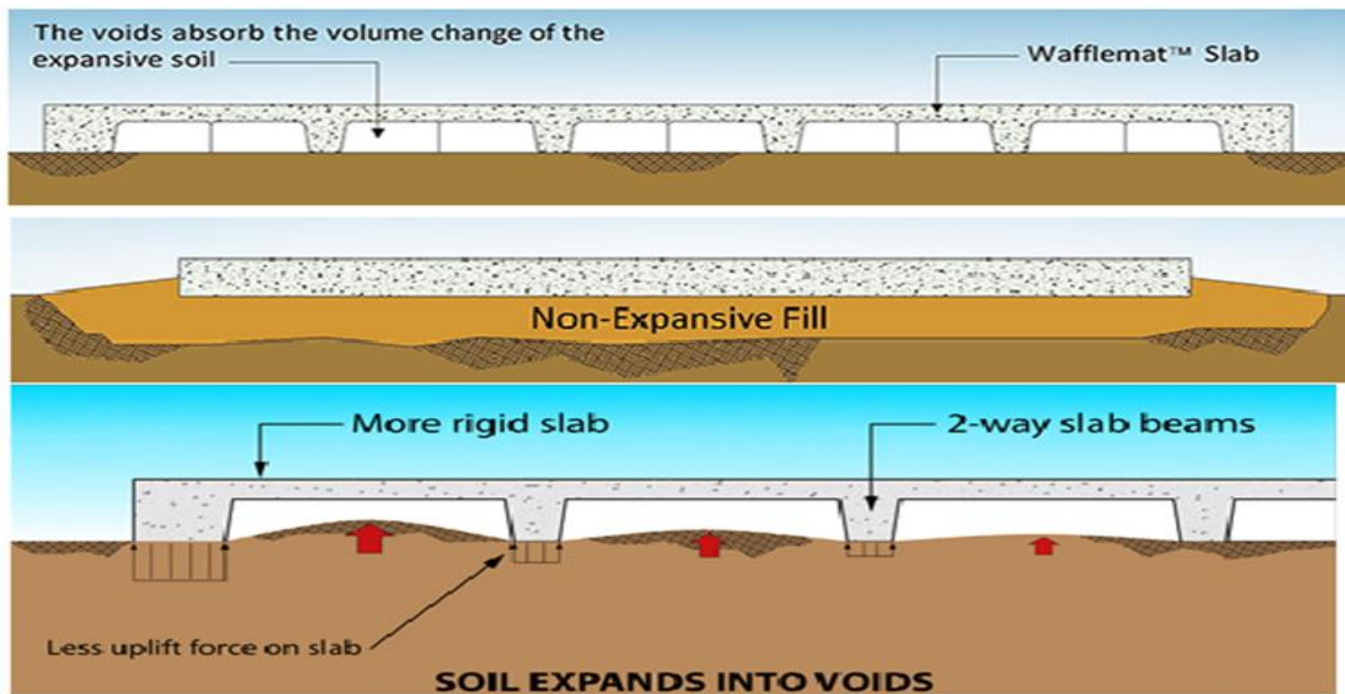
b) **Altering the soil properties (modification of expansive soil)**

- 1) Excavation and replacement of the active soil; by CNS soil (**cohesive non-swelling soil**)
- 2) Chemical treatment of the active soil (for example lime treatment or mixing the soil with pfa [pulverized fuel ash]) to reduce its expansive potential; and
- 3) Prewetting of the soil. The idea is to build the foundation on a soil that has already expanded rather than a soil that will expand after the foundation is in place

c) **Altering the method of construction**

- 1) Bypassing the expansive clay soils in the active zone by resorting to deep foundations
- 2) When using mat or spread footings, using “waffled” foundations. The “waffle” pattern has the advantages that it allows the soil to expand into the voids, and that the foundations are quite stiff. This reduces differential settlement and heave due to expansion and shrinkage
- 3) Anchoring the foundation at depth of little volume change (more than 2m)
- 4) construct highly flexible structures and foundations which can accommodate large settlement/movement
- 5) Counteracting swelling pressure
- 6) Stiffening of structural components by providing bands at various levels





Corrosive soil:

Schiff (1982) indicated that corrosion would be most likely in the following soil conditions

- High moisture content
- Poorly aerated
- Fine grained
- Black or grey colour
- Low electrical resistivity
- Low or negative redox potential
- Organic material present
- High chemical content
- Highly acidic
- Sulfides present
- Anaerobic microorganism present

Corrosion is the problem for structure where steel is used

Steel foundations in marine environment have a significant potential for corrosion especially those exposed to salt water. Areas where the elevation of the groundwater table fluctuates, such as tidal zones are potential areas for corrosion. Contaminated soils, such as sanitary landfills and shorelines near old outfalls, are also more likely to have corrosion problems

Preventive measures for corrosion

Steel structures:

- 1) use different construction material (concrete or wood --)
- 2) Increase the thickness of steel section by an amount equal to anticipated deterioration
- 3) Cover steel with protective coating (coal tar epoxy)
- 4) Provide cathodic protection

Concrete structures:

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Buried concrete is usually very resistant to corrosion and will remain intact for many years. However, presence of sulphates (SO_4) in soil can react with the cement to form calcium sulfo-aluminate (ettringite). As these crystals grow and expand, the concrete cracks and disintegrates.

- 1) Reduce the water cement ratio (reduce permeability)
- 2) Increase cement content (reduce permeability)
- 3) Use sulphate resisting cement
- 4) Coat the concrete with an asphalt emulsion

Ground improvement methods: Ground improvement are the techniques where in unsuitable soil is made suitable by changing its properties (physical / chemical) In general terms ground improvement may be considered to be “When the Engineer forces the soil to adapt to the project requirements by altering its natural state, rather than changing the Engineering design in response to the natural limitation of the soil.”

Techniques of ground improvement:

Reference	Criteria	Categories
Mitchell (1981)	Construction /function	1) In situ deep compaction of cohesion less soils 2) Pre-compression 3) Injection and grouting 4)Admixtures 5) Thermal) Reinforcement
Hausmann (1990)	Process	1)Mechanical 2) Hydraulic 3) Physical and chemical 4) Modification by inclusion and confinement
Ye et al. (1994)	Function	1)Replacement 2) Deep densification 3) Drainage and consolidation 4) Reinforcement 5)Thermal treatment 6) Chemical stabilization
Chu et al. (2009)	Soil type and inclusion	1)Ground improvement without admixtures in granular soil or fill material 2) Ground improvement without admixtures in cohesive soils 3) Ground improvement with admixtures or inclusions 4) Ground improvement with grouting type admixtures 5) Earth reinforcement
Schaefer and Berg (2012)	Application	1)Earthwork construction 2) Densification of cohesion less soil 3) Embankment over soft soils 4)Cutoff walls 5) Increased pavement performance 6) Sustainability 7) Soft ground drainage and consolidation 8) construction of vertical support elements 9) Lateral earth support 10) Liquefaction mitigation 11) Void filling
Jie Han (2015)	Function	1) Densification 2) Replacement 3) Drainage and consolidation 4) Chemical stabilization 5) Reinforcement 6) Thermal and biological treatment

Module 4: Shallow foundations (5 Lectures)

Assumptions and limitation of rigid design analysis, safe bearing pressure, settlement of footings. Design of isolated, combined, strap footing (rigid analysis) Raft foundation (Elastic analysis), I.S. Code of practice for design of raft foundation

Introduction:

The foundation in which depth of foundation is less than or equal to width of foundation is known as 'shallow foundation'. It is commonly used for structure carrying moderate loads and subsoil is having sufficient strength.

Types of Shallow Foundation:

The foundation in which depth of foundation is less than or equal to width of footing is called shallow foundation which is further subdivided into following types:

- Isolated footing.
- Combined footing.
- Strap footing.
- Strip or continuous footing (wall footing).
- Mat or raft footing.

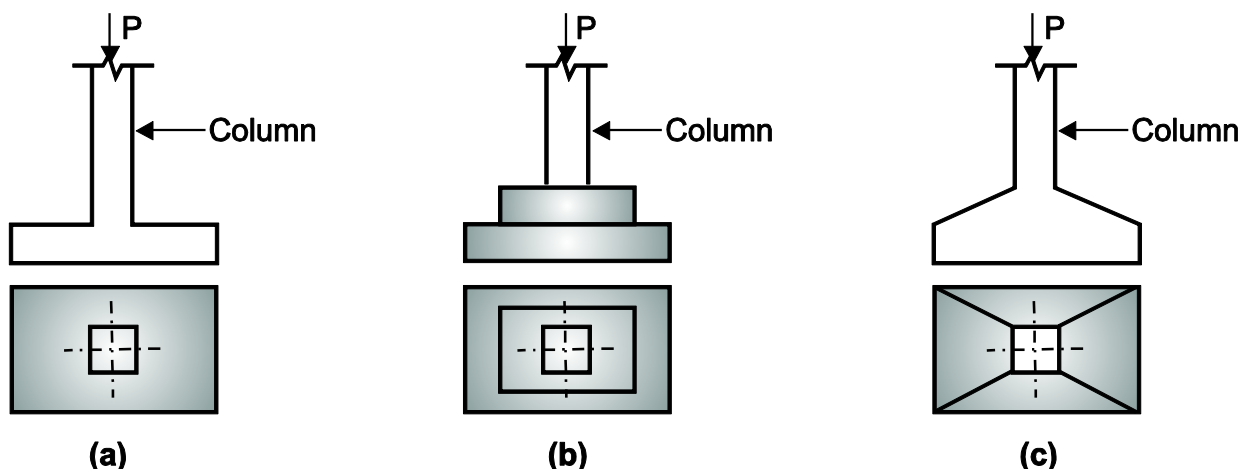
Footing which is used for supporting single column is called '**isolated footing**' and can be provided in any of following way: Spread footing, stepped footing or slopped footing.

Footing which supports two adjacent columns is called '**combined footing**' and is generally rectangular or trapezoidal.

Footing in which two isolated footings are connected by a structural member (strap beam) is called '**strap footing**'.

Footing which is used for supporting wall or row of columns is called '**strip or continuous footing**'.

Footing which supports walls and columns under part or all of the structure, is termed as '**mat or raft footing**'



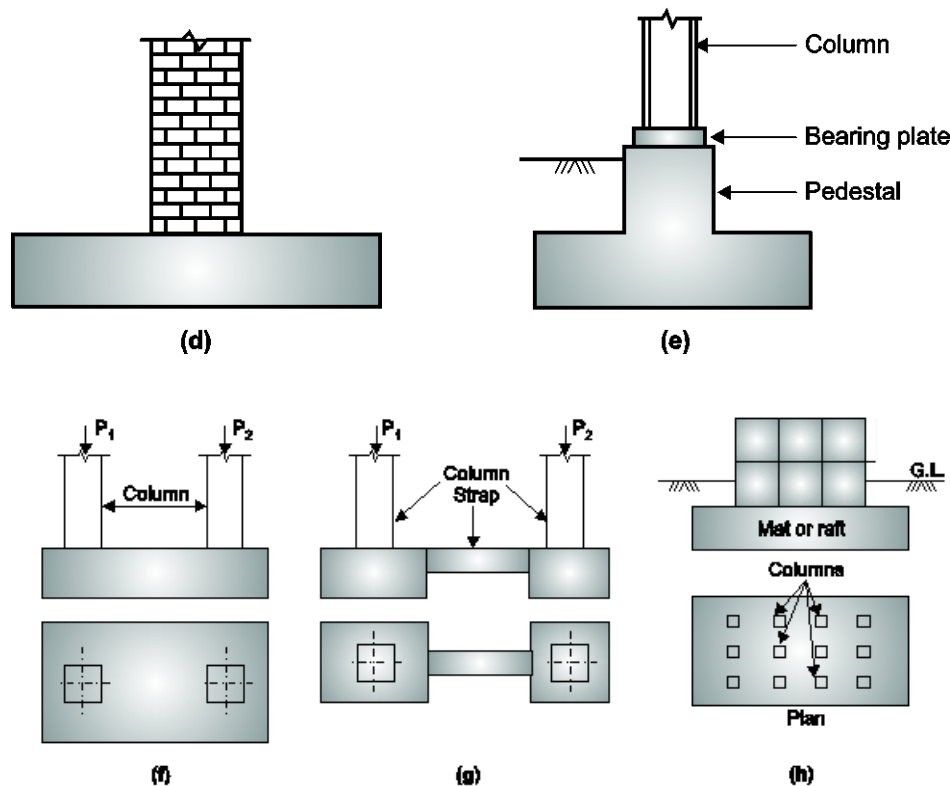


Fig. : Common type of footings : (a) single spread footing, (b) stepped footing, (c) Sloped footing, (d) Wall footing, (e) Footing with pedestal, (f) Combined rectangular footing, (g) Strap footing, (h) Mat or raft foundation

Design of shallow foundation:

Design of foundation is the process wherein we have to determine

- Size of footing
- Shape of footing
- Depth of footing
- Reinforcement required to resist shear and bending induced in the footing

Determination of first three parameters is called geometrical design of footing and determination of reinforcement required is called structural design of footing.

Design of Isolated footing

SIZE OF FOOTING: Area required for a footing is calculated by using following considerations

a) Shear failure consideration (bearing capacity):

Area = Load on the column + self weight of footing Safe bearing capacity of footing

Self-weight of footing is taken as 10% of load on the column

b) Settlement consideration: (Housel's method)

Housel has given an equation to calculate load carrying capacity of footing i.e.

$$Q = A \cdot m + P \cdot n$$

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Where A and P are area and perimeter of footing; m and n are constants which are determined by using plate load test data for two different size plates.

Let P – load for which footing is to be designed

S – Permissible settlement for footing and soil system

Q1 – Load on plate 1 of area A1 and perimeter P1 for a settlement of S

Q2 – Load on plate 2 of area A2 and perimeter P2 for a settlement of S

By substituting these values in Housel's equation we get following equations

$$Q_1 = m A_1 + n P_1 \text{ -----(1)}$$

$$Q_2 = m A_2 + n P_2 \text{ -----(2)}$$

Solving these two equations we get values of constants m and n

Substituting these values of m and n again in Housel's equation [for example for square footing]

$$Q = B^2 * m + 4B * n$$

Solving above equation we get size of footing (B)

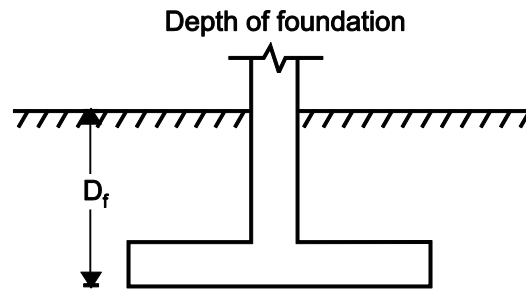
SHAPE OF FOOTING:

Usually square footing is preferred When load is axial or eccentric about both axes
square footing is preferred When load is eccentric about both axes then rectangular footing is preferred

Depth of footing:

Vertical distance between the ground surface and base of footing is called '**depth of footing**'.

Depth of footing depends on number of factors. Thus, for particular type of structure it is taken as maximum amongst the following :



1. Loading criteria:

$$D = \frac{q}{r} \left[\frac{1 - \sin \phi}{1 + \sin \phi} \right]^2$$

Where q – SBC of soil, r – unit weight of soil and ϕ – angle of internal friction of soil

2. Depth of volume change: (Expansive soil)

$$D > 1.5\text{m},$$

$$D > \text{Root depth},$$

$$D > \text{Level of artificially increased temperature}$$

3. Frost depth:

$$D > 1.5\text{m in cold areas, } D > \text{frost depth}$$

4. Erosion depth:

$$D > \text{Erosion depth},$$

$$D > 0.3\text{m single/double storey},$$

$$D > 0.6\text{m for heavy construction}$$

5. Scour depth:

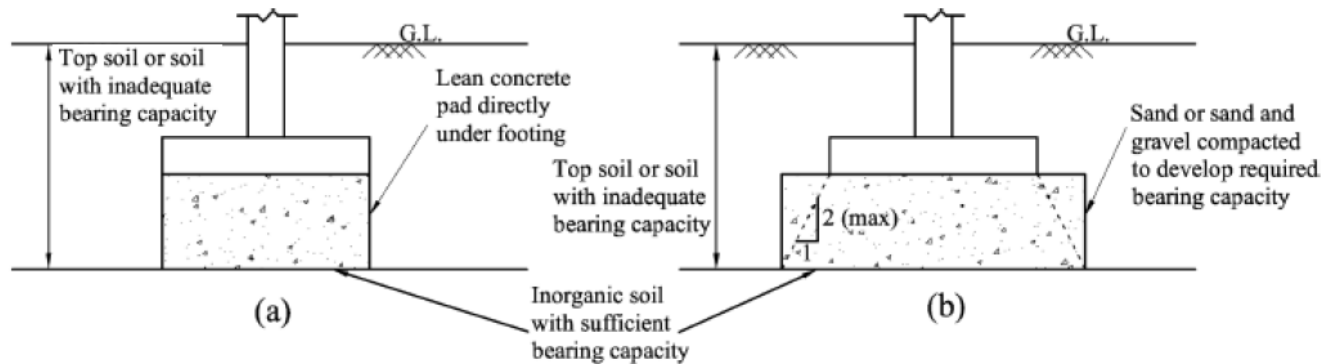
$$D > \text{Scour depth}$$

6. Settlement consideration:

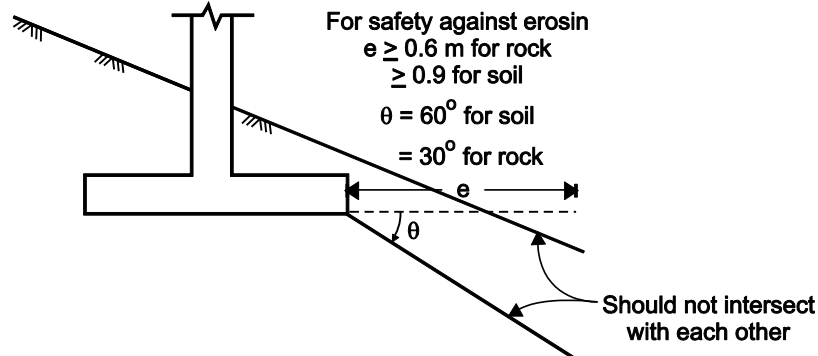
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$D >$ Depth of top organic soil (compressible soil),

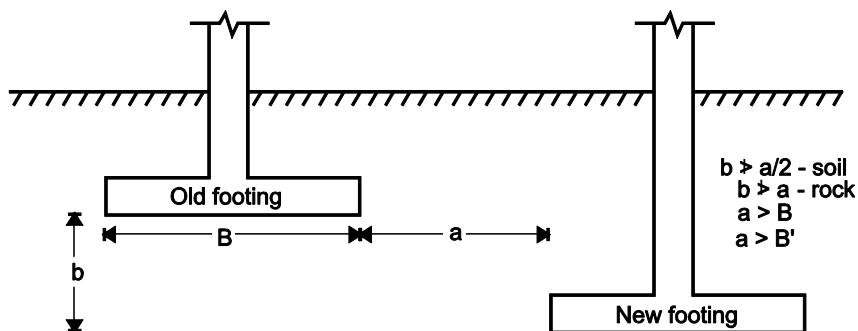
If depth of organic soil is considerable then one of following alternative can be used

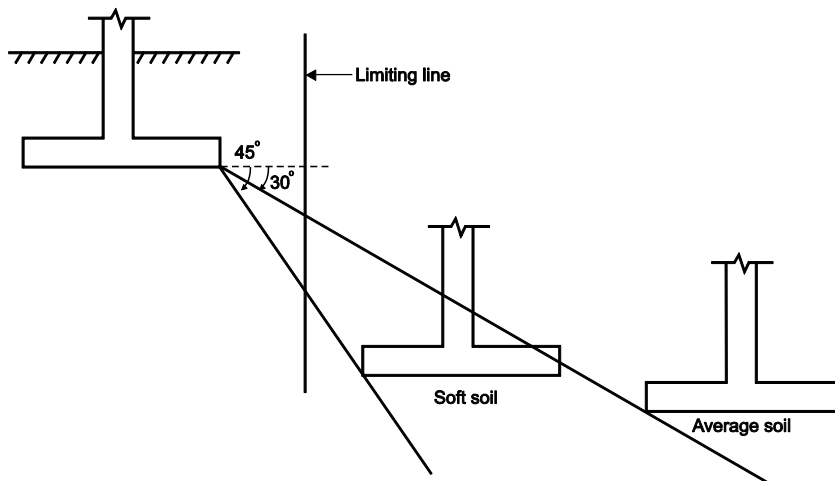


7. Topographical Consideration:



8) Adjoining Structure:



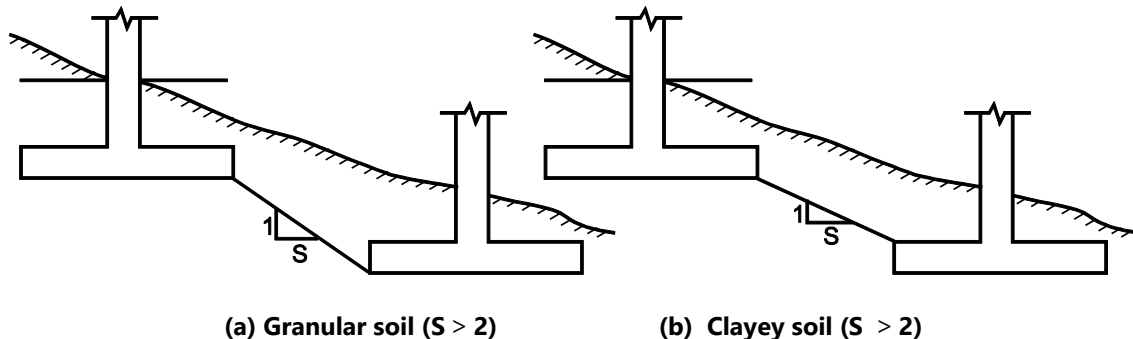


As far as possible new footing should not extend beyond limiting line.

New footing should be carried below old footing to minimize effect of new on old footing.

As far as possible new footing should not intersect reference line (red line).

9) Sloping Ground and Adjoining Structure:



10) Position of Ground Water Table: Ground water table affects in three ways:

- (a) Construction below Ground water table is difficult.
- (b) It reduces bearing capacity of soil.
- (c) Uplift and damp proofing of structure.

$D_f < \text{Depth of ground water table}$

Design of combined footing:

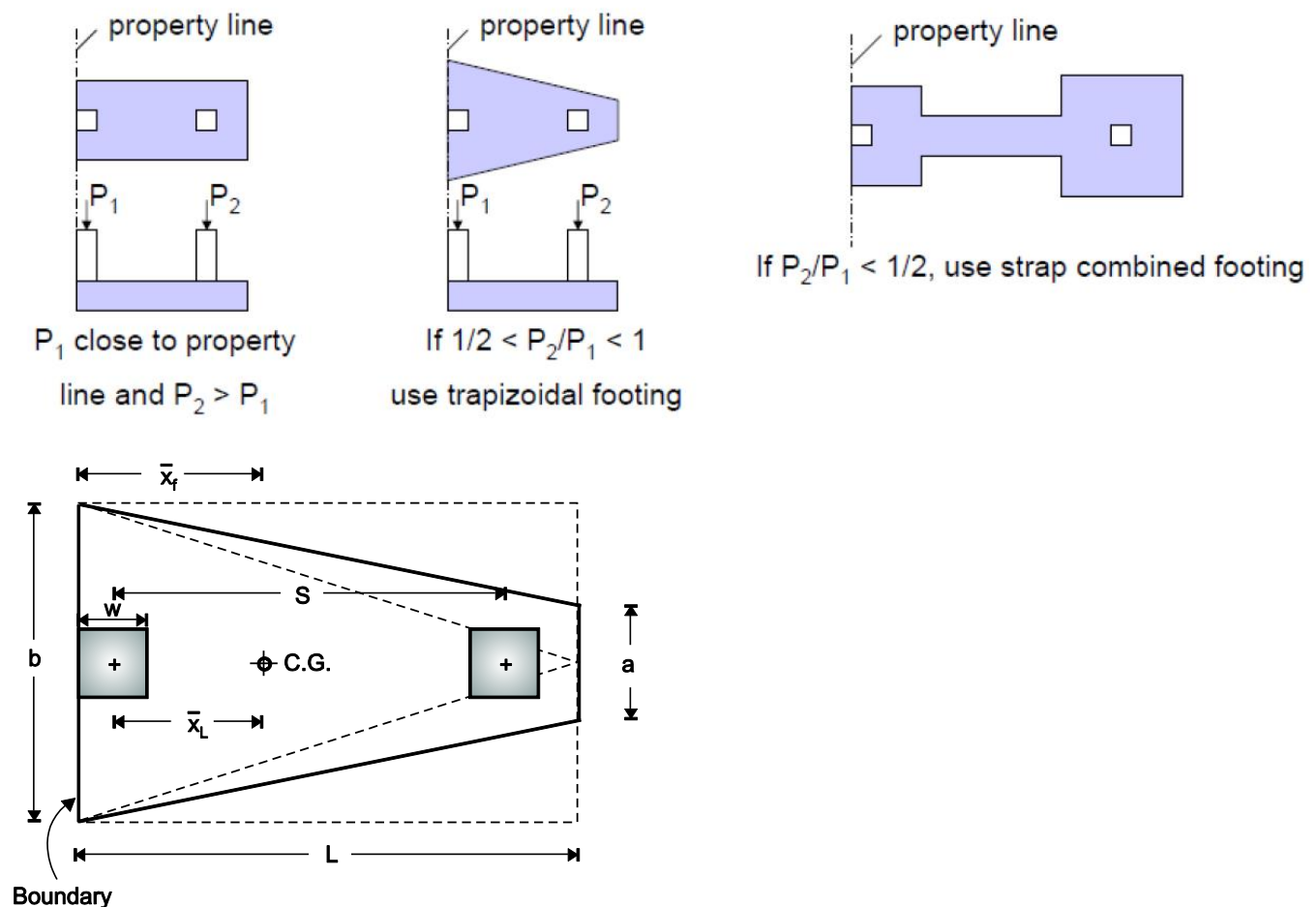
COMBINED FOOTINGS: The use of combined footings is appropriate either when two columns are spaced so closely that individual footing are not practicable or when a wall column is so close to the property line that it is impossible to center an individual footing under the column.

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Basic principle of design of foundation is that line of action of resultant load should coincide with the geometrical center of footing, so that the pressure distribution is uniform.

A combined footing is so proportioned that the centroid of the area in contact with the soil lies on the line of action of the resultant of the loads applied to the footing; consequently, the distribution of soil pressure is reasonably uniform. In addition, the dimensions of the footing are chosen such that the allowable soil pressure is not exceeded. When these criteria are satisfied, the footing should neither settle nor rotate excessively. A combined footing may be of rectangular shape or of trapezoidal shape in plan. These are usually constructed using reinforced concrete.

If area required for a column footing is not overlapping with area zone of adjacent footing or not crossing the boundary of plot, in such case one can use isolated column footing. However, if any one of above criteria is not satisfied then either, we have to use combined footing or strap footing. Combined footing is used when adjacent columns are close to each other, so that area of individual column footing overlaps or are very close. In order to match load center and geometrical center of footing either provide rectangular or trapezoidal combined footing.



Consider two adjacent columns having center to center spacing S , width of exterior column is w .

Let, $\bar{x}_L$ —Distance of resultant of column load from center of external column

$\bar{x}_f$ – Distance of centroid of footing

a and b - Sides of footing

L – Length of footing

$$\bar{x}_f = \bar{x}_L + \frac{w}{2} \quad \dots \text{(From load consideration)}$$

$$\bar{x}_f = \frac{2a + b}{a + b} \cdot \frac{L}{3} \quad \text{and} \quad A_f = \frac{a + b}{2} \cdot L \quad \dots \text{(From geometry of footing)}$$

If $a = b$, then footing will be rectangular and $\bar{x}_f = \frac{L}{2}$.

If $a = 0$, then footing will be triangular (practically not possible) and $\bar{x}_f = \frac{L}{3}$.

Thus, geometrical criteria's for selecting appropriate type of footing are as mentioned below.

- (i) If $\bar{x}_f$ as calculated from load consideration is more than or equal to $\frac{L}{2}$, then provide rectangular footing.
- (ii) If $\bar{x}_f < \frac{L}{3}$, then provide strap footing.
- (iii) If $\bar{x}_f$ lies between $\frac{L}{3}$ and $\frac{L}{2}$, then provide trapezoidal footing.

Design of footing (foundation engineering) means to fix the geometrical parameter of footing in plan and to draw its shear force and bending moment diagram.

A) Rectangular Combined Footing:

Footing can be rectangular under following situations:

- Two adjacent column carries equal load and the area of footing coincides.
- Restriction over width of footing.
- If interior column carries larger load than external column, which is close to boundary.

Procedure for Rectangular Combined Footing :

- Given data :
1. Allowable soil pressure.
 2. Loads carried by column.
 3. Distance of external column from boundary.

Required : 1. To fix dimension of footing.

Procedure/Steps :

- (i) Determine resultant column load,

$$R = Q_1 + Q_2$$

- (ii) Determine area of footing required,

$$A_f = \frac{R}{q_a}$$

- (iii) Locate position of resultant load with respect to external column,

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$$x_L = \frac{Q_2 \cdot L}{R}$$

(iv) Determine position of centre of gravity of footing and length of footing,

$$x_f = x_L + \frac{w}{2}$$

$$L = 2x_f$$

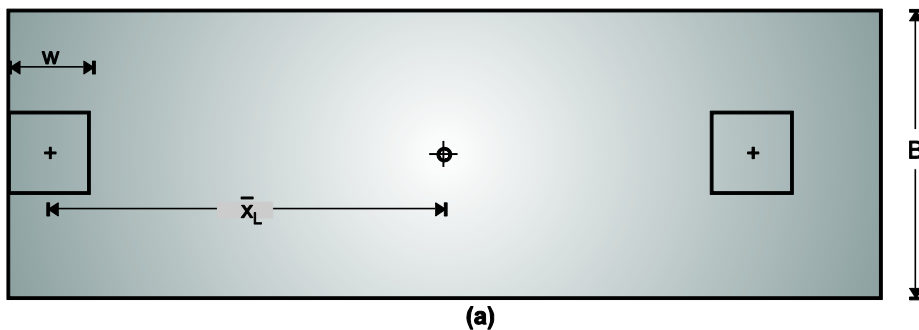
(v) Determine width of footing

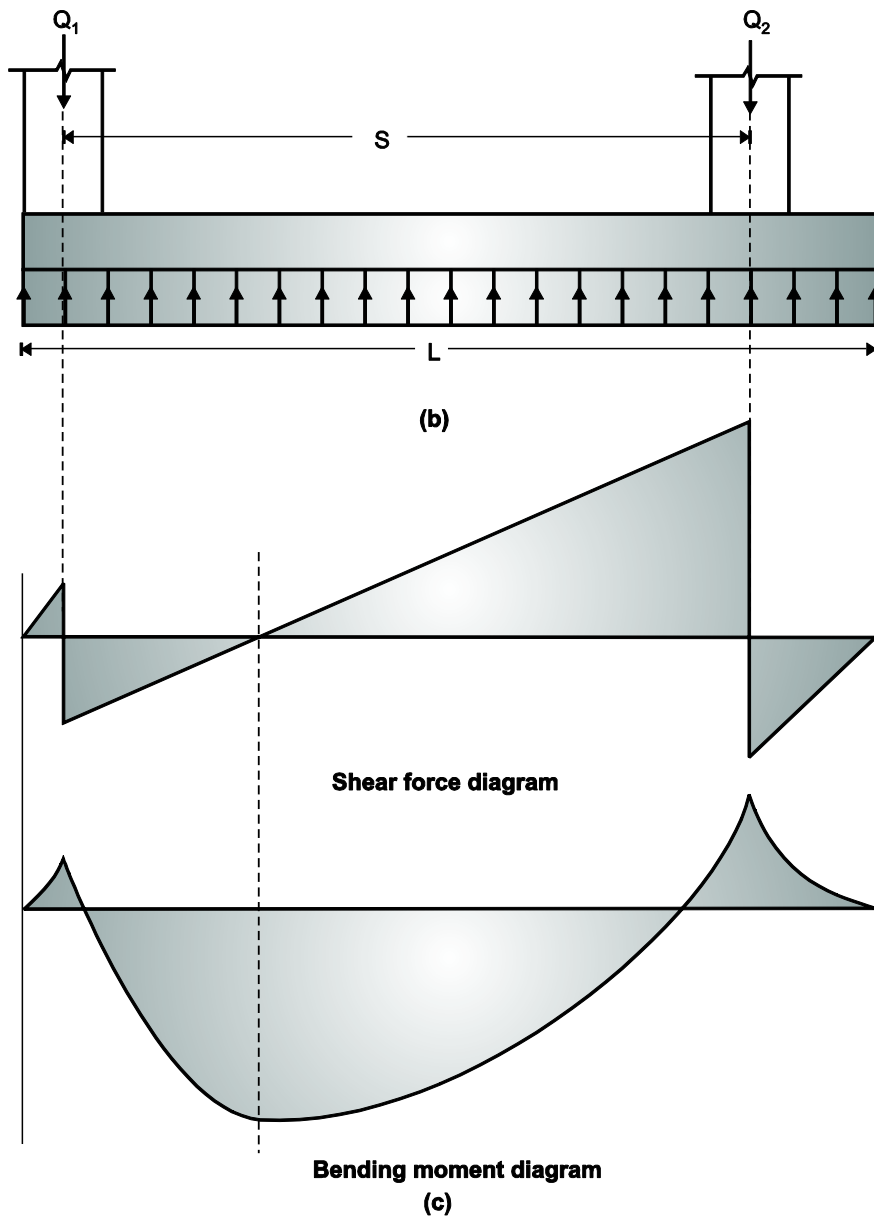
$$B = \frac{A_f}{L}$$

(vi) Determine upward soil pressure per unit width of footing,

$$p = \frac{q_a}{B}$$

(vii) Draw shear force and bending moment diagram along the length using column loads as point load and soil pressure as uniformly distributed load.





Design of Trapezoidal Combined Footing (Conventional Method):

When the two column loads are unequal, the exterior column carrying higher load and when the property line is quite close to the exterior column, a trapezoidal combined footing is used. It may be used even when the interior column carries higher load; but the width of trapezoid will be higher in the inner side. The location of the resultant of the column loads establishes the position of the centroid of the trapezoid. The length is usually limited by the property line at one end and adjacent construction, if any, at the other.

The width at either end of the trapezoid can be determined from the solution of two simultaneous equations—one expressing the location of the centroid of the trapezoid and the other equating the sum of the column loads to the product of the allowable soil pressure and the area of the footing. The resulting pressure distribution is linear or uniformly varying (and not uniform) as shown in Fig.

Given: Loads on columns, spacing between columns, allowable soil pressure and distance of external column from boundary and length of footing.

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Required: Size of footing, shear force diagram (SFD) and bending moment diagram (BMD).

Steps:

(1) Determine resultant load, $R = Q_1 + Q_2$.

(2) Find area of footing, $A_f = \frac{R}{q_a}$

(3) Locate position of resultant load, $\bar{x}_L = \frac{Q_2 \cdot S}{R}$.

(4) Find center of gravity (C.G.) of footing, $\bar{x}_F = \bar{x}_L + \frac{w}{2}$.

(5) Compare $\bar{x}_f$ with $\frac{L}{2}$ and $\frac{L}{3}$, if $\frac{L}{3} < \bar{x}_f < \frac{L}{2}$ provide trapezoidal footing.

(6) If a, b and L are dimensions of footing.

$$A_f = \frac{a + b}{2} \cdot L \quad \text{and} \quad \bar{x}_f = \frac{2a + b}{a + b} \frac{L}{3}$$

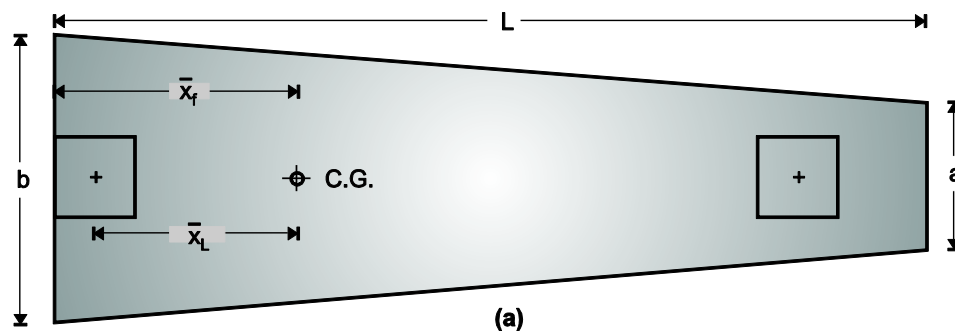
Substitute value of A_f and $\bar{x}_f$ from step (2) and (3) and solve above equation for a and b.

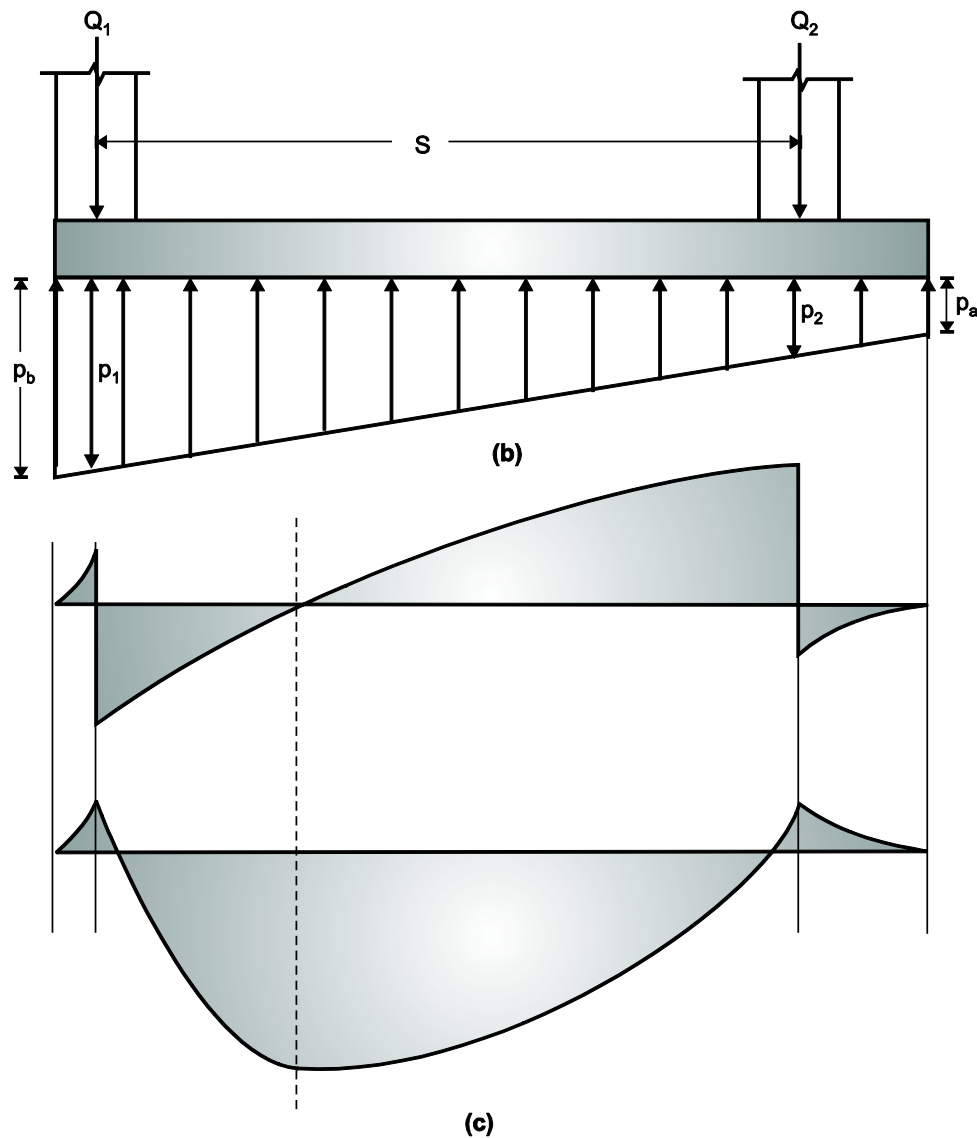
(7) Compute the intensity of upward soil pressure at each edge of footing and also under load.

$$p_b = (q_a) b \quad \text{and} \quad p_a = (q_a) a$$

p_1 and p_2 can be determined by interpolation.

(8) Compute shear force (S.F.) and bending moment (B.M.) at critical points and hence plot shear force and bending moment diagram.





Note: Simplified formulas for computing pressure intensity and bending moment. Due to trapezoidal loading.

$$c_x = c_2 - \frac{c_2 - c_1}{(x + y)} (x)$$

$$= c_1 + (c_2 - c_1) \frac{y}{x + y}$$

$$M_x = (2c_2 + c_x) \frac{x^2}{6}$$

$$= (2c_1 + c_x) \frac{y^2}{6}$$

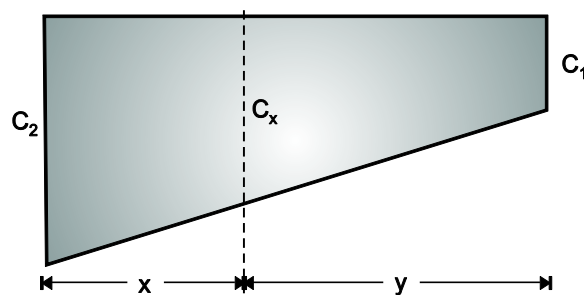


Fig. 2.24

Design of strap footing:

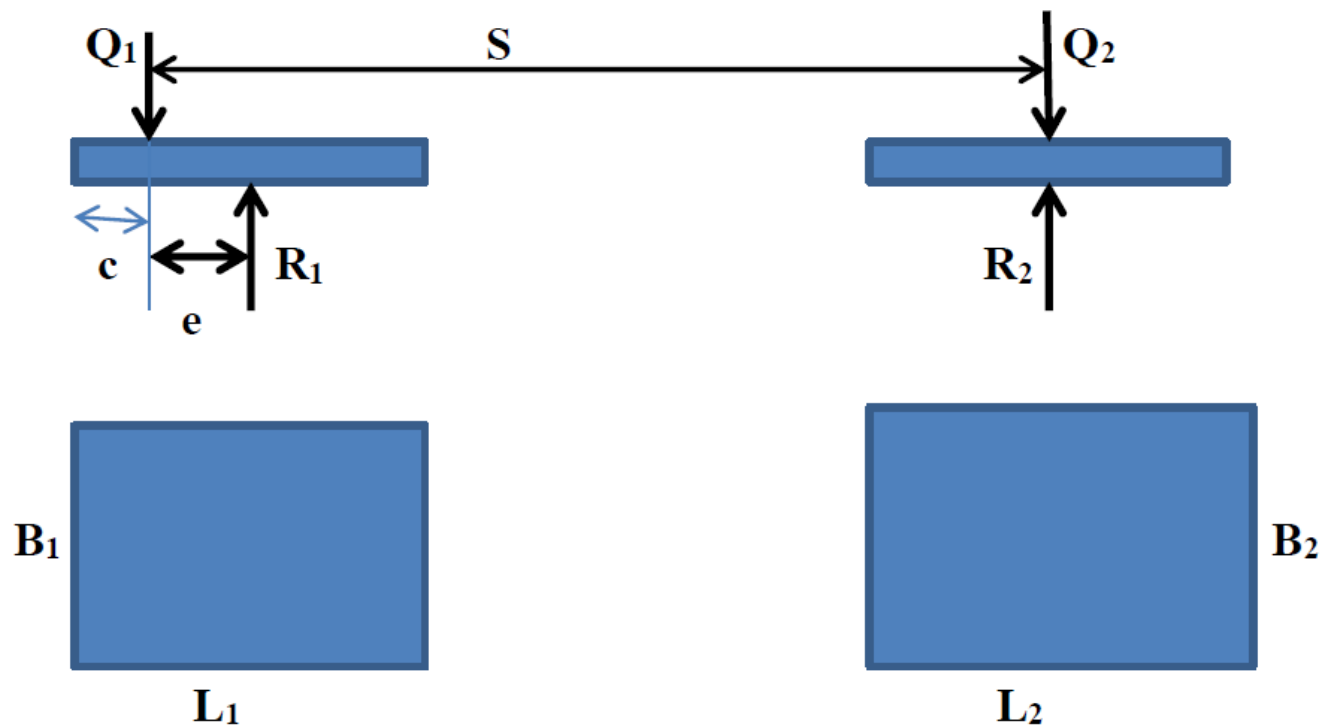
STRAP FOOTINGS:

Basis for Design of Strap Footings

Strap footings are designed based on the following assumptions:

- 1) The strap footing is considered to be infinitely stiff. It serves to transfer the column loads onto the soil with equal and uniform soil pressure under both the footings.
- 2) The strap is a pure flexure member and does not directly take soil reaction. The soil below the strap will be loosened up in order that the strap does not rest on the soil and exert pressure.

With these assumptions, the procedure of design is simple. With reference to Fig. it may be given as follows:



Design of strap footing:

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- 1) Assume suitable value of eccentricity e (0.5m – 1.5m)
- 2) Calculate soil reaction under each footing

$$R_1 = Q_1 \left\{ \frac{S}{S-e} \right\} \quad \text{and} \quad R_2 = Q_1 + Q_2 - R_1$$

- 3) Calculate area of footing required to transmit soil reaction safely

$$A_1 = \frac{R_1}{SBC} \quad \text{and} \quad A_2 = \frac{R_2}{SBC}$$

- 4) Calculate dimensions of footing

$$L_1 = 2(c + e); \quad B_1 = \frac{A_1}{L_1}; \quad B_2 = B_1 \quad \text{and} \quad L_2 = \frac{A_2}{B_2}$$

- 5) Check that L and B ratio should not be too high and also

$$S > 0.5L_1 + 0.5L_2 + e$$

- 6) If above conditions are not satisfied then revise the design by assuming some other value of e

Mat foundation:

A mat foundation is a large concrete slab used to interface one column, or more than one column in several lines, with the base soil. It may encompass the entire foundation area or only a portion. A mat may be used to support on-grade storage tanks or several pieces of industrial equipment. Mats are commonly used beneath silo clusters, chimneys, and various tower structures.

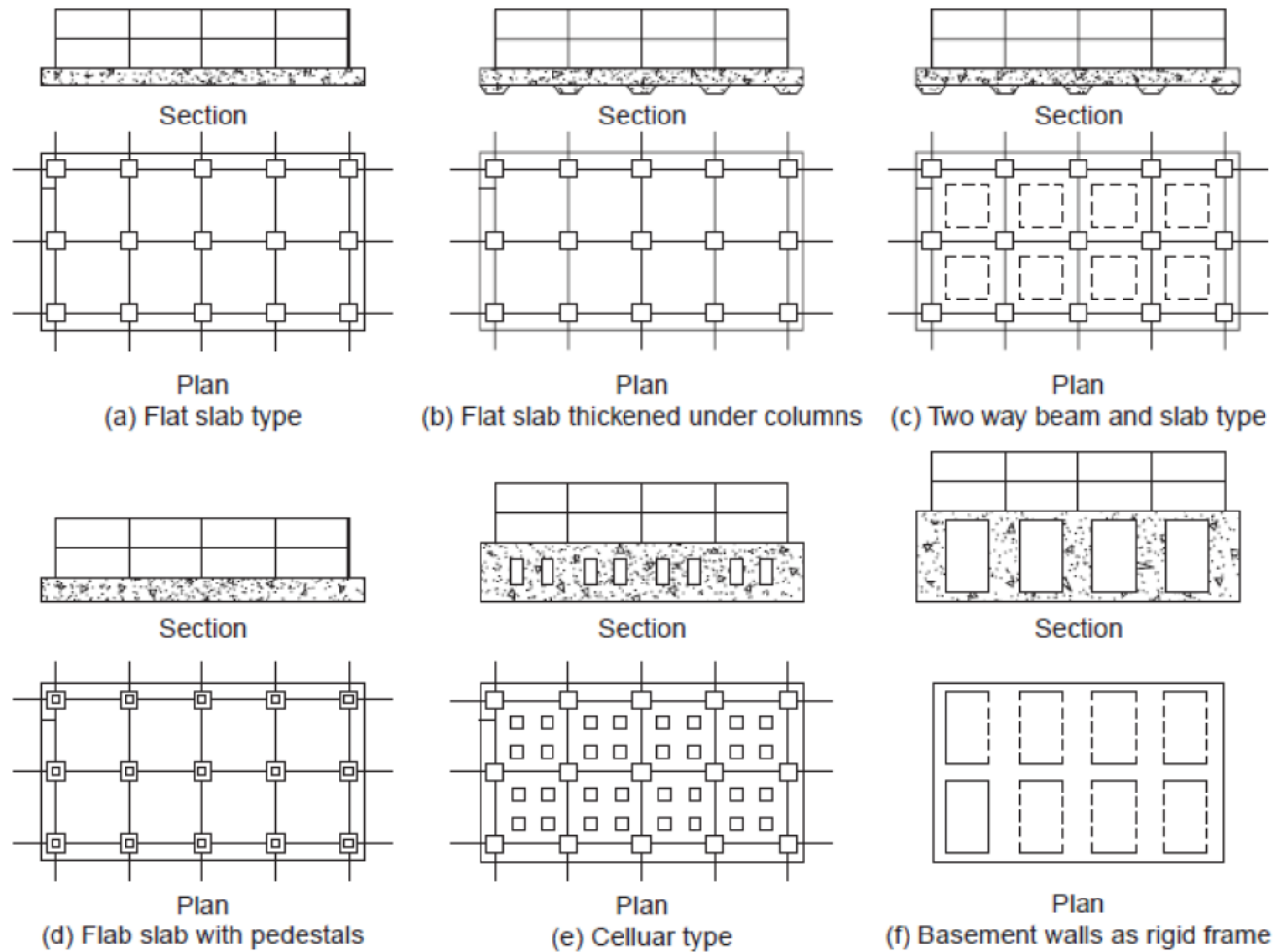
A mat foundation may be used where the base soil has a low bearing capacity and/or the column loads are so large that more than 50 percent of the area is covered by conventional spread footings. It is common to use mat foundations for deep basements both to spread the column loads to a more uniform pressure distribution and to provide the floor slab for the basement.

Conditions for Mat Foundations:

- Structural loads are so high and soil conditions are so poor that spread footings would be exceptionally large
- Soil is very erratic or prone to differential settlements, or soils that are expansive or subject to differential heaves
- Structural loads are erratic
- Unevenly distributed lateral loads (cause differential horizontal movements in spread footings or pile caps).
- Uplift loads are larger than spread footings can accommodate; weight of the mat is a factor here
- Bottom of the structure is located below the GWT, so waterproofing is an important concern. Mat foundations are monolithic, so easier to waterproof. Weight of mat also resists hydrostatic uplift forces

Ribbed mats, consisting of stiffening beams placed below a flat slab are useful in unstable soils such as expansive, collapsible or soft materials where differential movements can be significant (exceeding 12mm).

Different types of Raft:



Flat slab type : This type of mat is used when loads on column are approximately, soil is homogeneous and columns are equispaced.

Slab thickened under column: This type of mat is used when columns carry heavy loads

Slab and beam type mat: This type of mat is used when column carries loads with moments

Cellular mat : This mat is used when structure carries large lateral loads along with vertical load

Design of Rafts: Rafts can be designed by using following methods

A) Elastic method: (Flexible method)

B) Rigid method: (Conventional method/IS code method)

A) Elastic method: In this method it is assumed that the soil behaves like an infinite number of individual springs, not affected by adjacent springs. The elastic constant of the spring is equal to the modulus of subgrade reaction of the soil. Further it is assumed that spring will resist both tension and compression. This concept is given by Winkler so this method is also called Winkler approach.

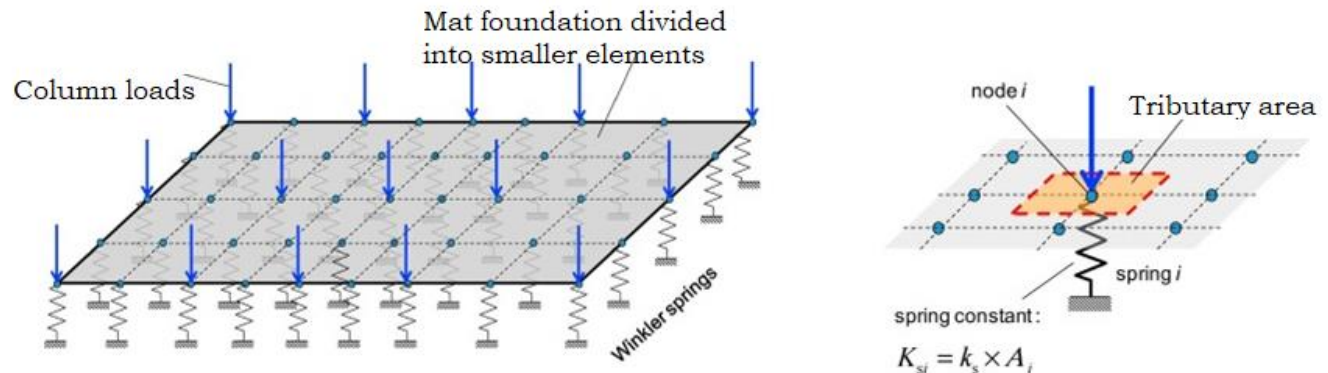


Fig : Mat supported on number of piles (Winkler approach).

Design of Mat foundation by IS code method (Rigid method):

RIGID BEAM METHOD: (CONVENTIONAL METHOD)

This method can be used when the settlements are small. According to IS 2950 (1981), this method may be used when raft satisfies the condition $\lambda L < 1.75$ where λ is the characteristic coefficient and L is the column spacing.

The characteristic coefficient is calculated by

$$\lambda = \left[\frac{kB}{4E_c I} \right]^{0.25}$$

Where k – modulus of subgrade reaction corrected for width B (kN/m³)

B – Width of raft

E_c – Modulus of elasticity of concrete = $5\sqrt{f_{ck}} \times 10^6$ kPa

I – Moment of inertia of raft in m⁴

Procedure for rigid method:

Let us consider a raft of dimension B X L as shown in fig.

$P_1, P_2, P_3, \dots$ – Loads on the columns

R – Resultant column load

I_x – MI of raft about X axis = $BL^3/12$

I_y – MI of raft about Y axis = $LB^3/12$

M_x – Moment about X axis = $R e_y + M_x(\text{lateral load})$

M_y – Moment about Y axis = $R e_x + M_y(\text{lateral load})$

e_x – eccentricity of resultant load about centroid of footing about Y axis

e_y – eccentricity of resultant load about centroid of footing about X axis

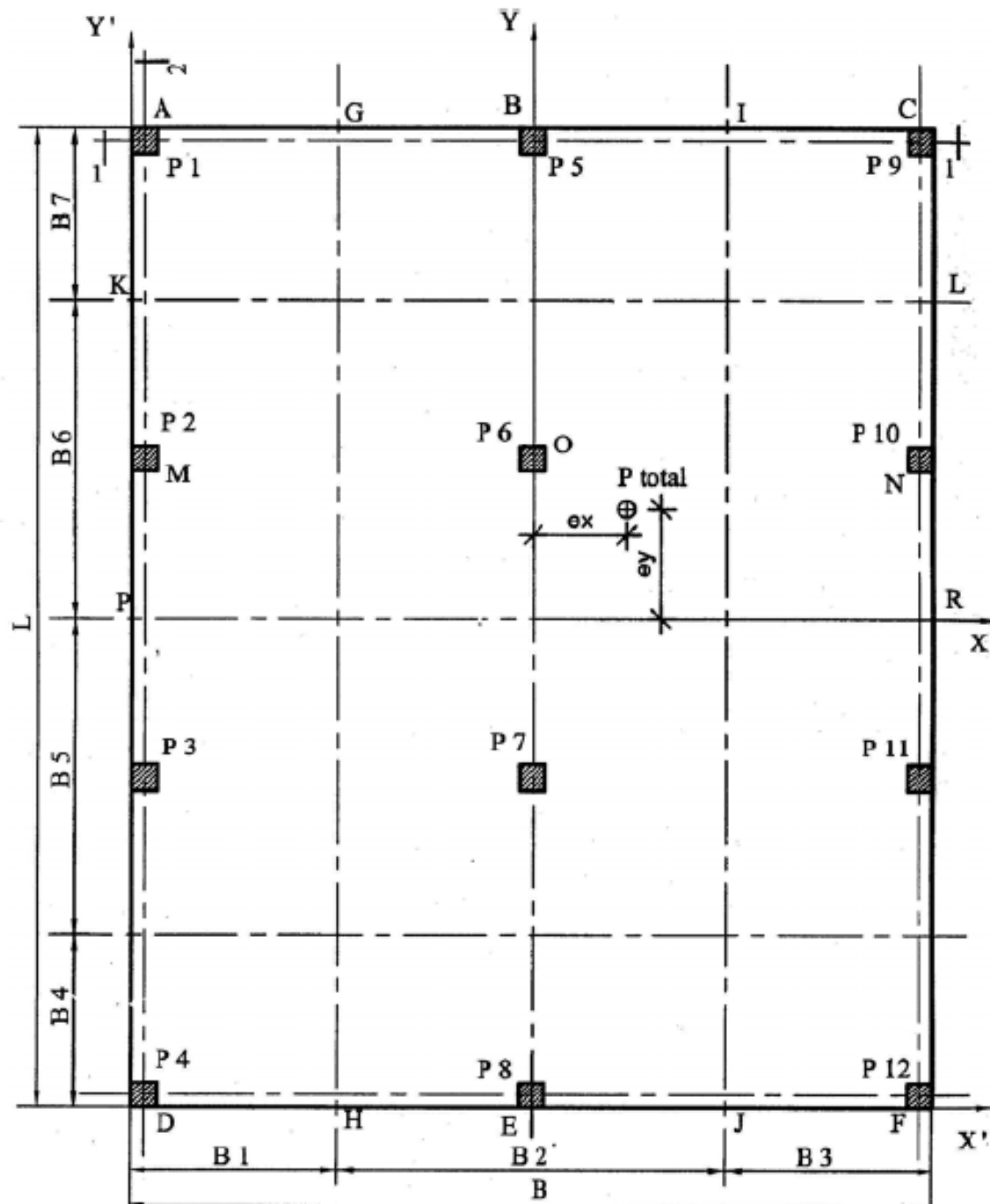
1) Choose reference axis in such a way that entire raft lies in the first co ordinate

2) Determine position of the resultant column load with respect these axis

$$X' = \frac{P_1 x_1 + P_2 x_2 + P_3 x_3 + \dots}{R} \quad \text{and} \quad Y' = \frac{P_1 y_1 + P_2 y_2 + P_3 y_3 + \dots}{R}$$

3) Determine eccentricity with respect to centre of footing

$$e_x = X' - \frac{B}{2} \quad \text{and} \quad e_y = Y' - \frac{L}{2}$$



4) Divide the raft into number of strips parallel to Y axis (B_1, B_2, B_3) as well as parallel to X axis B_4, B_5, B_6 and B_7 as shown in figure.

5) Draw SFD and BMD for all such strips

6) Sample calculation for one such strip GBIJEH is shown below

Soil pressure along the Centre of this strip is assumed to be constant along the width of strip

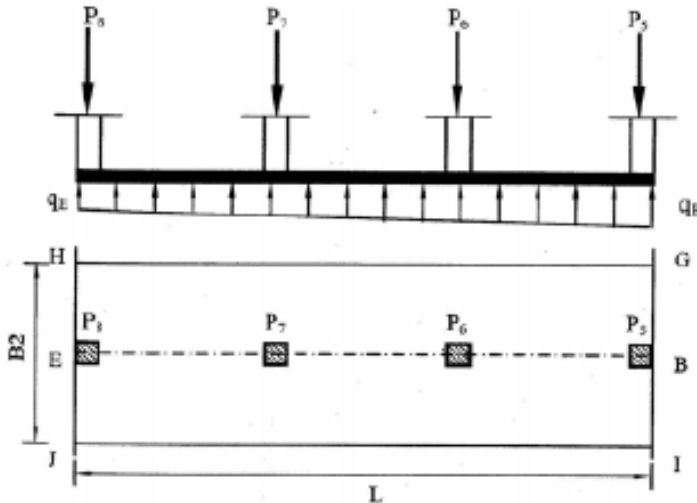
Pressure at point B and E can be calculated by using the equation

$$q = \frac{R}{A} \pm \frac{M_y}{I_y} x \pm \frac{M_x}{I_x} y$$

Calculate average pressure on the strip

$$q_{av} = \frac{q_B + q_E}{2}$$

This average pressure is used for the analysis of strip



7) The strip has to satisfy equilibrium condition

Total downward load = $Q_D = P_5 + P_6 + P_7 + P_8$

Total upward load = $Q_U = q_{av} (L B_2)$

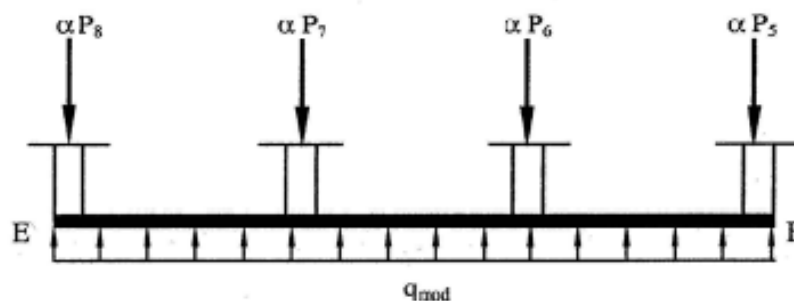
For equilibrium $Q_D = Q_U$; However this condition is not satisfied so we need to modify the pressure as well as soil pressure

Average load on strip $Q_{av} = \frac{Q_U + Q_D}{2}$

Modified soil pressure per unit width = $w_m = \frac{Q_{av}}{L}$

Each column need to be multiplied by a factor α so that total upward load matches with column loads

$$\alpha (P_5 + P_6 + P_7 + P_8) = Q_{av}$$



8) Draw shear and BM diagram for the strip

Assumptions and limitation of rigid design analysis:

Rigid method is based on following *assumptions*

1. The footing or mat is infinitely rigid, and therefore, the deflection of the footing or mat does not influence the pressure distribution. Thus magnitude and distribution of bearing pressure depends only on load
2. The soil pressure is distributed in a straight line or a plane surface such that the centroid of the soil pressure coincides with the line of action of the resultant force of all the loads acting on the foundation.

Limitation of rigid analysis:

In reality no foundation is rigid some flexibility will be there which depends on thickness of foundation. In case of Mat foundation ratio of width to thickness is quite large as compared to spread footing thus assumption of rigidity does not hold good in case of raft foundation. Portion of mat beneath columns and bearing walls settle more than the portion with less load, which means the bearing pressure will be greater beneath the heavily loaded zones. Rigid method does not account for redistribution of bearing pressure (contact pressure), it does not produce reliable estimates of the shears, moments and deformation of the mat. Actual distribution of contact pressure below the footing depends on type of footing, depth of footing and type of soil

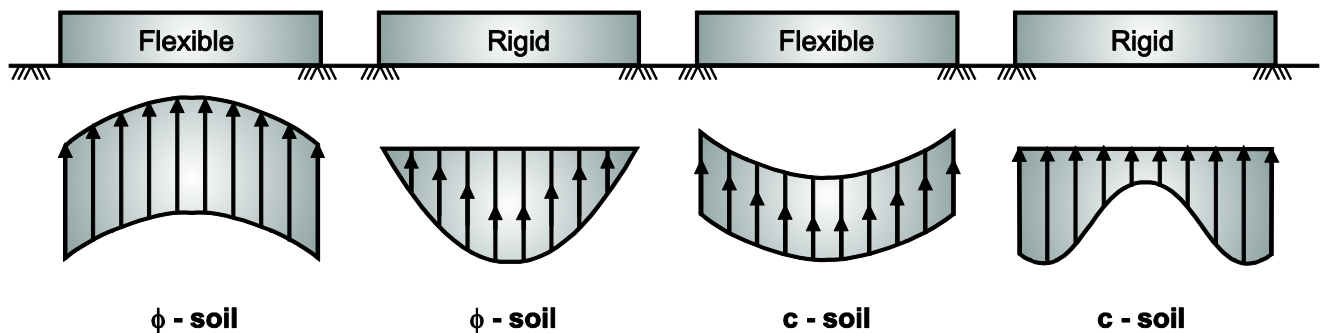


Fig: Contact pressure diagram for surface footing :

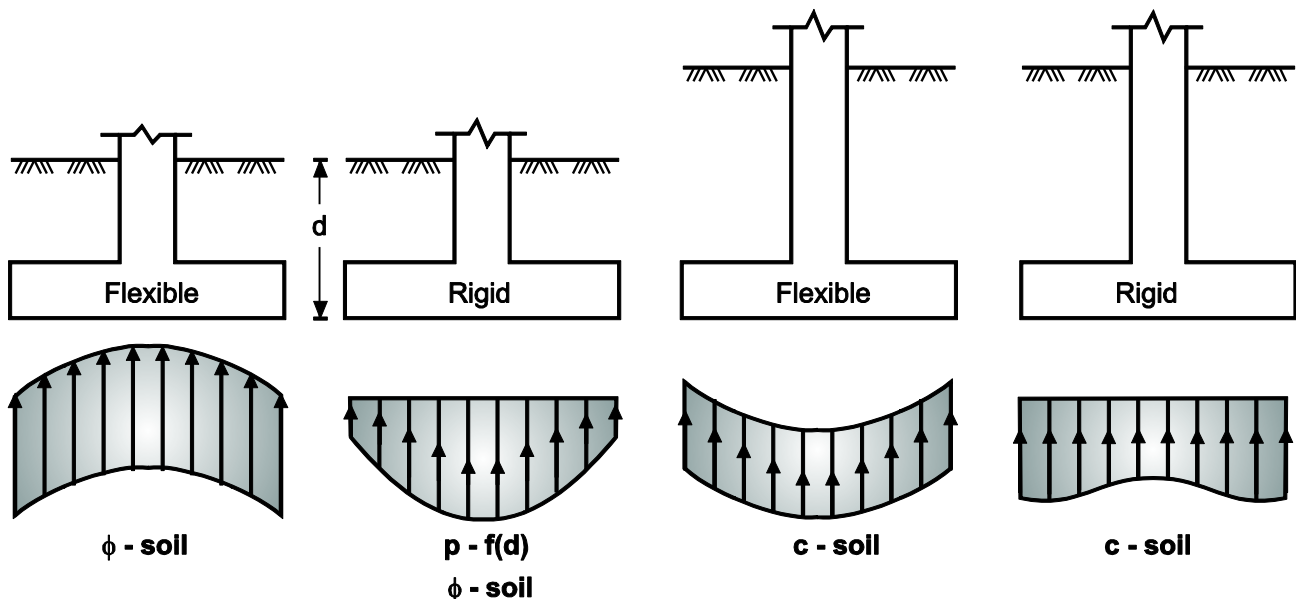


Fig: Contact pressure diagram for deep footing:

In general, pressure distribution is assumed to be uniform for axial load and linearly varying for eccentric loads. The reason for such disparity between theory and practice is due to:

1. Footings are neither absolutely rigid nor completely flexible.
2. There is always certain embedment to footings. The effect of the embedment is to make soil pressure uniform.
3. The settlement and also the bearing capacity of footing are governed by stress distribution in soil zone extending below the base of footing and contact pressure does not influence them.

In case of flexible footing, pressure is uniform. It will produce a dish shaped settlement in clay and in case of granular soil 'E' increases with confining pressure and further E varies across width of loaded area being more near the center than near the edges, which causes dish shaped settlement.

For a rigid footing the settlement is uniform over the contact area. Since uniform contact pressure produce dish shaped pattern thus the contact pressure must be more near the edges and less near the center for clayey soil ($E = \text{constant}$), and is zero at edges to maximum at center for sandy soil, because E increases towards footing center.

Module 5: Deep foundation (7 Lectures)

Pile foundation: Classification, pile driving, load carrying capacity of piles, single pile capacity, dynamic formulae, static formulae, pile load tests, penetration tests, negative skin friction, under reamed piles, group action of piles.

Caisson foundation: Box, pneumatic, open caissons, forces, grip length, well sinking, practical difficulties and remedial measures.

Sheet piles: Classification, design of cantilever sheet pile in cohesionless and cohesive soils. Design of anchored sheet pile by free earth support method, cellular cofferdam: Types, cell fill stability considerations.

Note: Take unit IV as it is for pile foundation and Caisson foundation (from our book)

A pile is a relatively slender member made of timber, concrete, steel or composite material usually either driven into the soil or placed/bored into the soil to provide vertical or lateral support or for other engineering functions. The piles transfer load by bearing on competent material or through the friction between the soil and the pile (skin friction).

When to use pile foundation?

In general pile foundation is likely to be more expensive than spread or mat foundation. Hence are used when no alternative is left to support the structural loads; Pile foundations are used when the underlying soils are incapable of resisting the loads from the structure. The piling is placed in the ground through poor quality materials to bear on competent soils and is commonly used in following situations.

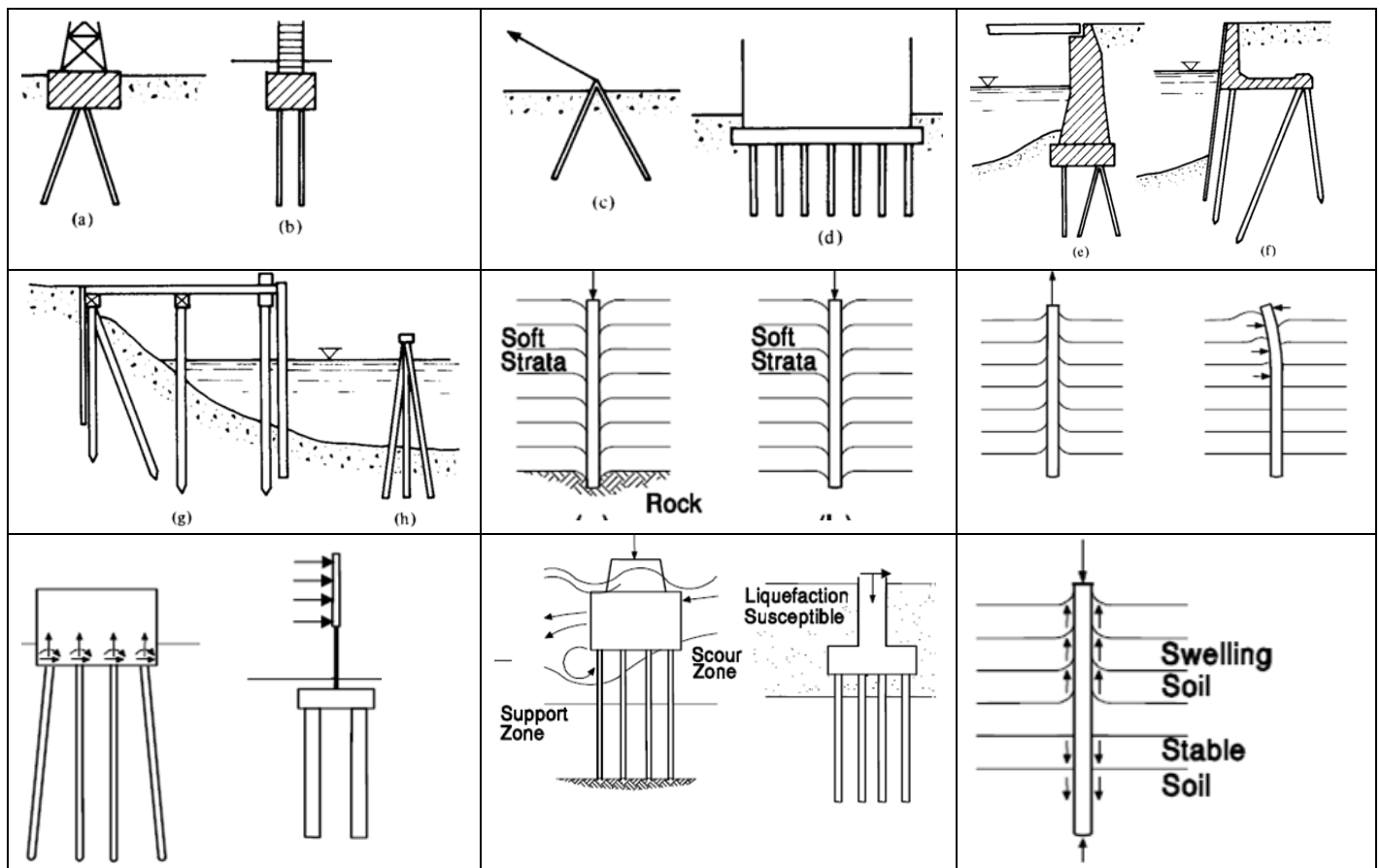
- Structure carries large loads and a good stratum is available at large depth.
- Structure carries large lateral loads.
- Structure carries eccentric loads.
- Structure carries inclines loads.
- Structure is highly sensitive to settlement.
- Structure is to be constructed in expansive soil with high swelling potential.
- For offshore structures.
- Bridge piers and abutments when there is large scouring.
- To anchor down the structures subjected to uplift due to water pressure or excessive moments (tension piles or uplift piles).
- For compacting loose sandy soil
- To provide anchorage against horizontal forces from sheetpiling walls or other pulling forces (anchor piles).

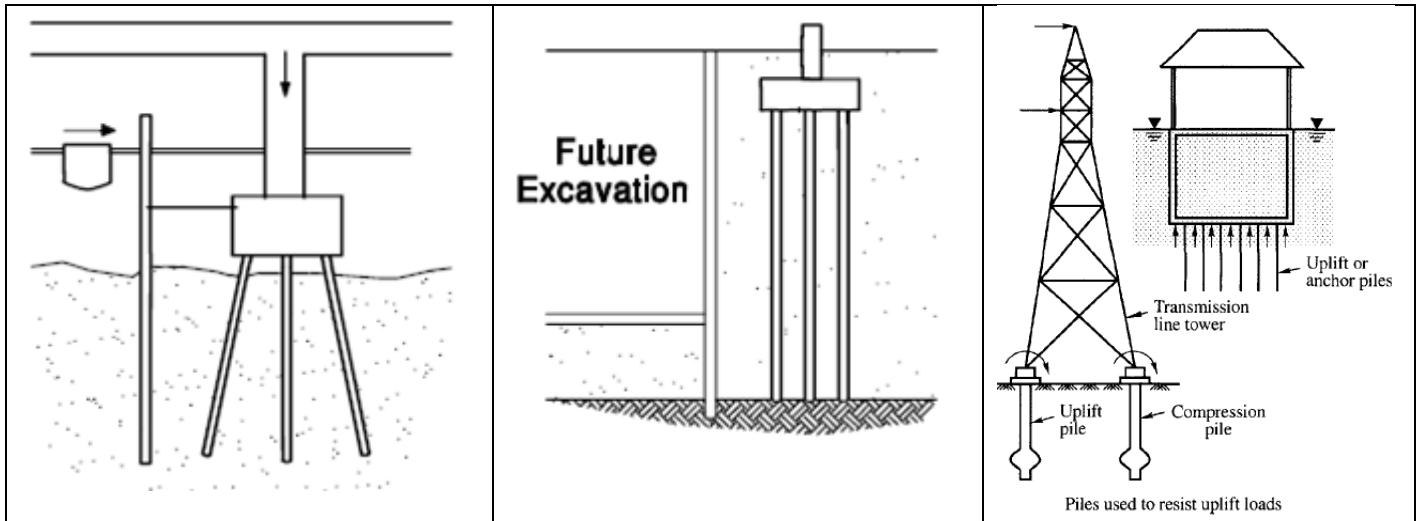
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- To shield water front structures against impact from ships or other floating objects (fender piles and dolphins).

Classification of pile:

Pile foundations can be categorized into two general types: displacement piles and replacement piles. A displacement pile is a pile that is driven or vibrated into the ground and displaces the surrounding soil during installation. (Which in turn is classified into two types based on volume of soil displaced by piles i.e. small displacement and large displacement piles) Whereas a replacement pile is a pile that is placed or constructed within a previously drilled borehole and replaces the excavated soil.





Large displacement piles comprise solid-section piles or hollow-section piles with a closed end, which are driven or jacked into the ground and thus displace the soil. All types of driven and cast-in-place piles come into this category.

Small-displacement piles are also driven or jacked into the ground but have a relatively small cross-sectional area. They include rolled steel H- or I-sections, and pipe or box sections driven with an open end such that the soil enters the hollow section. Where these pile types plug with soil during driving they become large displacement types.

Replacement piles are formed by first removing the soil by boring using a wide range of drilling techniques. Concrete may be placed into an unlined or lined hole, or the lining may be withdrawn as the concrete is placed. Preformed elements of timber, concrete, or steel may be placed in drilled holes.

Piles can be classified based on various criteria's as listed below

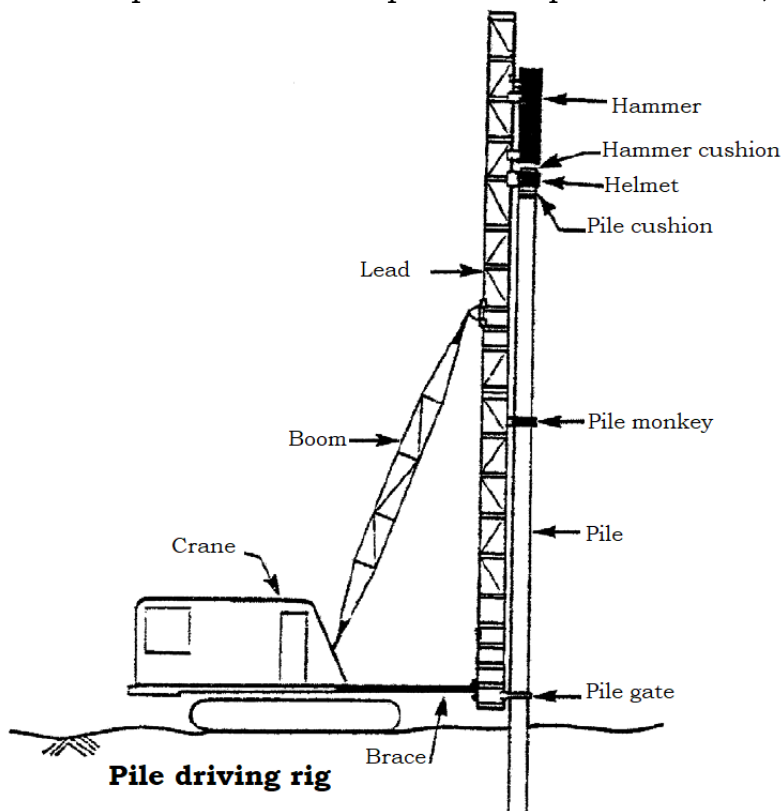
Classification Based on							
1) Material	Timber	Steel	Concrete		Composite		
2)Shape in elevation	Cylindrical	Tapered	Under-reamed				
3)Shape in Plan	Circular	Square	Hexagon	Octagon	H-Pile	I Pile	Box
	Pipe						
4) Method of Intallation	Bored	Driven	Jacked	Vibrated	Jetted	Tremie	
5) Mode of load transfer	Friction	Bearing	Combined				
6)Method of forming	Cast - in - situ	Pre- cast					
7) Length	Short	Long	Slender				
8) Volume of soil displaced	High	Low	Non				

Pile driving: In case of precast piles (concrete/steel/timber). The piles are casted and carried to the site and then is driven into the ground at desired location. The rig used for driving the pile is as shown below. Various kinds of rammers are used to drive the pile into ground. Hammers are lifted up by some distance and is then dropped on the pile, due to impact of the hammer pile will penetrate into the ground. The process is continued till pile penetrate to desired depth.

The following procedure is generally used during the driving of piles.

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- A stake is driven at the location of the pile.
- The pile is straightened and kept upright on the location marked.
- The plumbness of the pile is checked.
- The pile-driving hammer is lowered to the top of the pile.
- Few light blows are given and checked for plumbness.
- Full driving starts.
- The rate of penetration is recorded. (Rate of penetration=Number of blows per mm)
- The pile is driven to the planned depth. In some cases, pile driving is stopped when the required rate of penetration is achieved.
- The inspector should keep an eye on rate of penetration of the pile.
Unusually high blows per mm may indicate boulders or bedrock. Some piles (especially timber piles) get damaged if they hit a boulder.
- After the pile is driven as per the required criteria, the pile is cut off from the top.



Important elements of the driving system include the **leads**, the **hammer cushion**, the **helmet**, and for concrete piles, the **pile cushion**.

Typical components of a pile driving system are shown in Figure. The leads are used to align the hammer and the pile such that every hammer blow is delivered concentrically to the pile system. The helmet holds the top of the pile in proper alignment and prevents rotation of the pile during driving.

When the hammers are used to install a pile, the following important factors must be considered:

1. Weight and size of the pile
2. Driving resistance and net transferred energy
3. Available space at site
4. Crane facility
5. Noise control or restrictions

CAPACITY OF A PILE:

Maximum load a pile can carry without failure is called its capacity and can be determined by following methods

- 1) Static method. 2) Dynamic method.
- 3) By pile load test. 4) Correlation with penetration test data

Static method:

When a load is applied at the top of a pile, the pile will tend to move vertically downward relative to the surrounding soil. This will cause shear stresses to develop between the soil and surface of the shaft. As a result, applied load is distributed as friction load along a certain length of the pile measured from the top. As the load at the top is increased, the friction load distribution will extend more and more towards the tip of the pile, till at a certain load level, the entire length of the pile is involved in generating the frictional resistance.

In general $Q_U = Q_P + Q_f - W_P$

Where Q_U – Ultimate capacity of pile

Q_P - Point Bearing load

Q_f - Friction component of load

W_P – Self weight of pile

Magnitude of W_P is very less compared to the other terms so in most of the cases it is neglected.

$$Q_U = Q_P + Q_f$$

$$Q_U = q_p A_p + f_s A_s \quad \text{Where}$$

$$q_p = cN_c + qN_q + 0.5\gamma BN_\gamma$$

$$f_s = \alpha c_1 + Kq_{av}\tan\delta$$

various terms in above equations are

q_p – ultimate bearing capacity of the soil at tip of pile

f_s – unit skin friction developed along the shaft of pile (< 100kPa)

N_c , N_q and N_γ are bearing capacity factors

B – diameter / size of pile

K – coefficient of earth pressure

δ - angle of wall friction

q – effective overburden pressure at the tip of pile

q_{av} – average effective overburden pressure over the embedded length of pile

c – unit cohesion at the tip of pile

c_1 - average unit cohesion for the depth of embedment of pile

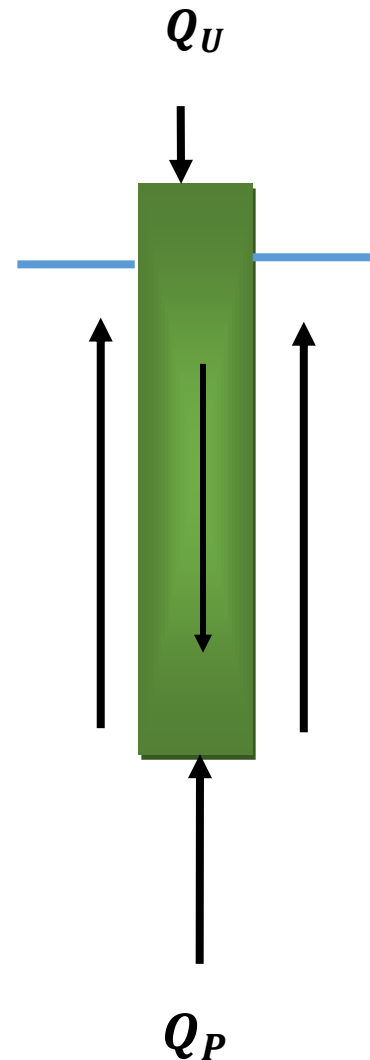
α – adhesion coefficient

The value of third term in the equation $q_p = cN_c + qN_q + 0.5\gamma BN_\gamma$ is negligible in comparison with first two terms thus the equation for point bearing can be written as $q_p \cong cN_c + qN_q$

Where q – effective surcharge and is calculated as

$$\text{as } q = \gamma' L \text{ for } L < D_c \quad \text{and} \quad q = \gamma' D_c \text{ for } L > D_c$$

D_c – critical depth



This is the ultimate skin friction resistance of the pile Q_f . It is only when the load at the top of the pile exceeds Q_f that the load in excess of Q_f begin to be transferred to the soil at the base of the pile. This load is known as point bearing load, which goes on increasing till soil at the base of the pile fails by punching shear failure. Load in bearing corresponding to this stage is called ultimate point load.

Value of N_c is taken as 9 and the value of N_q and D_c can be taken as below
Value of N_q and critical depth will be taken from the table given below.

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State of sandy soil	Density index	Value of N_q		Value of critical depth D_c
		Driven pile	Bored pile	
Loose	0.2- 0.4	60	25	6B
Medium	0.4 - 0.75	100	60	8B
Dense	> 0.75	180	100	15B

Value of K , δ and α can be taken as below

Values of K and δ				Values of α			
Pile material	δ	Value of K		Consistency of soil	N value	α	
		Loose sand	Dense sand			Bored pile	Driven cast in situ pile
Steel	20°	0.5	1.0	Soft to very soft	< 4	0.7	1.0
Concrete	0.75φ	1.0	2.0	medium	4 - 8	0.5	0.7
Timber	0.67 φ	1.5	4.0	stiff	8-15	0.4	0.4
				Stiff to hard	> 15	0.3	0.3

Value of α depends on undrained strength of the soil. Smaller the strength of the soil softer the consistency of soil and greater will be the tendency for the soil to adhere to the pile hence for such soil α will be close to 1.0 and for very stiff clays the adhesion coefficient will be as low as 0.3. Although value of α is less for stiff soil but still its contribution in skin friction is considerable due to its inherent high shear strength.

Dynamic Method:

Dynamic pile driving formulas are based upon the theory that ultimate carrying capacity is equal to the ultimate driving resistance. These formulas are derived starting with the relation:

Energy input = energy used + energy lost

The energy used equals the driving resistance times the pile movement. Thus by knowing the energy input and estimating energy losses, driving resistance can be calculated from observed pile movements. Numerous dynamic formulas have been proposed. They range from the simpler Engineering News Formula, to the more complex Hiley Formula.

a) Engineering News Record formula

$$Q_U = \frac{W_h H}{S + C} \quad \begin{array}{l} C = 2.5\text{cm for drop hammer and} \\ 0.25\text{cm for single and double acting hammer} \end{array}$$

b) Modified ENR formula

$$Q_U = \frac{1.25\eta_h E_h}{S + 0.25} \times \frac{W_h + e^2 W_p}{W_h + W_p}$$

c) Hiley's Formula

$$Q_u = \frac{\eta_h \eta_b W_h H}{S + 0.5C}$$

Where η_b – Efficiency of hammer blow (ratio of energy after impact to striking energy of ram)

$$\eta_b = \frac{W_h + e^2 W_p}{W_h + W_p} \quad \text{for } W_h > e W_p$$

$$\eta_b = \frac{W_h + e^2 W_p}{W_h + W_p} - \left[\frac{W_h - e W_p}{W_h + W_p} \right]^2 \quad \text{for } W_h < e W_p$$

where

Q_u – Ultimate capacity of pile

W_h – Weight of hammer in kg

W_p – weight of pile helmet and follower

H – Height of drop of hammer in cm

S – penetration or set in cm per blow

e – coefficient of restitution

C – total compression = $C_1 + C_2 + C_3$

C_1 – temporary elastic compression of dolly and packing

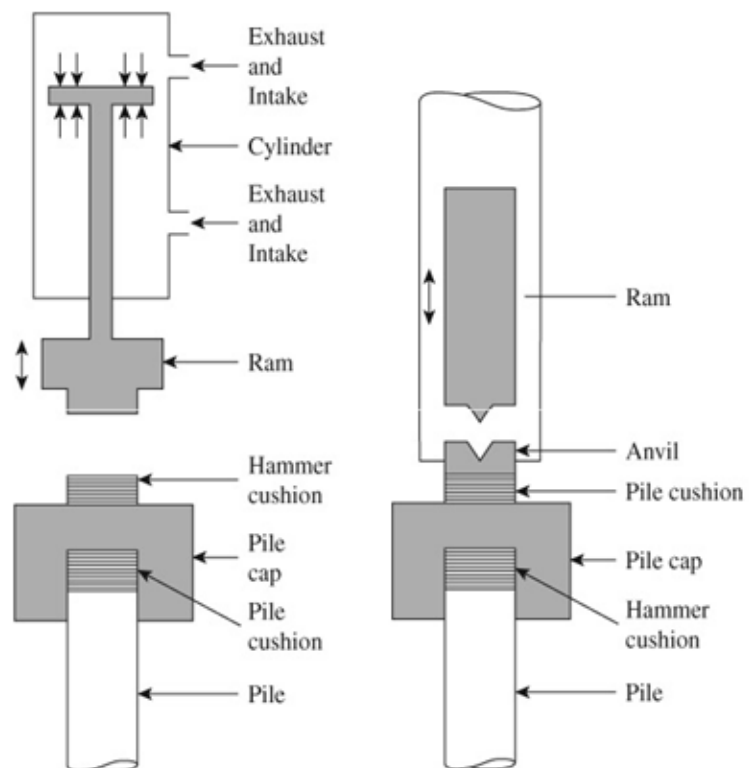
C_2 – temporary elastic compression of pile

C_3 – temporary compression of soil

η_h – efficiency of hammer

Efficiency of hammer can be taken as

Type	Efficiency %
Drop hammer	75 – 100
Single acting hammer	75 – 85
Double acting	85
Diesel hammer	85 – 100



Pile Load Test

Actual load test on piles is the most effective and accurate method of determining the load capacity.

The test is performed in one of the following ways.

1. Continuous loading – Load applied is gradually increased.
2. Cyclic loading – Load applied is removed and again applied (increased).

The test is conducted on a test pile or working pile.

The pile which is used only for testing (i.e. to find load carrying capacity); the same pile is not used for supporting the structure after the test, is called '**test pile**'.

The pile which is used for testing and subsequently used for supporting structural loads is called '**working pile**'.

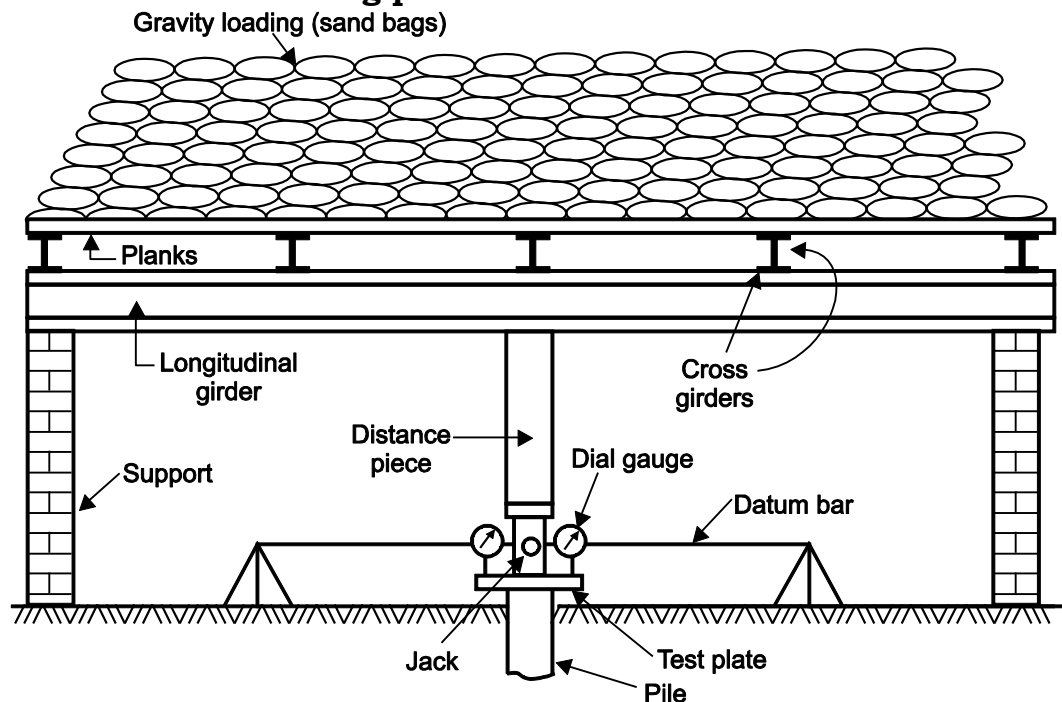


Fig: Pile load test setup

Continuous Loading Test

Arrangement for the test is as shown in Fig. and is used to determine the settlement of pile at a given load and also to determine ultimate load which the pile can take at failure.

Load is applied in equal increments of about 20% of estimated allowable load. Corresponding settlements are recorded with the help of three dial gauges symmetrically arranged over the test plate.

Each load is kept for sufficient time till the rate of settlement becomes less than 0.02 mm/hr. Test piles are loaded until ultimate load is reached. Test load is increased up to a value of 2.5 times allowable load (estimated) or to a load which cause settlement equal to 10% of pile diameter whichever occurs earlier. The results are plotted in the form of load-settlement curve.

Allowable load is determined from load settlement curve load.

1. $\frac{1}{3}$ to $\frac{1}{2}$ of load which causes a settlement equal to 10% of pile diameter.
2. $\frac{2}{3}$ of load which causes settlement of 12 mm.
3. $\frac{2}{3}$ of load which causes a net settlement of 6 mm.

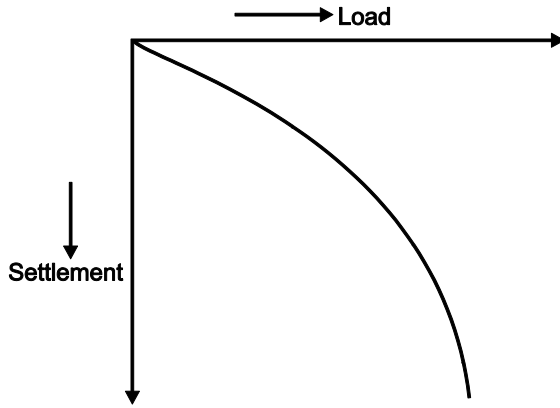


Fig : Load settlement curve

Cyclic Pile Load Test

In this test, the load is raised upto a particular value then released to zero; again raised to higher value and released to zero. Settlements are recorded at each increment and decrement of load. This test is performed to separate out the point bearing load and frictional load.

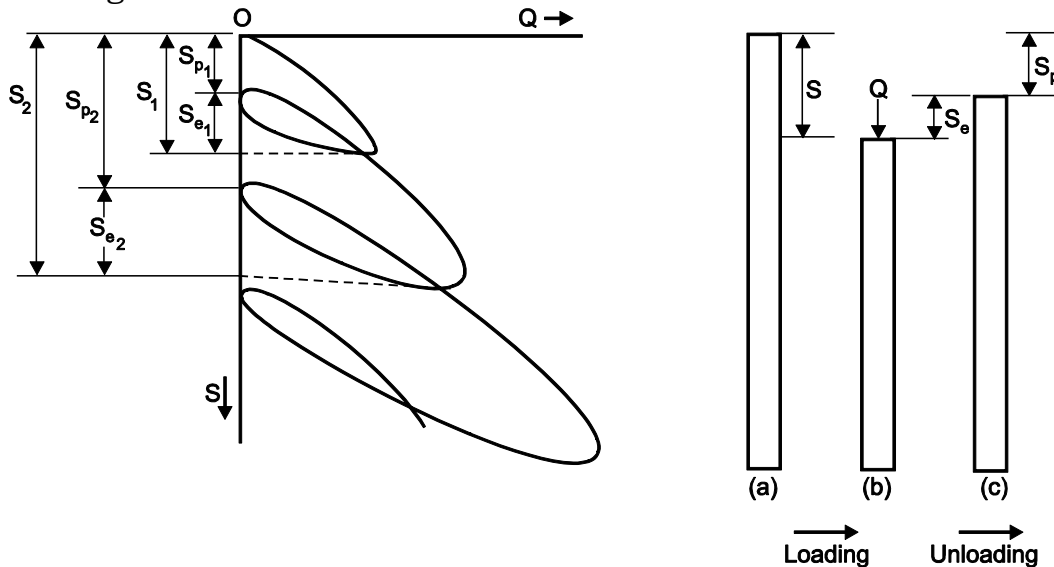


Fig Elastic and plastic settlement of pile

Fig. shows a pile which is being loaded. During loading the total settlement is "S" and then it is unloaded. While unloading, there is an elastic recovery 'Se' and residual settlement which is called 'plastic settlement'. If again we load the pile and unload it we will get elastic and plastic settlement which is as shown in graph. Thus, for each load we have elastic and plastic compression.

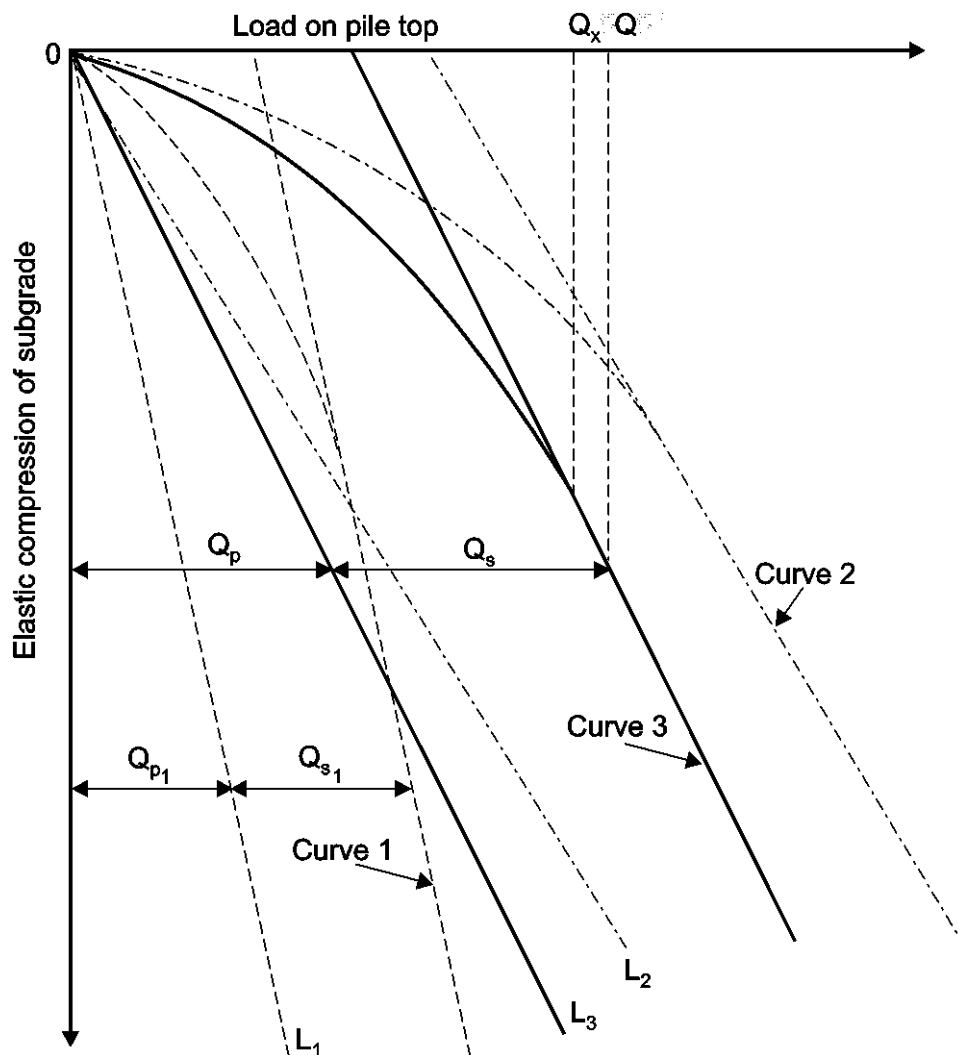


Fig: separation of skin friction and point bearing

Procedure for Cyclic Pile Load Test :

1. Draw a graph between applied load and elastic settlement $(Q_1, S_{e1}), (Q_2, S_{e2}) \dots (Q_n, S_{en})$, as shown by curve '1' in graph.
2. Draw straight line "L1" parallel to straight portion of curve 1 starting from origin.
3. This line approximately divides the load into Q_p and Q_s .
4. Find out skin load corresponding to each load. Say $Q_{s1}, Q_{s2}, Q_{s3} \dots Q_{sn}$.
5. Find elastic compression of pile by formula for each load:

$$\Delta L = \frac{\left(Q - \frac{Q_s}{2}\right) \cdot L}{A \cdot E}$$

Where, A – Cross-sectional area of pile

E – Young's modulus of elasticity of pile material

6. Find modified values of elastic compression

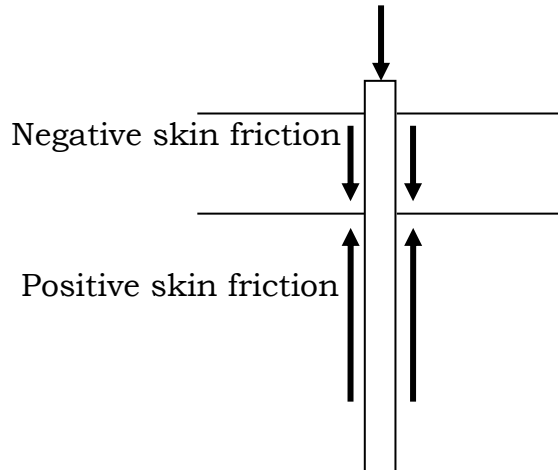
$$\bar{S}_e = S_e - \Delta L$$

7. Plot graph between $\bar{S}_e$ and Q .
8. Repeat the procedure (steps 4, 5 and 6) till two subsequent values of Q_s are close to each other.

Negative Skin friction:

Negative skin friction is downward drag acting on the piles due to relative movement between the piles and the surrounding soil. When piles are driven through compressible soils, and site has newly placed fill or will be filled in the future, the possibility of negative skin friction should be investigated. Soft to medium clays, soft silt, peat, mud, etc are compressible soils. Lowering of ground water table in such compressible soils may also bring about negative skin friction. This occurs on the part of the shaft along which the downward movement of the surrounding soil exceeds the settlement of the pile. Negative skin friction could result from consolidation of a soft deposit caused by dewatering or the placement of fill. The dissipation of excess pore water pressure arising from pile driving in soft clay can also result in consolidation of the clay.

Negative skin friction is calculated similar to positive skin friction $Q_{ns} = Q_s = f_s A_s$



The magnitude of negative skin friction that can be transferred to a pile depends on (Bjerrum, 1973):

- (a) pile material,
- (b) method of pile construction,
- (c) nature of soil, and
- (d) amount and rate of relative movement between the soil and the pile.

Development of negative skin friction on pile surface is not desirable, as it will impose additional load on the pile so pile when subjected to negative skin friction need to be designed for external load and negative skin friction, thus as far as possible such condition is avoided either by compacting the soil before driving the pile or the compressible soil is replaced by incompressible soil.

UNDERREAMED PILE:

An 'under-reamed' pile is one with an enlarged base or a bulb; the bulb is called 'under-ream'. There could be one or more under-reams in a pile; in the former case, it is called a single under-reamed pile and in the latter, it is said to be a multi-under-reamed pile.

Under-reamed piles are cast-in-situ piles, which may be installed both in sandy and in clayey soils. The sides may be stabilized, if necessary, by the use of bentonite slurry, sometimes called 'drilling mud'. The under-reams are formed by special under-reaming equipment. The ratio of bulb size to the pile shaft size may be 2 to 3; usually a value of 2.5 is used. The bearing capacity of the pile increases because of the increased base area; the more the number of under-reams the more the capacity. Field tests indicate that an under-reamed pile is more economical than a straight bored pile for a given load.

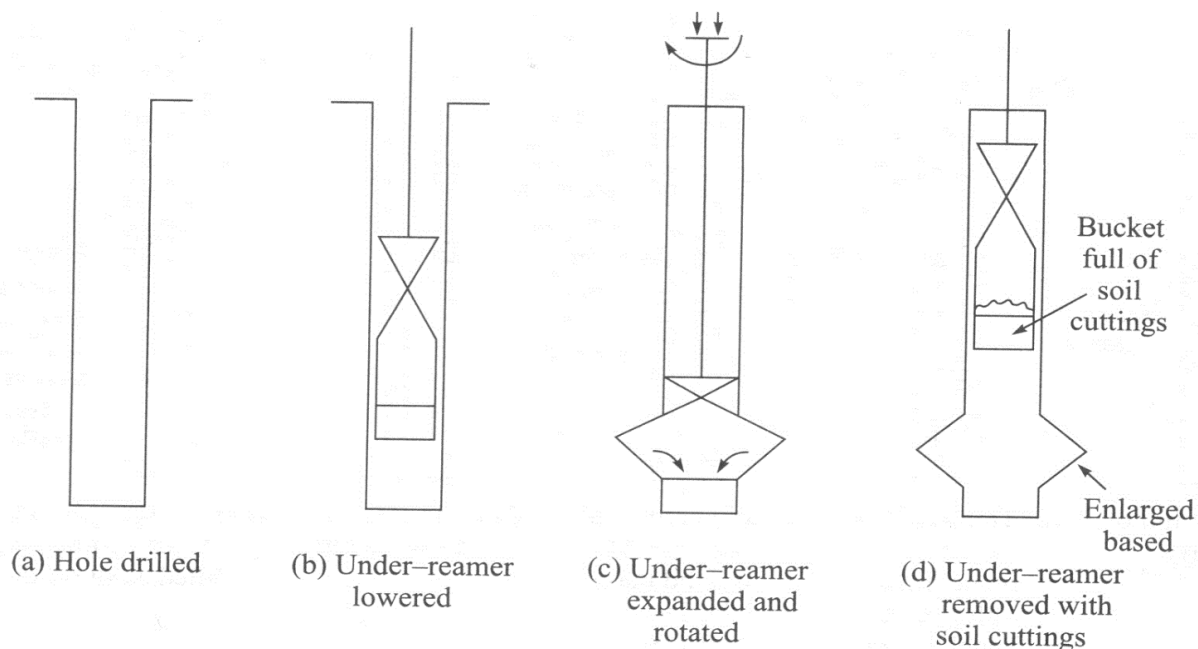
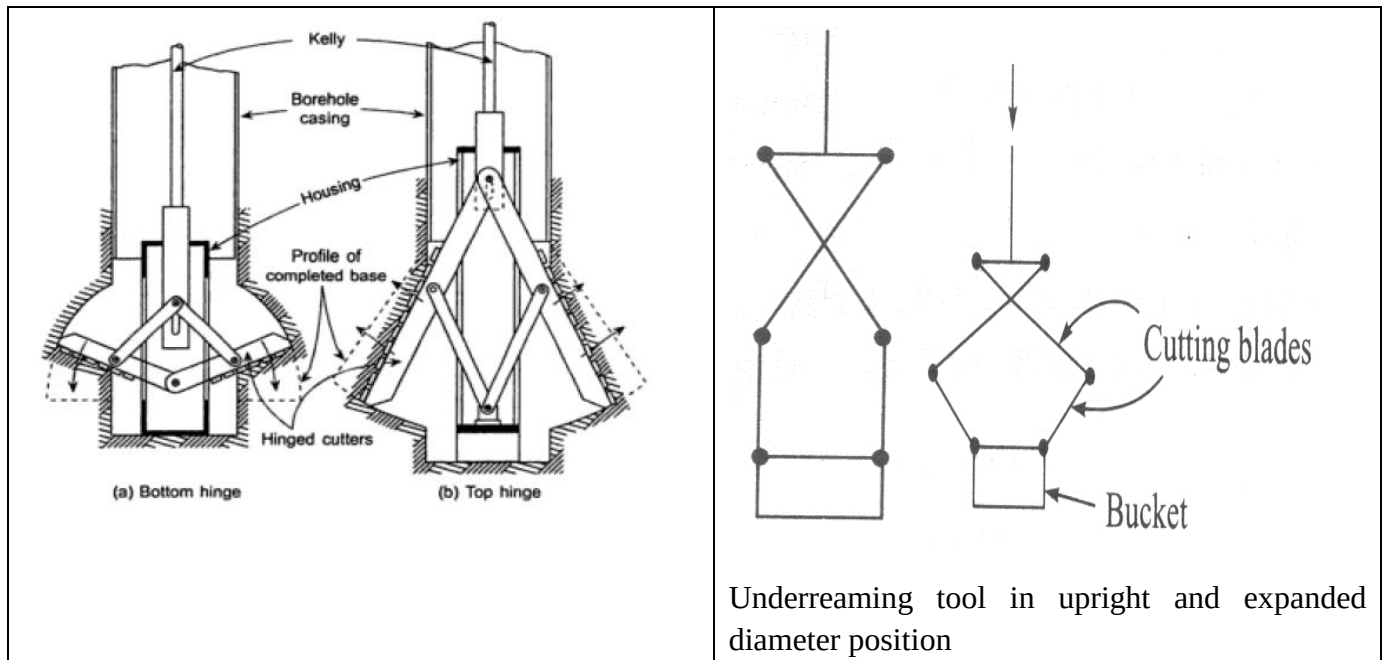


Fig: Construction of under-reamed pile

Procedure :

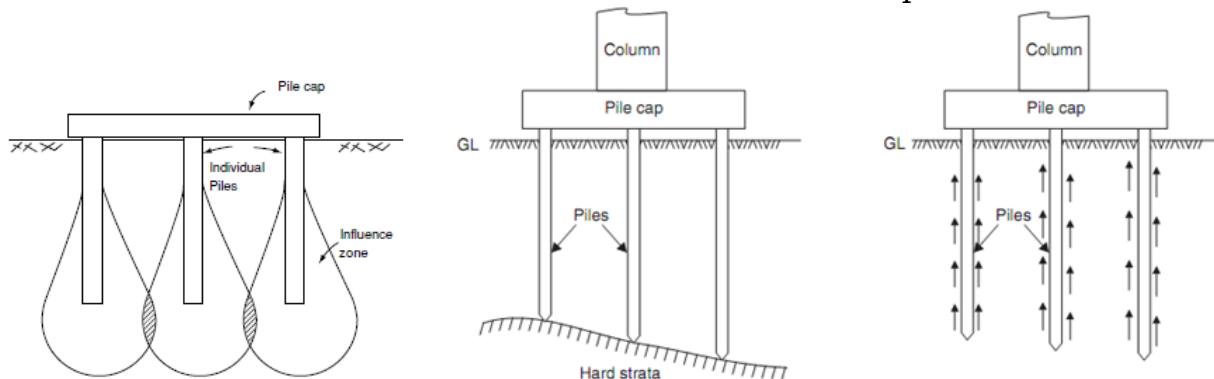
- A bore is drilled by means of a hand operated auger up to required depth.
- Under-reamer is lowered into the hole. It is then pressed down and rotated
- The blades widen under pressure and cut the soil from sides by rotation.
- When the bucket is full, under reamer is pulled out and the soil is removed.
- Tool is again lowered in to the hole and process is repeated.

Precautions:

- Verticality of the borehole need to be ensured by guiding frame
- Casing pipe may be used in loose top soil
- Obstructions to boring need to be broken down with cutting tool. In case obstruction cannot be handled, pile location need to be shifted
- For boring in granular soil drilling mud is used to keep the side stable
- Depth of borehole should be checked before inserting under-reamer otherwise the position of bulb will change
- Spread should be checked before concreting
- For underwater construction concrete of higher slump should be used by using tremie pipe
- Proper compaction of concrete need to be done in bulb

Pile Caps: Structure is never supported on single pile. Generally, there will be a minimum of two or three piles under a foundation element or footing to allow for misalignments and other inadvertent eccentricities. Building codes may stipulate the minimum number of piles under a building element.

Superstructure never rest directly on the pile, concrete slab is cast on the piles to form a base for supporting the structure such slab is called pile cap whose function is to receive the load from structure and transfer it to the piles.



Pile groups: Structure is never supported on single pile. Piles are always used in groups to support the structure. General, there will be minimum of two or three piles under a foundation element or footing to allow for misalignments and other inadvertent eccentricities. Building code may stipulate the minimum number of piles under a building element.

Capacity of pile group is not necessarily equal to sum of capacity of individual piles, because installation and loading of pile bring about changes in

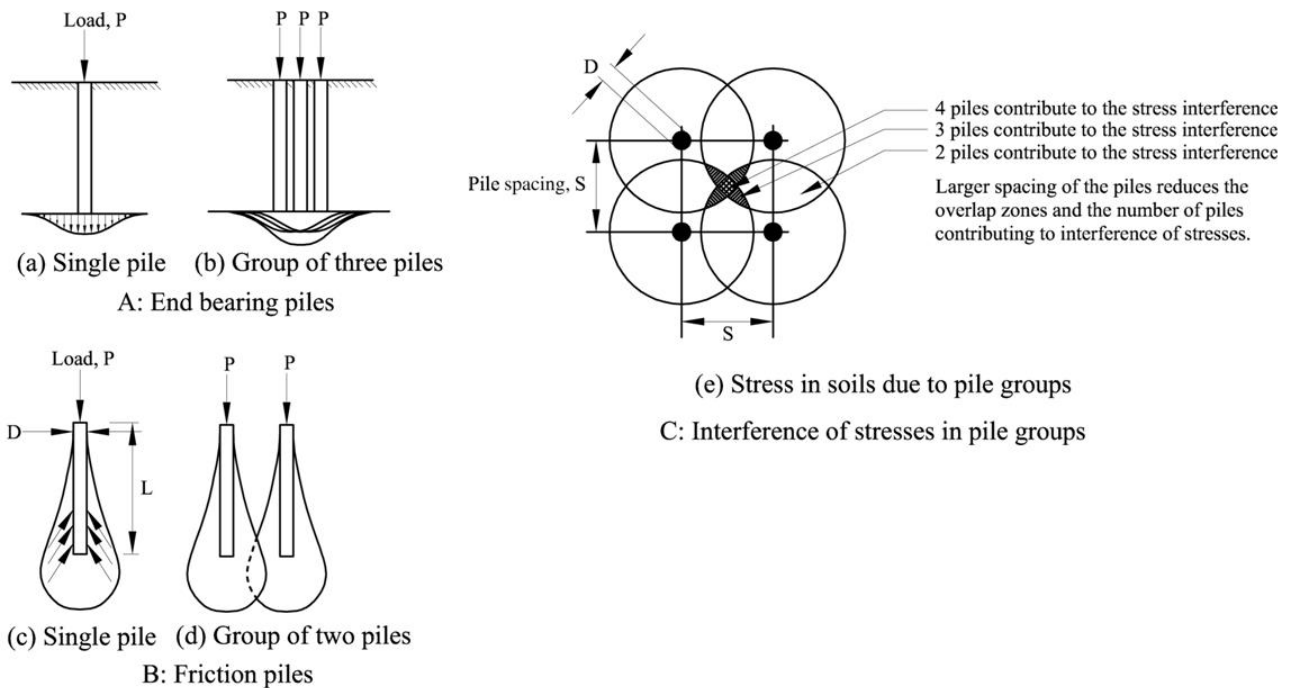
- Stress condition
- Consistency of clayey soil
- Density index of granular soil

Piles installed in a group to form a foundation will, when loaded, give rise to interaction between individual piles as well as between the structure and the piles. The pile-soil-pile interaction arises as a result of overlapping of stress (or strain) fields and could affect both the capacity and the settlement of the piles. The piled foundation as a whole also interacts with the structure by virtue of the difference in stiffness. This foundation-structure interaction affects the distribution of loads in the piles, together with forces and movements experienced by the structure.

The analysis of the behaviour of a pile group is a complex soil-structure interaction problem. The behaviour of a pile group foundation will be influenced by:

- method of pile installation, e.g. displacement or replacement piles,
- dominant mode of load transfer, i.e. shaft friction or end bearing,
- nature of founding materials,
- three-dimensional geometry of the pile group configuration,
- presence or otherwise of a ground-bearing cap, and
- relative stiffness of the structure, the piles and the ground.

Spacing of Piles: The typical stresses/soil pressures produced from shaft friction or end bearing of a single pile and a group of piles are shown in Figure. If piles are used in groups, there may be an overlap of stresses, as shown in Figures. If the spacing is too close the overlap is large, and the soil may fail in shear or settlement will be very large. Though the overlapping zone of stresses obviously decreases with increased pile spacing, it may not be feasible since the pile cap size becomes too large and hence expensive.



Spacing of the piles should be such that neither it increases the cost of pile cap nor there is excessive overlapping of stresses. It depends on following factors

- Type of soil
- Method of installation
- Overlapping of stresses
- Efficiency of pile group
- Cost of foundation

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Optimal spacing will be 2.5D - 3.5D for vertical loads and may be suitably increased for lateral dynamic loads.

Efficiency of pile group :

It is defined as the $E_g = \frac{\text{group capacity}}{\text{number of piles} \times \text{individual pile capacity}}$

If piles are driven into compressible cohesive soil or into dense cohesionless material underlain by compressible soil, then the ultimate axial compression capacity of a pile group may be less than that of the sum of the ultimate axial compression capacities of the individual piles. In this case, the pile group has a group efficiency of less than 1.

In cohesionless soils, the ultimate axial compression capacity of a pile group is generally greater than the sum of the ultimate axial compression capacities of the individual piles comprising the group.

In this case, the pile group has a group efficiency greater than 1. (granular soil densifies in the vicinity of a driven pile).

Effect of spacing of piles on group efficiency (AASHO Guidelines)

Pile Spacing c/c	Pile group efficiency	
	Clayey soil	Sandy soil
3D	0.67	0.67
4D	0.78	0.74
5D	0.89	0.80
6D	1.00	0.87
7D	1.00	0.93
8D or more	1.00	1.00

Empirical equations to calculate efficiency of pile group :

1) Converse Lanarre's formula

$$E_g = 1 - \left\{ \frac{m(n-1) + n(m-1)}{mn} \right\} \frac{\theta}{90}$$

2) Seiler Keeney formula

$$E_g = \left\{ 1 - 0.479 \left(\frac{S}{S^2 - 0.093} \right) \left(\frac{m+n-2}{m+n-1} \right) \right\} + \frac{0.3}{m+n}$$

3) Los Angeles formula

$$E_g = 1 - \left\{ m(n-1) + n(m-1) + \sqrt{2}(m-1)(n-1) \right\} \frac{D}{\pi m n S}$$

4) **Felds rule** : Value of each pile is reduced by $1/16^{\text{th}}$ on account of effect of nearest pile in each diagonal or straight row

Where D – diameter of pile

S – Spacing of pile

m and n – number of rows or columns of piles

θ – Angle in degrees = $\tan^{-1} \left(\frac{D}{S} \right)$

Calculation of capacity of pile groups : (clayey soil)

Capacity of the pile group can be calculated by considering group to fail by individual action and by block action

a) By Individual action :

Group capacity = no. of piles in group X capacity of single pile

b) By block action

Group capacity = $P_g L C + A_g X 9C - \gamma . L . A_g$

Above two values are compared and whichever is least is taken as actual capacity of pile group.

Caisson foundation:

The word caisson is derived from the French word **caisse** meaning a box. Caisson is a boxlike structure which is sunk through ground or water to remove water and soil during the process of excavation of foundation and subsequently becomes integral part of the substructure. It usually refers to a substructure element used in wet construction sites such as rivers, lakes, docks etc.

Classification of caisson:

Caissons are classified into three types

- 1) Box caisson (Open at top and closed at bottom)
- 2) Pneumatic caisson (Open at bottom and closed at top)
- 3) Open caisson or Well (Open both at top and bottom)

Uses of Caissons:

- As foundation for bridge piers and abutments in rivers, lakes and seas, breakwaters and other shore works.
- In other protection works and large water front structures, such as pump house, which are subjected to huge vertical and horizontal forces.
- For large and multi-storey buildings and other structures, occasionally caissons are used.

BOX CAISSON: This is a watertight vessel which is open at top and closed at bottom; the box could be made out of timber, RCC or steel. The box is constructed on the river bank then it is floated and towed to the position. It is then filled with D.L. and sunk at site where it serves as foundation floating caisson does not penetrate the soil. It simply rests on a hard, level surface; thus, the load-carrying capacity depends solely on the resistance at the base as there is no frictional resistance at the sides. A concrete cap is cast on its top to receive the loads from the superstructure. To prevent scour, rip rap is placed around the base.

The merits of floating caissons

- Since floating caissons are precast, good quality can be ensured.
- The installation of a floating caisson is quick and convenient.
- Floating caissons are less expensive than other types; they may also be transported at a low cost by floating.
- It can be used where construction of other types of caissons are not possible.

The demerits of floating caissons

- The foundation bed has to be levelled before installing the caisson.
- The base of the caisson must be protected against scour.
- The load carrying capacity is smaller than that of other types of comparable size.

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- This type is suitable only if a good supporting stratum is available at shallow elevation; otherwise, it becomes costly owing to deep excavation, as the saturated soil tends to flow into the excavation.

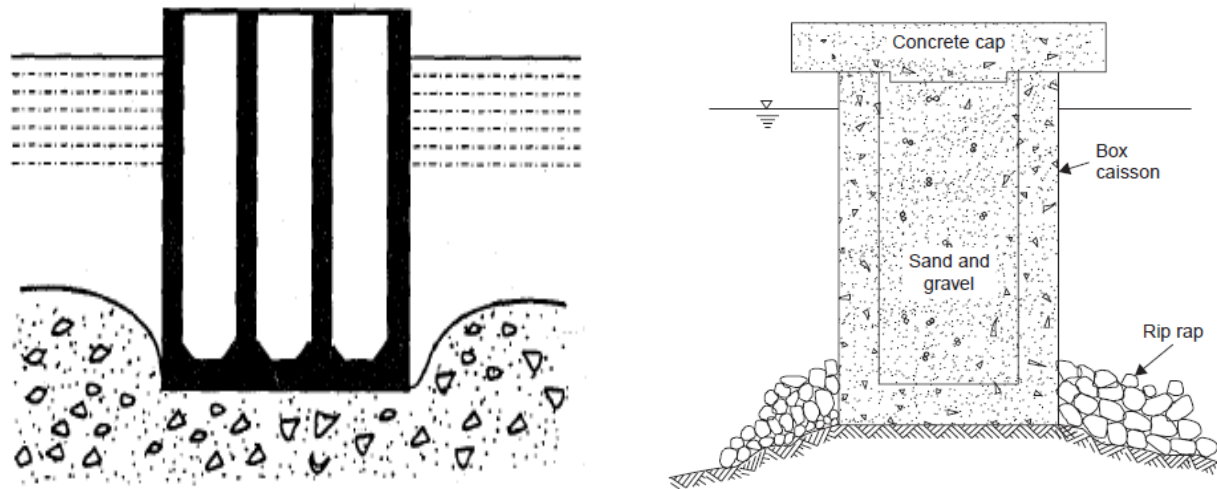


Fig: Box caisson

Construction:



Sequence of construction of box caissons are as given below

- (1) Floating caissons are cast away from site on land.
- (2) Floated to desired location.
- (3) Caissons are filled with sand or gravel to facilitate sinking.
- (4) The desired base location is excavated and levelled as per requirement.
- (5) Caisson is then sunk to desired location and depth.

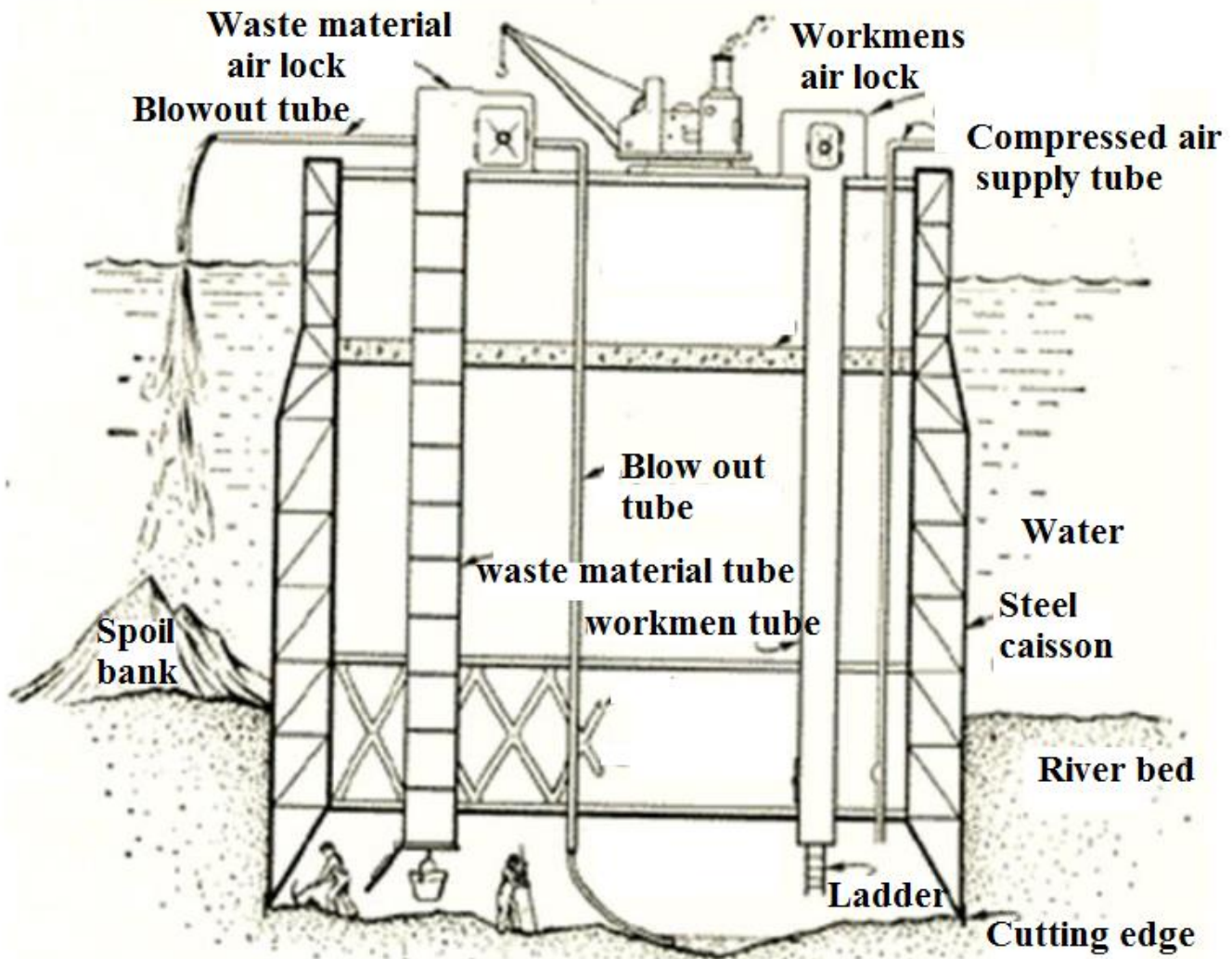
(6) Rip-rap is provided.

(7) Concrete cap is casted at the top to carry loads from superstructure.

Floating caissons are constructed by using RCC or steel. Internal strutting and diaphragm walls may be used if it is to be floated and placed in rough waters.

PNEUMATIC CAISSON:

Pneumatic caisson is open at bottom and closed at top between these two air pressure is maintained so that water does not enter the caisson so as to facilitate excavation and concreting.



Component parts of caisson: A pneumatic caisson consists of the following component parts:

(i) Working Chamber(ii) Air Shaft(iii) Air Lock(iv) Miscellaneous Equipment

(i) **Working Chamber:** This is made of structural steel, about 3 m high, with a strong roof, and is absolutely air tight. The air pressure in the Chamber is raised above atmospheric and is kept at a certain specified value to prevent entry of water and soil into it. This pressure varies with the depth at which excavation is proceeding at any time. The outside surface is made smooth to reduce friction. A cutting edge is provided at the bottom to facilitate the sinking process.

The air pressure must be sufficient to balance the full hydrostatic pressure due to water outside. However, there is a maximum limit to the air pressure in view of the physiological characteristics of human beings; a pressure greater than 0.4 N/mm^2 (400 kN/m^2) is beyond the endurance limit of human beings. Therefore, the maximum depth of water through which a pneumatic caisson can be sunk successfully is about 40 m. working under a pressure greater than 0.4 N/mm^2 may cause a special sickness called 'Caisson disease'.

(ii) Air Shaft: This is a vertical passage which connects the working chamber with an airlock. It is meant to provide access to the working chamber for workmen. It is also used to transport the excavated material to the ground surface. In large caissons, separate shafts may be provided for men and materials. Air-shafts are made of Steel; the joints being provided with rubber gaskets to make them leak proof. Each Shaft is provided with its own air lock at its top. As the caisson sinks, the air shaft is extended to keep the airlock always above the water level.

(iii) Air Lock: This is a steel chamber provided at the upper end of the air shaft above the water level. Its function is to permit the workmen to go in or come out of the Caisson without releasing the air pressure in the working chamber.

The chamber of the airlock is provided with two air-tight doors, one of which opens to the shaft and the other to the atmosphere outside. When a workman enters the airlock through the outside door the pressure in the chamber is kept at atmospheric value. This is gradually raised till it becomes equal to that in the working chamber, and the workman allowed to enter the airshaft through the door to it, and to descend into the working chamber. The procedure is just reversed when one has to come out. However, the decompression must be effected much more slowly to prevent caisson disease. A minimum of half-an-hour is necessary for the pressure to be reduced from 0.3 N/mm^2 to atmospheric pressure. Fresh air is circulated into the working chamber by opening a valve in the airlock in order to prevent the air inside becoming stale. The workmen should not be made to work inside the working chamber under compressed air pressure for more than two hours at a stretch

(iv) Miscellaneous Equipment: Certain miscellaneous equipment such as motors, pressure pumps, and compressors are usually located outside at bed level. Pressure in the working chamber is maintained through an air pipe connected to a compressor. At least one stand-by unit for all equipment should be provided to cope with any emergency.

Merits of Pneumatic Caissons:

- Control over the work and foundation preparation are better, since all work is done in the dry.
- Obstruction from boulders or logs may be readily removed since direct visual inspection of the bottom near the cutting edge is possible.
- Concrete placed in the dry is more capable of attaining better quality and reliability.
- Plumbness of Caisson is easier to control than with other types.

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- Soil can be inspected, samples taken, and bearing capacity ascertained more reliably, if necessary, by *in-situ* testing.
- No settlement of adjoining structures need be apprehended since no lowering of ground water table is expected to occur.
- Large depths of foundation can be achieved to bed rock through difficult strata for major civil engineering works.

Demerits of Pneumatic Caissons:

- Pneumatic Caissons are highly expensive and hence should be used only when other types of caissons are not feasible.
- The depth of penetration is limited to 30 m to 40 m below water table.
- A lot of inconvenience is caused to the workmen while working under compressed air pressure, and they may be afflicted with caisson disease.
- Extreme care is required for the proper working of the system; even a small degree of slackness may lead to an accident.

Construction:

Pneumatic caissons may be:

- (1) Constructed at site or
- (2) Floated to the site and placed from barges.
- (3) The **sand island method** can also be used.

Steps Involved in Construction of Pneumatic Caisson:

Cutting edge is very carefully positioned to desired location.

- Introduction of compressed air is needed into working chamber to keep off mud and water.
- Working chamber is dewatered and made dry.
- Workmen are allowed to enter into shafts through air locks.
- As workmen carryout excavation, caisson starts sinking.
- The air pressure is raised to make up with the pressure due to the head of water as the sinking goes on.
- Excavated material is transferred to the top with the help of buckets through shafts.
- Blow-out pipes can be used in case of granular soils to transfer excavated material.
- When caisson reaches its final and desired position, working chamber is filled with concrete.
- Till concrete gets hardened, the air pressure in the chamber is kept constant.
- Finally, shafts are also filled with concrete after dismantling shaft tubes.

CAISSON DISEASE: In case of sinking process of pneumatic caisson, workers or workmen have to work in working chamber under compressed air. If the compressed air pressure is more than 0.35 N/mm^2 to 0.4 N/mm^2 , then workmen may suffer from the following pains:

- Workmen may suffer from giddiness
- There is a pain in ears of workmen

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- There is breaking of ear drums of workmen
- There is bursting of blood vessels in the nose or ears of workmen

The above mentioned pains are not that serious or fatal, but workmen are actually suffering during decompression and effect causing depression is called caisson disease the symptom of the disease were dizziness, double vision, headache, trouble to speaking, pains in legs etc. The disease results in the loss of consciousness, paralysis or sometimes even death.

Excessive oxygen gets absorbed in the blood and tissues during decompression is more troublesome to workmen. Absorbed oxygen gas is thrown out of blood in the form of bubbles which can block in vessels and may cause bursting of vessels.

- If bubbles are developed in joints it causes bends
- If the bubbles are developed in spinal cord, it causes paralysis and if the bubbles are developed in heart, it causes heart attack

Precautions to avoid caisson disease:

- Person should not work continuously for more than four hours per day
- Temperature in the working chamber should be roughly around 25° C
- Person with strong heart, relatively normal blood pressure and good circulation should be employed
- Worker should undergo complete check up from a physician once in a month
- Rate of compression and decompression should be gradual without causing any discomfort



Fig: working chamber of pneumatic caissons

OPEN CAISSON OR WELL:

Open Caisson is open both at the top and the bottom during construction. It is provided with a cutting edge at the bottom to facilitate sinking. When the caisson has reached the desired location, a fairly thick concrete seal is provided. The thickness of the seal may range from 1.5 to 4.5 m.

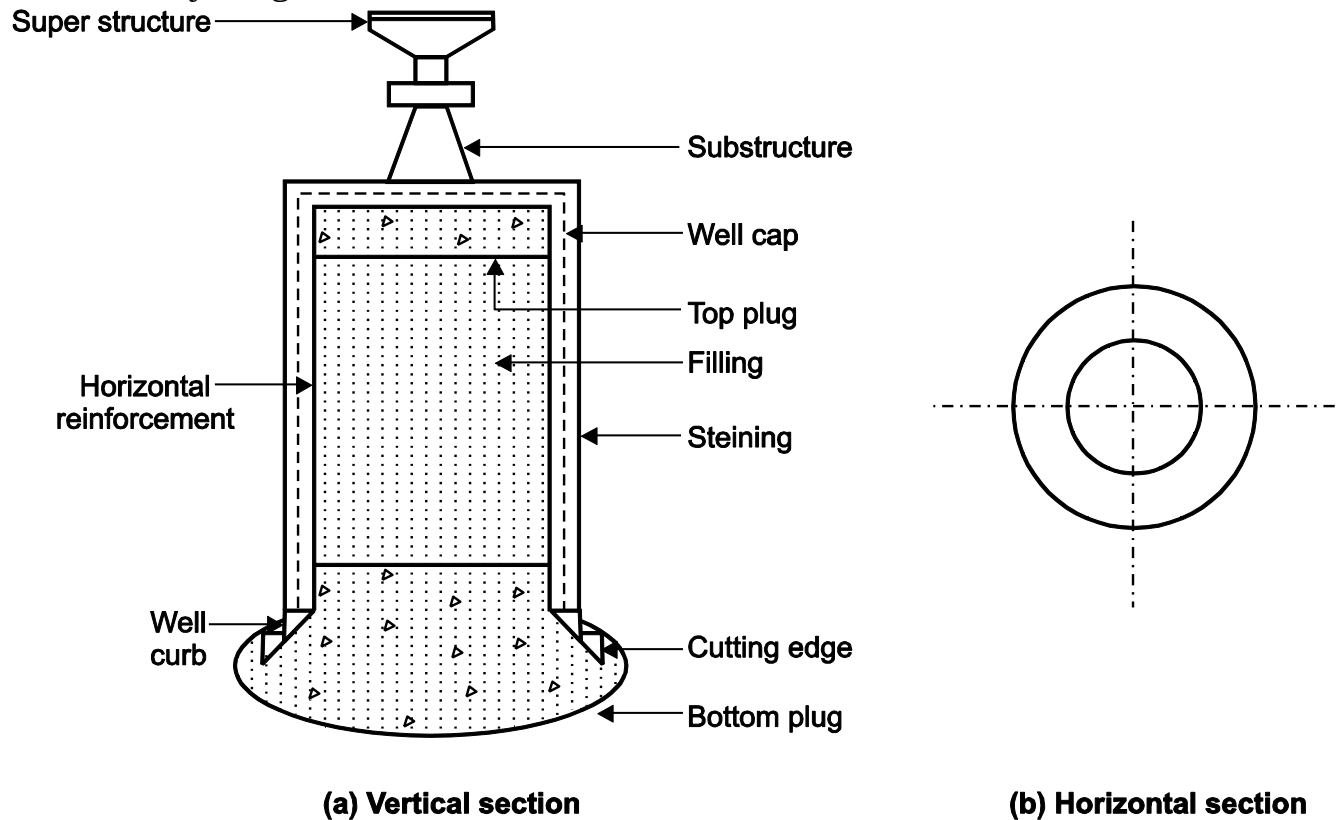


Fig: Open caisson

Advantages of Open Caissons are:

- It is feasible to extend to large depths.
- The cost of construction is relatively low.

Disadvantages:

- The bottom of the Caissons cannot be inspected and thoroughly cleaned.
- The concrete seal is necessarily placed in water, and may not be satisfactory.
- The help of divers may be necessary for excavation near the haunches at the cutting edge.
- If obstructions of boulders or logs are encountered, the work is slowed down significantly.

The elements (component parts) of a well foundation:

- 1) Cutting edge
- 2) Curb
- 3) Concrete seal or Bottom Plug
- 4) Steining
- 5) Top Plug, and
- 6) Well Cap

(i)Cutting Edge: The function of the cutting edge is to facilitate easy penetration or sinking into the soil to the desired depth. As it has to cut through the soil, it should be as sharp as possible, and strong enough to resist the high stresses to which it is subjected during the sinking process. Hence it usually consists of an angle iron with or without an additional plate of structural steel

(ii)Steining: The steining forms the bulk of the well foundation and may be constructed with brick or stone masonry, or with plain or reinforced concrete occasionally. The thickness of the steining is made uniform throughout its depth. It is considered desirable to provide vertical reinforcements to take care of the tensile stresses which might occur when the well is suspended from top during any stage of sinking.

(iii)Curb: The well curb is a transition member between the sharp cutting edge and the thick seining. It is thus tapering in shape. It is usually made of reinforced concrete as it is subjected to severe stresses during the sinking process.

(iv)Concrete Seal or Bottom Plug: After the well foundation is sunk to the desired depth so as to rest on a firm stratum, a thick layer of concrete is provided at the bottom inside the well, generally under water. This layer is called the concrete seal or bottom plug, which serves as the base for the well foundation. This is primarily meant to distribute the loads on to a large area of the foundation, and hence may be omitted when the well is made to rest on hard rock.

(v)Top Plug: After the well foundation is sunk to the desired depth, the inside of the well is filled with sand either partly or fully, and a top layer of concrete is placed. This is known as ‘top plug’.

(vi)Well Cap: The well cap serves as a bearing pad to the superstructure, which may be a pier or an abutment. It distributes the superstructure load onto the well steining uniformly.

Construction of open caisson:

- Construction of well curb.
- Construction of well steining.
- Sinking process.

Construction of Well Curb: In case of a dry river bed the well curb is to be built is placed at the correct position after excavating the bed for about 150 mm for seating. Sand island is constructed if water is present up to the depth of 5 m.

Wooden sleepers can be placed below cutting edge to distribute load evenly. The shuttering of well curb is erected, then reinforcement for the curb is placed in position. Concreting of curb is done without gap. Curb is allowed to cure for 7 days and after 7 days shuttering and sleepers are removed.

Construction of Well Steining: The well steining is constructed with the height of 1.5 m at a time. Sinking process is continued after concrete is set for 24 hrs. When well reaches 6 m depth below ground, then steining can be raised 3 m at a time.

Sinking Process: It is an important step in the construction of well foundation, sinking process is started when curb is cast and first stage of steining is ready after curing.

- The area at which the well is to be sunk shall be excavated to the approximate top elevation of the completed well foundation, In case spring level is higher than the top elevation of the completed well, the excavation shall be maintained in a dry condition by utilizing cofferdams and un-watering methods.
- The well curb shall be constructed in place in the proper position, or if a pre-cast curb is used it shall be set in proper level position at the surface below which well sinking is to be carried. If a precast curb is used it shall be set in proper level position. Precast curb shall not be handled until they have been cured for at least 14 days. The vertical bars for masonry reinforcement (wherever reinforced brick work has been specified or specially directed by the Engineer-in-Charge) shall be attached to the steel angle cutting edge by means of nuts and washers
- After placing of precast curbs; or not earlier than three days after placing of Concrete for cast-in-place curbs, brick Masonry walls with an approximate height of 1.5 meters shall be constructed on the curbs
- The well shall be sunk by dredging (defined as removal of material from inside the well) until the top of the masonry is approximately 0.7 meters above the ground, whereupon straight edges for another lift of masonry shall be set and a second lift of masonry up to 3 meters in height constructed. Care shall be taken that the exterior faces of succeeding masonry lifts are constructed parallel to the axis of the well rather than plumb so as to indicate any uneven sinking of the well and permit remedial action to be taken. Each successive lift of masonry shall be allowed to set for three days before sinking of the well is resumed.

In case spring level is high, then operation shall be maintained in a dry condition using coffer dams and dewatering methods.

- Dredging shall be accomplished in such a manner that the hole within the well shall not be extended below the cutting edge by more than 1.25 meter and that, when the well is sunk to its final position, the material outside of the well will not have been disturbed.
- Each well shall be frequently checked for plumb by means of plumb lines and mason's level or other approved means. Corrective action, consisting of dredging from the high side until the well rights itself, shall be taken immediately if the well is found to be sinking unevenly. If required, weights shall be added at the top of the well masonry on the high side. The corrective force shall be applied concurrently with sinking of the well.

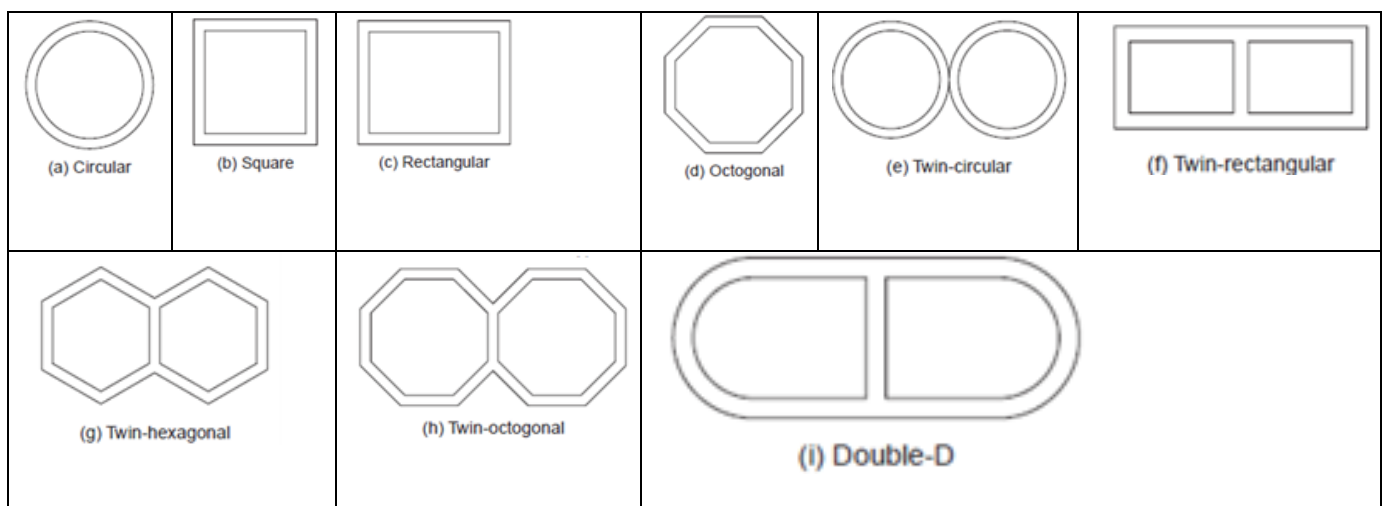
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- Each well shall also be frequently checked for longitudinal and lateral drift during sinking by the use of a suitable sighting device
- If the well does not sink as the dredging is advanced, a greater height of masonry, weighting or running shall be employed. Running, defined as the practice of removing water from within the well to reduce buoyancy and thereby increase the effective weight of the well
- When a well has been sunk to its indicated elevation, the bottom interior of the well shall be carefully sounded to detect the presence of any material within the space to be occupied by the bottom plug, and any material so detected shall be removed. The false masonry if constructed to increase the weight for sinking shall also be removed.
- Without un-watering, the bottom plug concrete shall be placed by means of a tremie

Sand filling: After the bottom plug concrete has set at least one day, and without unwatering the well, the sand filling shall be placed in lifts not exceeding one meter in depth and with a 24-hour elapse of time between placement of the lifts to permit settlement of the fill.

After the sand filling is placed to the demarcated elevation within the well, the top plug concrete shall be placed and screeding level at the elevation of the top of the brick masonry. If the well is tilted within the permissible limits, the masonry at the top of the well shall be constructed, so that the top surface of the masonry around the perimeter of the well meets the specified elevation.

Shapes of well:



Circular well: It is the simplest to construct, easy to sink and has uniform strength in all directions. Simple to correct the tilt for circle. Only disadvantage for this shape is the limitation of its size which restrict its use to bridge with smaller pier.

Twin circular well: Two independent wells placed close to each other and provided with a common well cap over which the pier can be built. Spacing between wells and its diameter can be decided based on the width and length of pier. Both the wells have to be sunk together to avoid possibility of tilt and shift

Double D – Well: This is the most common type of foundation in major bridges with multiple lane. This shape facilitates easy casting and sinking

Grip length: A well foundation should be sunk below the maximum scour depth such that there is adequate lateral stability. The depth of the bottom of the well below the maximum scour level is known as the ‘Grip Length’.

The scour depth can be ascertained by one of the following approaches:

(a) Actual sounding at or near the proposed site immediately after a flood, at any rate before there is any time for silting up appreciably.

(b) Theoretical methods taking into account the characteristics of flow like the direction, depth, and velocity, and those of the river bed material.

In case the first approach of taking soundings is not feasible, the second approach may be used and the normal depth of scour may be calculated by Lacey’s formula

$$d = 0.473 \left\{ \frac{Q}{f} \right\}^{1/3}$$

Where d = normal scour depth, measured below high flood level (m),

Q = design discharge (m^3/s), and

f = Lacey’s silt factor

Lacey’s silt factor is given by $f = 1.76\sqrt{d_m}$ where d_m – mean diameter of the particles (mm).

The grip length for wells of railway bridges is taken as 50% of maximum scour depth, generally, while for road bridges 30% of maximum scour depth is considered adequate. According to IS: 3955-1967, the depth should not be less than 1.33 times the maximum scour depth. The depth of the base of the well below the scour level is kept not less than 2 m for piers and abutments with arches, and 1.2 m for piers and abutments supporting other types of structures.

Note: Maximum scouring depth can be taken as twice of normal scour depth.

Comparison of pneumatic caisson and open caisson

Pneumatic caisson	Open Caisson
Labour cost is more	Labour cost is low
Excavation can be possible in any type of soil	Excavation in hard strata is difficult
Foundation can extend to max depth of 40m below water surface	No limit for depth of foundation
Special compressed air is needed in working chamber for dredging	No compressed air is required
High risk to workers	No risk for worker
Bottom of this caisson is cleaned in dry condition	Cleaning is done under water
Special precautions are required to avoid disease	No precautions are required

Obstacles can easily be observed and tackled during excavation	It is not possible to see the obstacles
Direct access to bottom of caisson for manual excavation	No direct access to bottom of caisson

TILTS AND SHIFTS OF WELL:

Well normally should be sunk straight and at the correct position. However, it is not an easy task some displacement is bound to be there (translational and or rotational).

Shift of well represent overall lateral displacement of wells axis from its true position in plan; and tilt represent rotation of a well about its bottom. These displacement results in changing of span length of the bridge and eccentric loading on pier.

Causes of tilts:

- Non-uniform soil condition
- Presence of large boulder or obstruction
- quick condition
- Sudden sinking (blasting)
- Method of sinking

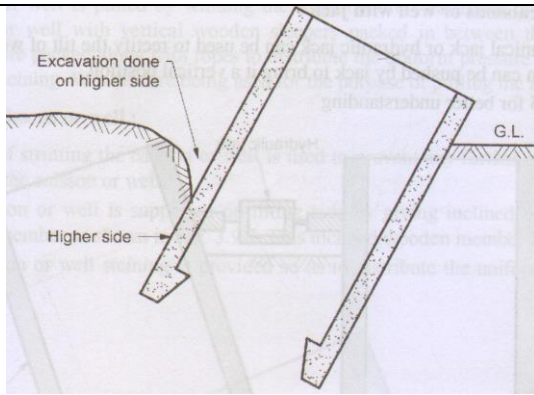
Precautions to avoid tilt:

- Outer surface of well curb and steining should be regular and smooth
- Diameter of curb should be kept about 4 – 8 cm larger than outer diameter of steining and well should be placed symmetrically
- Cutting edge of curb should be of uniform thickness and sharpness
- Dredging should be done uniformly on all sides in a circular well and in both pockets in case of twin wells

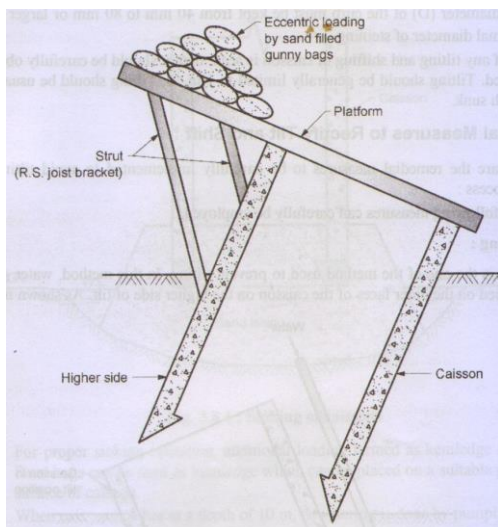
Rectification of tilt:

I.S. 3955-1967 recommends that tilt should be generally limited to 1 in 60 and shift should be restricted to 1% of depth of sunk. In case if these limits are exceeded then following measures need to be taken to rectify the tilt.

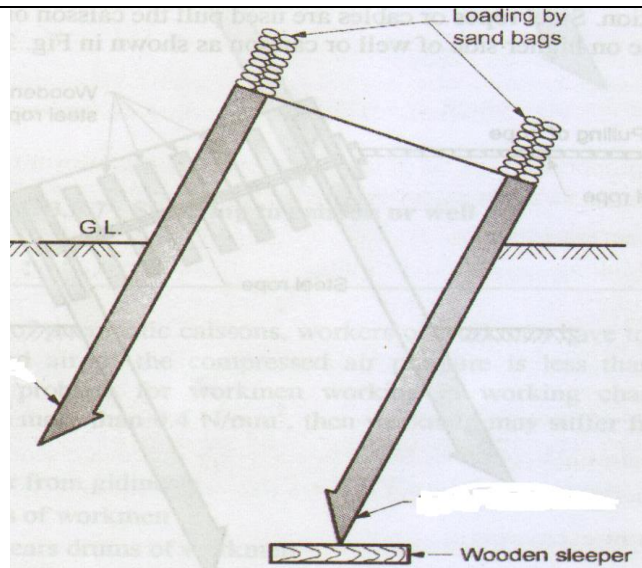
Regulation by excavation: Sinking of caisson on higher side due to excess excavation is more. This is all right in the early stages, otherwise dewatering of caisson or well is needed and open excavation may be done on higher side	Providing temporary obstacles below the cutting edge: Rectification of tilt can be done by inserting the wooden sleeper temporarily as an obstacle below the cutting edge on the lower side so as to prevent further tilt of the well or caisson.
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Eccentric loading : The caisson is normally given the additional loading called kentledge in order to have necessary sinking effort. In this method, eccentric loading or kentledge is applied in higher side so as to have greater sinking effort. For proper application of eccentric loading a platform with projection on higher side can be placed over the top of caisson. The eccentric load is kept on projected part of platform. Thus tilt can be rectified.

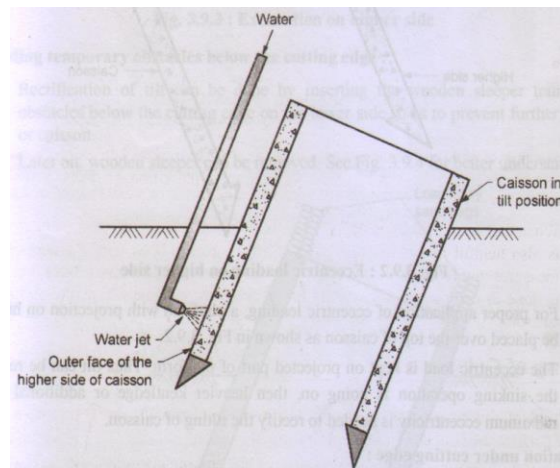


Pushing the caissons or well with



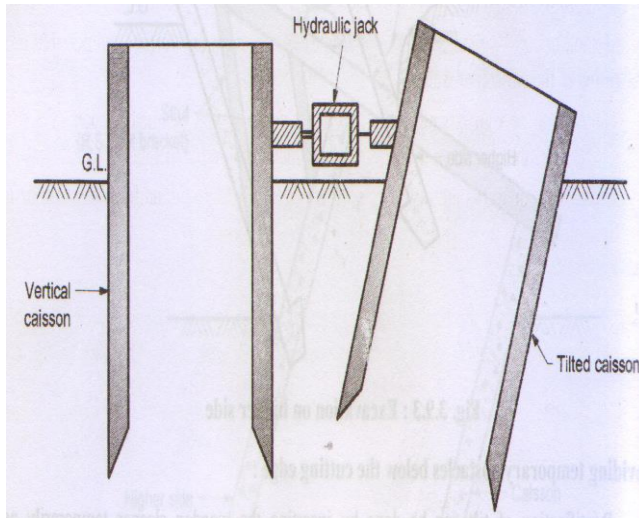
water jetting : This is the one of the method used to prevent tilting. In this method, water jet is forcedly applied on tilt.

• Application of water jetting on higher side reduces skin friction. Thus the tilting is rectified. This method is not more effective but gives the better result if used with the combination of other methods

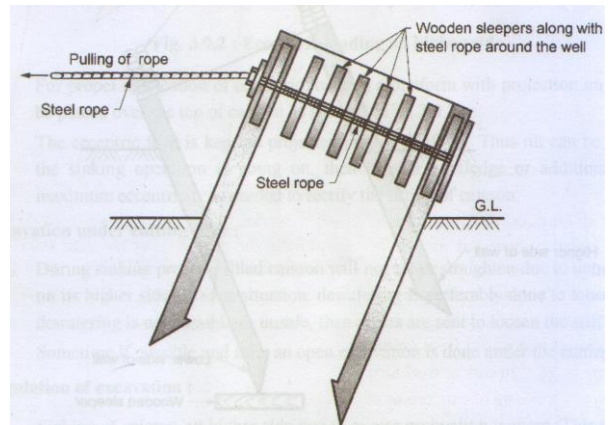


Pulling the well or caisson: This

jack: Mechanical jack or hydraulic jack can be used to rectify the tilt of well or caisson. Well or caisson can be pushed by jack to bring it a vertical position

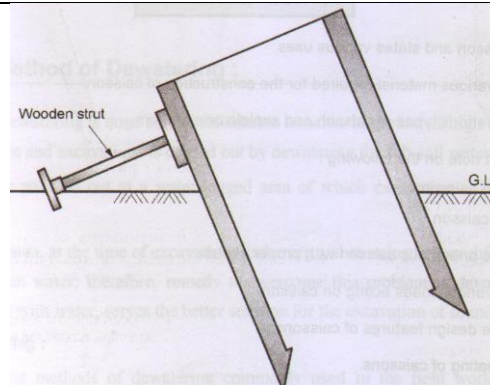


method is most suitable and effective in preliminary or early stages of sinking operation. Steel ropes or cables are used pull the caisson or well. Pulling of caisson or well is done on higher side of well or caisson



Strutting the caissons or well: •Method of strutting the caisson or well is used to prevent any further and possible rise in tilting of the caisson or well.

•The caisson or well is supported on the tilting side by giving inclined support of a strong wooden member. This inclined wooden member is called as a strut.



Advantage and disadvantage of caisson foundation compared to pile foundation

Advantage	Disadvantages
1) It can be carried through layers containing large boulders through which pile cannot penetrate	1) For structure on land of medium size caisson may be more expensive than pile
2) Does not produce vibration, heaving thus less damage to adjacent structures	2) Excavation of caisson in granular soil below water table is difficult
3) Cost of machinery as well as noise level are low compared to that in case of driven piles	3) Caving in of the soft soil may cause problems
4) Foundation layer can be inspected and tested physically and hence reliability of design is good	4) Seating of caisson on an irregular surfaces of bed rock is difficult
5) It can sustain larger lateral force and offer more effective resistance to destructive	

forces due to floating objects and scour	
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Forces acting on the well foundation: Well foundation will be subjected to following forces

1) Dead load	7)Earth pressure
2)Live load	8)Centrifugal force
3)Impact load	9)Buoyant force
4)Wind load	10)Seismic force
5)Water pressure	11)Temperature stress
6)Longitudinal force (tractive force/ braking force)	

Sheet piles: Classification, design of cantilever sheet pile in cohesionless and cohesive soils. Design of anchored sheet pile by free earth support method, cellular cofferdam: Types, cell fill stability considerations.

Sheet pile:

Sheet pile wall is a common type of earth retaining structure made of individual sheet piles driven in the ground. It consists of number of sheet piles driven side by side into the ground, thus by forming a continuous vertical wall. These walls are commonly used for:

1. Waterfront construction.
2. Temporary construction such as cofferdam.

These walls are not suitable in following situations:

1. When height of wall is more due to high flexural stresses.
2. If sub-soil is rocky strata, it prevents penetration of pile (pile derives its stability from depth of embedment.)

Sheet piles are prefabricated members made of timbers, R.C.C. or steel.

Timber sheet pilings are used for light lateral loads (low height), commonly used for braced excavation.

R.C.C. sheet pilings are precast concrete members, these members should be designed for handling and driving stress due to their heavy weights and also displace large volume of soil, hence are used rarely.

Steel sheet piles are most common type of piles that are being used and consists of structural members with interlocking either 'Finger and thumb' type or 'Ball and

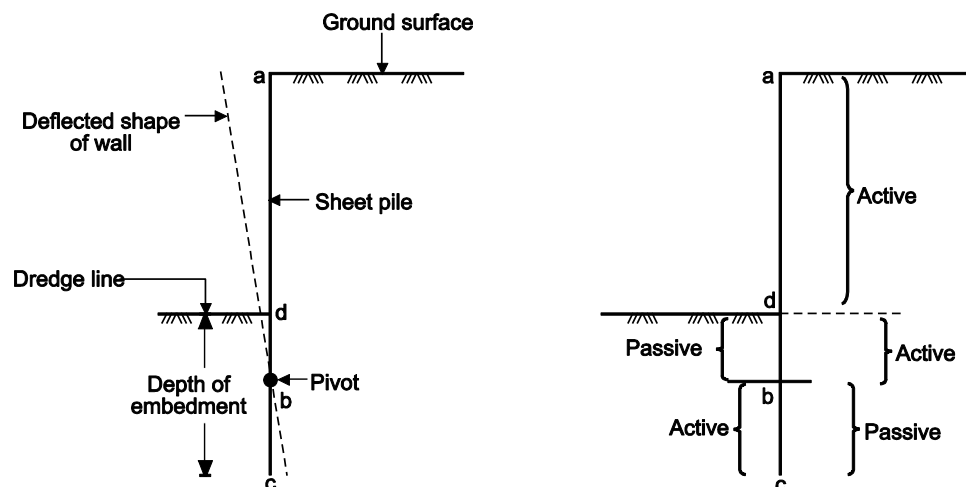
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socket type' to engage with one another. These sheet piles have following advantages over other.

- It is relatively light weight.
- It may be reused several times.
- It has high resistance to driving stress (in hard strata).
- More service life.
- Easy to alter the length of pile (welding/bolting).

Classification of sheet pile: Based on the way it resists the load sheetpile walls are classified into two types a) Cantilever sheet pile and anchored sheet pile walls. In case of cantilever sheet pile wall lateral load coming on it is resisted by cantilever action. In case of anchored sheet pile wall the load will be resisted by cantilever action along with anchorage to minimize lateral movement.

Cantilever Sheet Pile



Cantilever sheet pile wall derives its stability entirely from lateral resistance of soil into which it is driven. The sheet pile wall is adequately embedded into the soil below the dredge line.

These walls resist the load by cantilever action hence are suitable only for moderate heights because bending moment increases with the cube of cantilever height of wall.

Fig shows cantilever sheet pile and its deflected shape under the action of earth pressure. Portion 'ab' of wall is subjected to only active earth pressure. Portion 'db' is subjected to active pressure from one side and passive pressure from other side and portion 'dc' is subjected to passive pressure from one side and active from other side.

Determination of depth of embedment of sheet pile for resisting the lateral pressure is called design of sheet pile wall for which following cases are considered.

- a. Cantilever sheet pile wall in sandy soil above and below dredgeline.
- b. Cantilever sheet pile wall in clayey soil above and below dredgeline.
- c. Cantilever sheet pile wall with sandy soil above dredgeline and cohesive soil below dredgeline.

Case 1: Sandy Soil Above and Below Dredgeline :

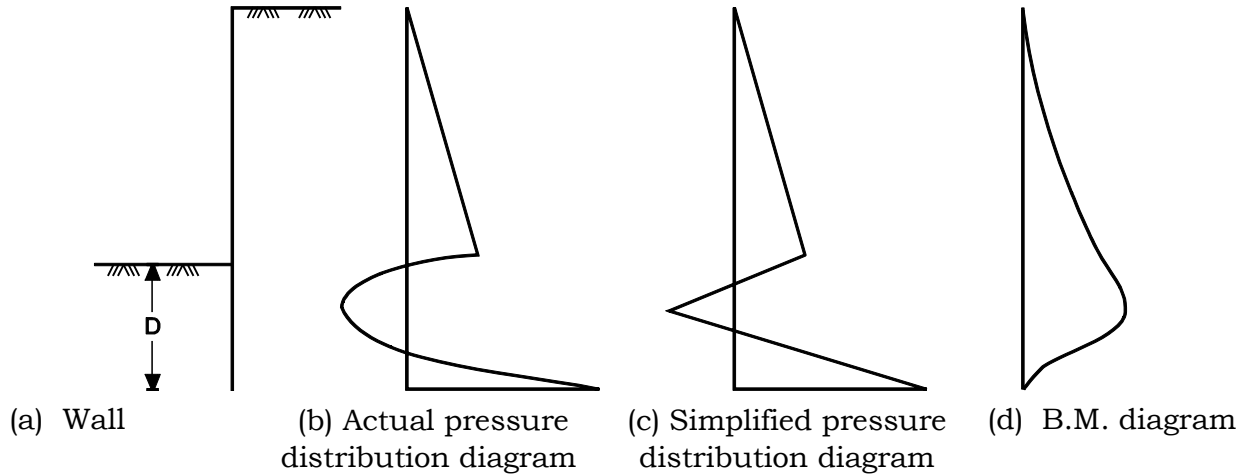


Fig. 5.16

Fig. 5.16 shows the sheet pile wall, active pressure distribution diagram acting on the wall, simplified pressure distribution diagram (which is used for design purpose) and bending moment diagram.

Depth of Embedment by Approximate Analysis : In approximate analysis the active earth pressure distribution on the back and passive earth pressure distribution on the front and concentrated passive pressure at the base of wall is considered for simplicity in analysis.

Taking moment about base

$$P_A \left(\frac{H + D}{3} \right) = P_p \cdot \frac{D}{3}$$

$$\frac{1}{2} K_a \cdot r (H + D)^2 \cdot \left(\frac{H + D}{3} \right) = \frac{1}{2} K_p \cdot r \cdot D^2 \cdot \frac{D}{3}$$

Above equation is solved for 'D' and then obtained value of 'D' is increased by 20-40%.

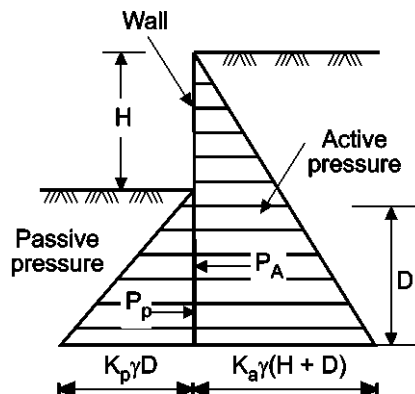


Fig: Pressure distribution diagram for approximate analysis

Case 2: Cantilever Sheet Piles in Cohesive Soil:

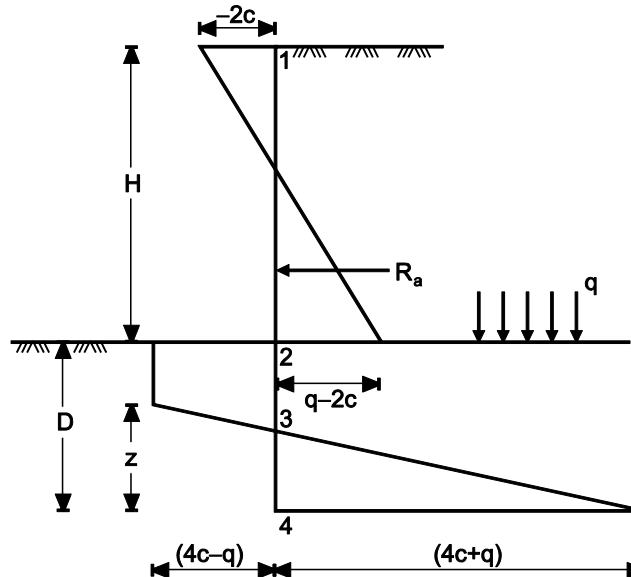


Fig. Pressure distribution diagram

Fig. shows the pressure distribution diagram for sheet pile wall embedded in cohesive soil and backfilled with cohesive soil.

Active and passive pressure in general are given by,

$$P_a = \sigma_v K_a - 2c\sqrt{K_a} \quad \text{and} \quad P_p = \sigma_v K_p + 2c\sqrt{K_p}$$

For purely cohesive soil $\phi = 0$, thus earth pressure coefficient = 1 (active and passive)

Thus $P_a = \sigma_v - 2c$ and $P_p = \sigma_v + 2c$

At point 1, vertical stress is $\sigma_v = 0$ thus

$$p_1 = 0 - 2c = -2c$$

At point 2 to the right (just above 2), $\sigma_v = rH = q$; thus $p_{2r} = q - 2c$

At point 2 to the left (just below 2), $\sigma_v = 0$; thus $p_{2l} = 0 + 2c$

$$\begin{aligned} \text{Net pressure at point 2} &= \text{passive pressure}(p_{2l}) - \text{active pressure}(p_{2r}) \\ &= 2c - (q - 2c) = 4c - q \end{aligned}$$

$$\begin{aligned} \text{At point "4", Net pressure} &= \text{Passive pressure} - \text{Active pressure} \\ &= (q + rD) + 2c - (rD - 2c) = 4c + q \end{aligned}$$

To determine the depth of embedment, we apply equilibrium equation.

We neglect the -ve pressure for calculation.

$$\sum H = 0$$

$$R_a + [(4c + q) + (4c - q)] \frac{\bar{z}}{2} - (4c - q) D = 0$$

$$\sum M = 0 \text{ about base}$$

$$R_a (D + \bar{y}) - \frac{D^2}{2} (4c - q) + [4c - q + 4c + q] \frac{\bar{z}}{3} \cdot \frac{\bar{z}}{2} = 0$$

Solving above equations), we get the embedment depth. Then increase the depth by 20-40%.

Case 3: Granular Soil Above and Cohesive Soil Below:

At point 1, $\sigma_v = 0$, $p_1 = \sigma_v K_a - 2c \sqrt{K_a}$

$$p_1 = \sigma_v - 2c = 0$$

At point 2 (just above 2) $\sigma_v = r \cdot H = q$

$$p_2 = rHK_a$$

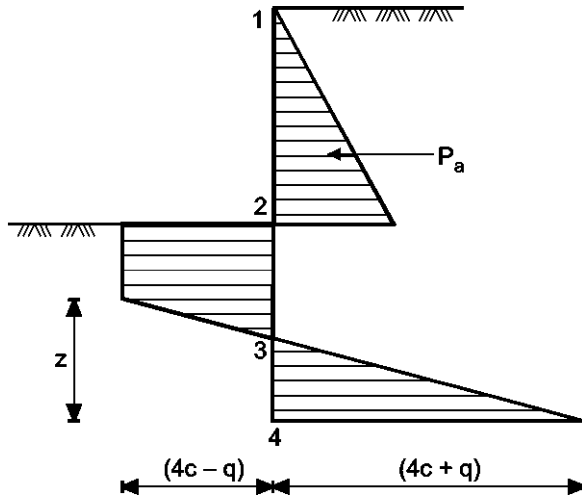


Fig : Pressure distribution diagram

At point 2,

Just below 2 to right, $p_{2r} = q - 2c$

Just below 2 to left, $p_{2l} = q + 2c = 2c$.

Net pressure at 2 to left of 2 = $p_{2l} - p_{2r} = 4c - q$

Net pressure at 4 = $p_{2r} - p_{2l} = 4c + q$

To determine the depth of embedment, we use the equilibrium equation.

$$\sum H = 0;$$

$$R_a + [(4c + q) + (4c - q)] \frac{\bar{z}}{2} = - (4c - q) D = 0$$

$$\sum M = 0; \text{ about base}$$

$$R_a (D + \bar{y}) - \frac{D^2}{2} (4c - q) + [(4c - q) + (4c + q)] \frac{\bar{z}}{2} \cdot \frac{\bar{z}}{3} = 0$$

Solving above equations value of D can be computed and the computed value of 'D' is increased by 20% - 40%.

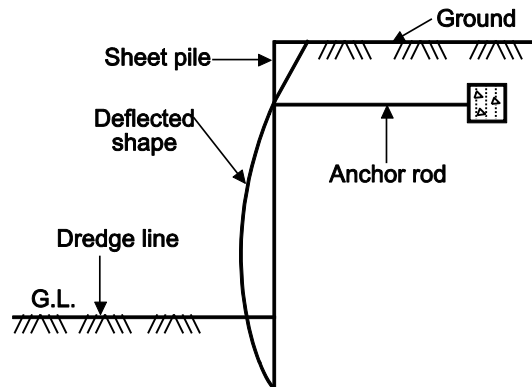
Anchored Sheet Piling:

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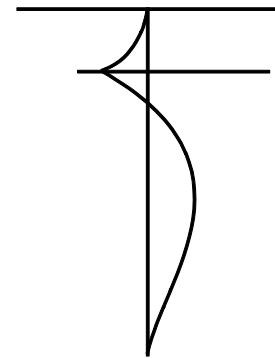
When the height of wall is more, cantilever sheet piles need penetration to considerable depth for stability. If sheet piles are anchored near their top, significant reduction in penetration depth can be made as it reduces the deflection near top.

If anchored sheet piles are driven to a limited depth the deflection of beam is somewhat similar to that of vertical beam. Simply supported such sheet piles are called anchored piles with free earth supports.

If anchored sheet piles are driven to greater depth the lower end of beam will behave just like partially fixed one. Such walls are called sheet pile wall with fixed earth support.

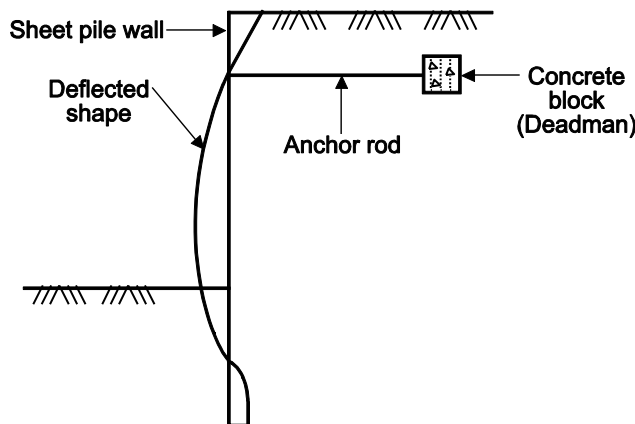


(a) Anchored sheet pile wall

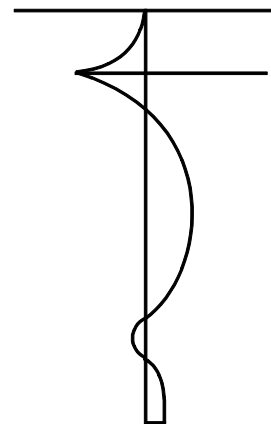


(b) Bending moment diagram

Free earth support method



(a) Anchored sheet pile wall



(b) Bending moment diagram

Fixed earth support method

Design of anchored sheet pile wall (bulk head) consists of determining :

- (1) Depth of embedment.
- (2) Magnitude of tensile force.

1. Free Earth Support Method:

Method is based on following assumptions:

1. Sheet pile is perfectly rigid as compared to surrounding soil.
2. Sheet pile is free to rotate at anchor rod level with failure occurring due to rotation about anchor rod.
3. Active and passive earth pressure acting on sheet pile wall.

(a) Granular Soil:

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Consider anchored sheet pile as shown in Fig. with granular soil above and below dredge line.

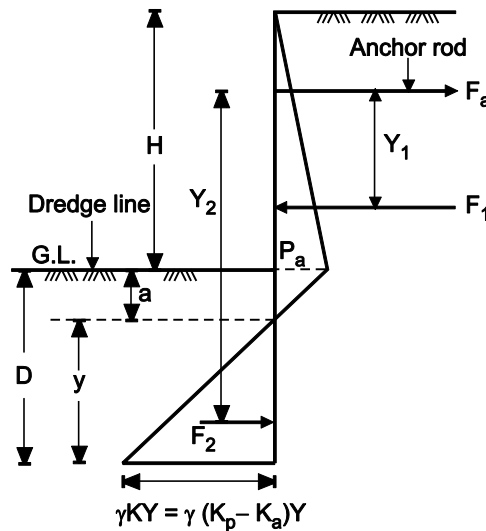


Fig.: Pressure distribution diagram on anchored sheet pile wall

Let,

r – Unit weight of soil

Φ – Angle of internal friction

K_a, K_p – Coefficient of earth pressure

P_a – Intensity of active earth pressure at dredge line ($= rHK_a$)

F_a – Force in anchor rod

F_1 – Resultant of active earth pressure

F_2 – Resultant of passive earth pressure

a – Depth of point of zero pressure below dredgeline

At depth 'a' below dredgeline :

$$\text{Active pressure} = P_a + r \cdot a \cdot K_a$$

$$\text{Passive pressure} = r \cdot a \cdot K_p$$

$$r \cdot a \cdot K_p = P_a + r \cdot a \cdot K_a \Rightarrow a = \frac{P_a}{(K_p - K_a)}$$

$$\sum M = 0 \text{ about anchor rod} \quad \text{fi} \quad F_1 Y_1 = F_2 Y_2$$

F_1 – Area of upper triangle

F_2 – Area of lower triangle

Y_1 – Distance of centre of gravity of upper triangle from anchor rod

Y_2 – Distance of centre of gravity of lower triangle from anchor rod

After substituting value of F_1, F_2, Y_1 and Y_2 in equation, we get an equation only in terms of 'y' which is solved for 'y'.

Depth of embedment, $D = y + a$.

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Value of 'D' is increased by 20-40% for safety purpose.

$$\text{Force in anchor rod} = F_a = F_1 - F_2$$

(b) Granular Soil Above Dredgeline and Cohesive Soil Below Dredgeline :

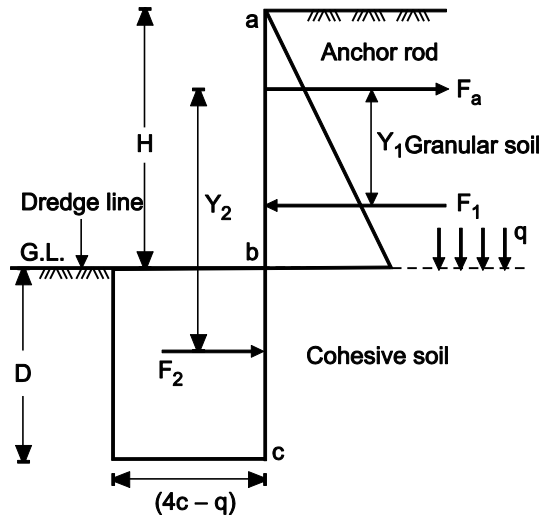


Fig : Distribution of earth pressure

Consider anchored sheet pile wall as shown in Fig.

Earth pressure at a = 0

$$\text{Earth pressure at b} = K_a \cdot r \cdot H = K_a \cdot q$$

$$\text{Active earth pressure just below 'b'} = q - 2c$$

$$\text{Passive earth pressure just below 'b'} = 2c$$

$$\text{Net pressure at dredgeline} = 2c - (q - 2c) = 4c - q$$

$$\text{At any depth 'z' below dredgeline : Active earth pressure} = (q + r \cdot z) - 2c$$

$$\text{Passive earth pressure} = rz + 2c$$

$$\begin{aligned} \text{Net pressure at depth z} &= \text{Passive pressure} - \text{Active pressure} \\ &= 4c - q \end{aligned}$$

$$\text{Applying } \sum M = 0 \text{ about anchor rod, } F_1 Y_1 = F_2 Y_2$$

which is solved for 'D', calculated value of 'D' is increased by 20-40%.

$$\text{Force in anchor rod is determined by applying } \sum H = 0 ; F_a = F_2 - F_1$$

COFFERDAMS

Introduction: Temporary structure generally constructed to prevent the water from entering in an area to facilitate construction projects in areas which are normally submerged such as bridges and piers. It is also constructed to remove soil from an area.

Cofferdam is usually constructed large enough to provide adequate working space, (space for the proposed structure and working area for the workers around it). Once the area is enclosed by cofferdam the area is dewatered by pumping. Although cofferdam is constructed watertight yet during its life certain water will leak through foundation of cofferdam thus needs some pumping

Ideal Requirements of a Cofferdam

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- a) It should be easy in construction at site of work.
- b) It should be stable against overturning.
- c) It should be watertight.
- d) It should occupy minimum possible area so as to provide sufficient space for containing permanent structures.
- e) It should provide least obstruction to the flow of water.
- f) It can be dismantled and reused easily.
- g) Its height should be more than HFL.
- h) Construction and maintenance cost should be low.

Uses of Cofferdam

Cofferdams are used in a variety of water environments, from small rivers to large harbors. Some of the more common uses for cofferdams include:

Hydroelectric Dam Construction: Cofferdams are used to divert away from the shoreline of a river to allow for the foundations of a dam to be constructed. In this application, generally half of river width is enclosed by the cofferdam at a time to maintain overall flow.

Bridge Construction: Cofferdams are used to divert water away from bridge foundation positions, either on the shore or within the waterway.

Ship Repair: Sometimes cofferdams are used to generate a "Dry dock" conditions for a ship in order for repair to proceed. This generally occurs when the ship cannot be moved to an actual dry dock, and it can also be more cost effective in some cases.

Sunken Vessel Recovery: Cofferdams can be used to expose sunken vessel in shallow waters to allow for recovery and repair if appropriate.

In Naval Architecture: Cofferdam can also refer to the watertight bulkheads within a ship.

Types of Cofferdam

- (1) Dikes:
 - (a) Earth dikes or Earth-fill cofferdam
 - (b) Rock dikes or Rock-fill cofferdam
 - (c) Sand bag dikes or Sandbag cofferdam.
- (2) Rock-fill crib cofferdam.
- (3) Single wall sheet pile cofferdam/single water timber sheet pile cofferdam.
- (4) Double wall cofferdam.

- (5) Braced cofferdam.
- (6) Cellular cofferdam.

Factors Affecting Selection of Type of Cofferdam

- (1) Extent of the area to be covered/enclosed.
- (2) Depth of water.
- (3) Possible wave-height.
- (4) Nature of velocity of flow (slow current/swift current).
- (5) Nature of bed on which cofferdam is to rest (pervious/impervious).
- (6) Scouring (bed erosion).
- (7) Local available material.
- (8) Availability of skilled labour.
- (9) Transport facility.
- (10) Risk of damage by floating debris or ice.

Suitability of Various Types of Cofferdam

(1) Earth-fill Cofferdam are suitable for low heads of water and if given surface protection, they can even be suitable for half tide work, or on sites exposed to flowing water.

(2) Single Wall Sheet-pile Cofferdam are suitable for restricted site area where cross bracings or ground anchors can be used.

(3) Double Wall Cofferdams or Cellular Sheet Piling are used for wide excavation where self-supporting dams are required.

(4) Rock or Earth-Filled Timber Cribs would be suitable for a remote site in undeveloped territory where heavy timber in log form is available and cost of importing and transporting steel sheet piling and its installation is comparatively very high.

(5) The foundation soil condition is an important factor in the choice of cofferdam types. For example, sheet piling is not suitable for ground containing numerous boulders and cellular cofferdam could not be carried in deep deposits of clay.

Brief Description of Different Cofferdams

(1) Earth fill Cofferdam: This is the simplest form of cofferdam and is suitable for sluggish rivers, lakes or other sheltered waters not subjected to high velocity of flow or wave action. These are usually restricted to low or moderate height of water (1.2 m – 1.5 m). It consists of an earth bank which is placed around the site to be enclosed as shown in Fig. The thickness of bank should be sufficient to furnish required stability to cofferdam and to make it fairly watertight. Materials to be used for

cofferdam depends on availability of material and site conditions usually a mixture of sand and clay used.

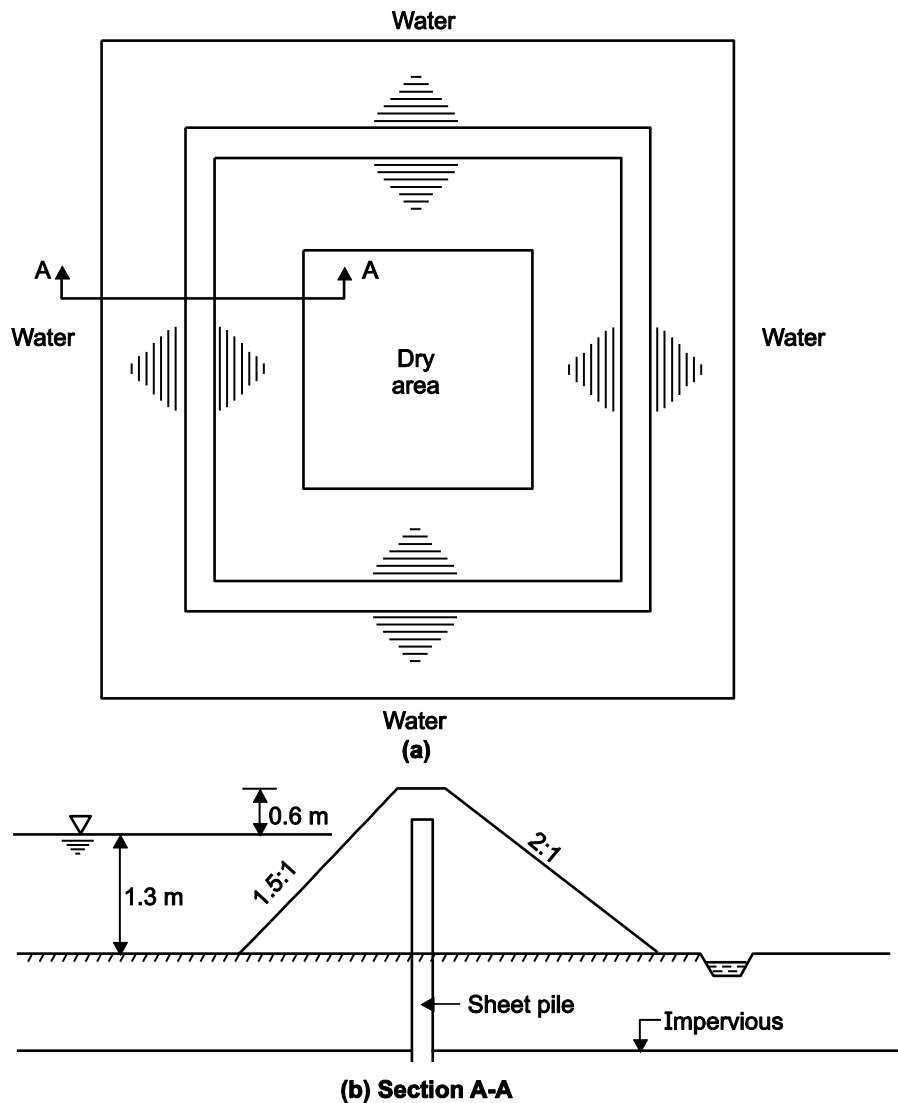


Fig: Earthfill cofferdam

(2) Rockfill Cofferdam: Rockfill cofferdams are similar in construction to earthfill cofferdam but due to inherent stability of material they can be formed with steeper slope than earthfill. They can be used for depth of water about 3 m and are suitable even in case of swift water. These cofferdams have the disadvantages of not being impervious. In case of low heights sufficient water tightness can be achieved by dumping impervious soil on outer face of cofferdam. Seepage will carry this material into the interstices of rock and fair degree of water tightness will gradually be attained. Rockfill cofferdam constructed properly can withstand over topping of water without any serious damage.

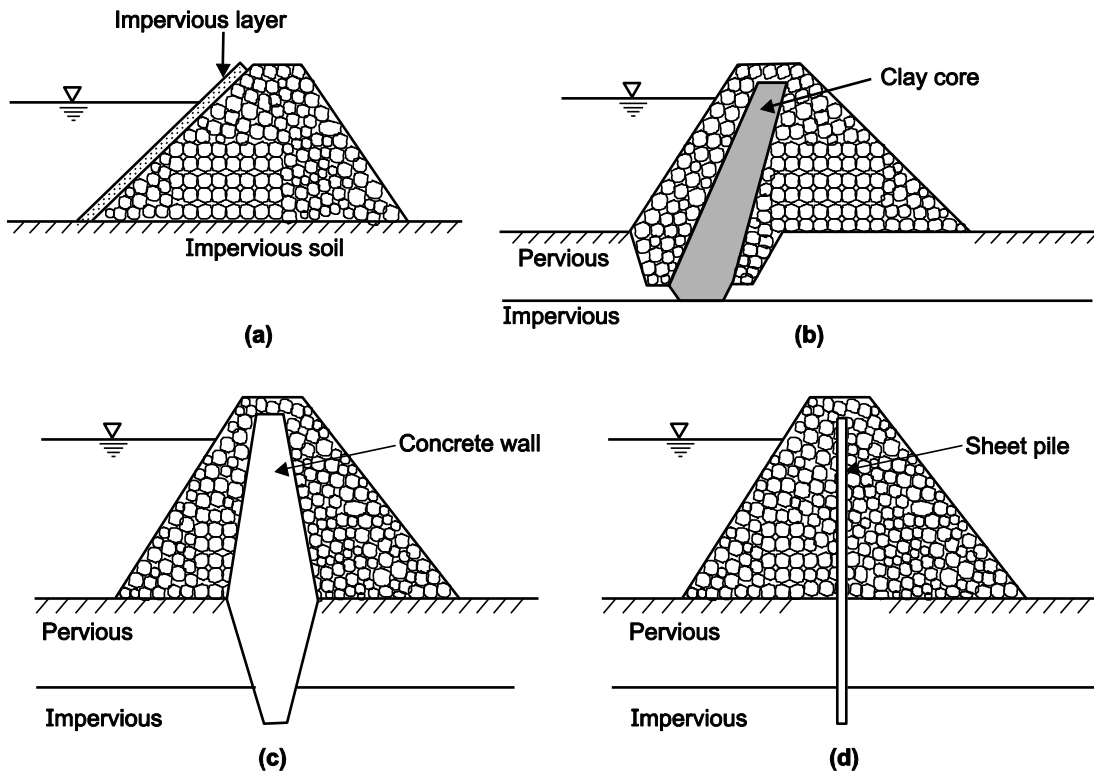


Fig : Rockfill cofferdam

(3) Sand Bag Cofferdam : In this type of cofferdam, mixture of sand and clay is filled in bags and such bags are placed to form a cofferdam.

Following points are noted :

- (a) It is desirable to use empty bags as small quantity of cement present in empty bag will help in achieving watertightness.
- (b) Bags should be partially filled. The bags can pack each other effectively and adjust with irregularities of soil at the bottom.
- (c) Number of bags required should be carefully worked out with due allowance to wastage.

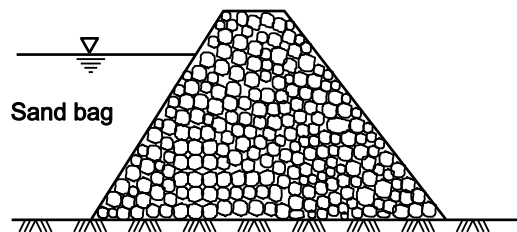


Fig.: Sand-fill cofferdam

(4) Rock-fill Crib Cofferdam: Rock-fill crib cofferdam consists of timber cribs. A crib is a box or a cell open at the bottom and it essentially consists of a framework of horizontal timber laid in alternate courses. The pockets thus formed are then filled with rock or gravels or earth to give stability to crib against overturning and sliding. The timber to be used for construction of cribs may consist of rough logs or old sleepers from railway. This cofferdam is ideally suited for following situations:

- (a) Water current is swift.
- (b) There is danger of overtopping.

- (c) Depth of water is moderate.
- (d) Working space is limited.
- (e) Bed of stream is a hard rock.

(5) Single Sheet-pile Cofferdam: In this type of cofferdam there is a single row of cantilever sheet-piles the piles are sometimes heavily braced. Joints in the sheet-piles are properly sealed.

Single sheet-pile cofferdams are generally used to enclose small foundation sites in water for bridging at shallow depth and is suitable for moderate velocity of flow of water and depth of water up to 4 m.

Proportioning of wall are as below :

- $h \geq 0.25 H$ (for coarse sand and gravel)
- $h \geq 0.5 H$ (for fine sand)
- $h \geq 0.85 H$ (for silt.)

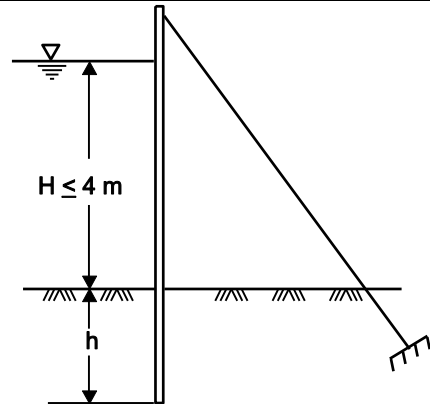


Fig : Single sheetpile cofferdam

(6) Double Sheetpile Cofferdam :

Double sheetpile cofferdam consists of two straight vertical walls of sheeting tied to each other and space between them is filled with granular soil. Double sheetpiling cofferdam higher than 2.5 m should be strutted sometimes, inner berm is provided to increase its stability.

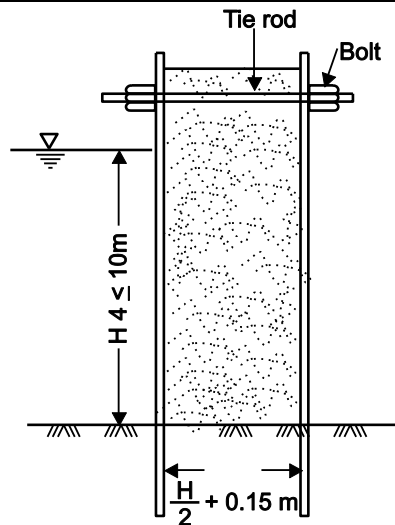


Fig: Double sheet pile cofferdam

(7) Braced Cofferdam: When depth of excavation is greater than 10 m two rows of sheet-pile with interspace filled with murum is also inadequate/uneconomical. In such case braced cofferdam is used. This cofferdam is formed by driving vertical sheeting into the ground. Different sections of vertical sheeting are held in position by horizontal beams called wales. The wales in turn are supported by struts. These cofferdams are suitable up to a depth of 15 m.

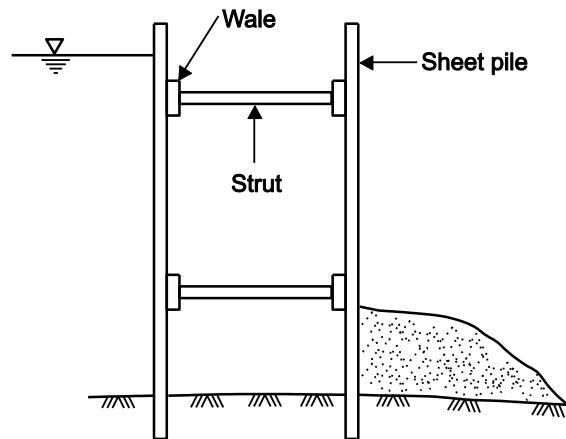


Fig: Braced cofferdam

(8) Cellular Cofferdams: The cofferdams consisting of cells are known as cellular cofferdams. These cofferdams are used as retaining structures. The retaining material is usually water whereas cell is filled with granular soil. These cells then interconnected enclose a working area in which construction operation or excavation operation can be carried out.

Functions of Cell Fill MATERIAL:

- It provides mass for stability of cofferdam.
- It reduces permeability of cofferdam.

Ideal requirement of a fill material.

- It should have large coefficient of permeability so that free drainage is possible.
- It should have large angle of internal friction.
- It should contain very least amount of fine material.
- It should offer resistance against scouring due to turbulence of flow.

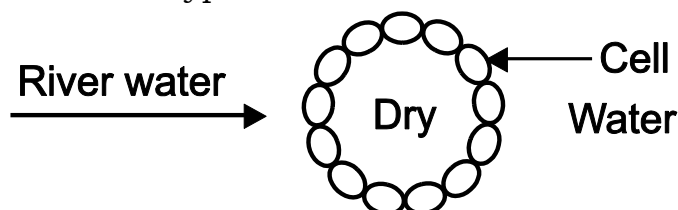
It is practically impossible to get cell fill satisfying all these requirements.

Advantages of Cellular COFFERDAM:

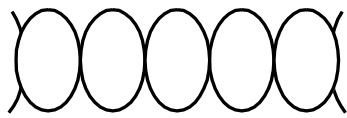
- Cellular cofferdam is more economical when enclosed area and/or water head is large.
- More watertight.
- Cellular cofferdam is smaller than other types of cofferdam for given enclosed area and head thus obstruct less area of flow for navigation.
- Has more salvage value compared to other types.
- Relatively simple construction.

Cellular cofferdam based on its shape and classified into following types:

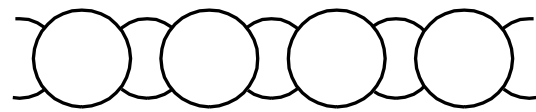
- Circular type cofferdam.
- Diaphragm type cofferdam.
- Cloverleaf type cofferdam.



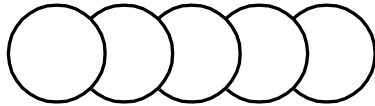
(a) Cellular cofferdam



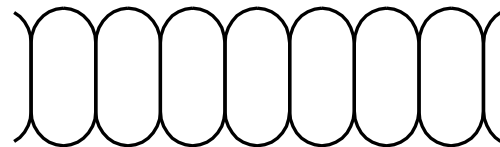
Diaphragm type



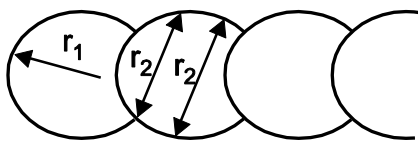
Circular type



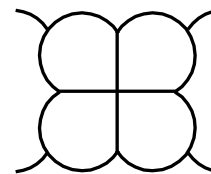
Modified circular type



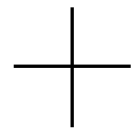
Diaphragm type



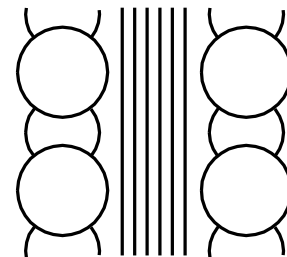
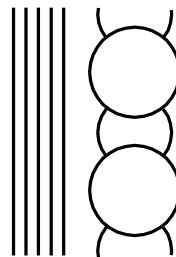
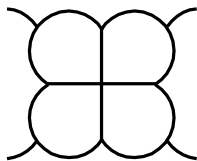
Diaphragm with circular crosswalls



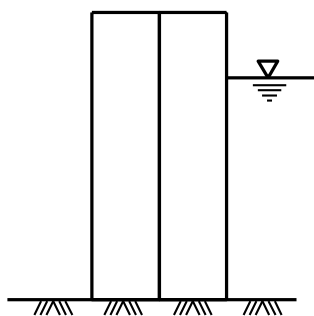
Clover leaf type



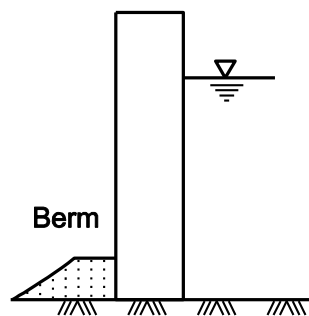
(b) Types of cellular cofferdam



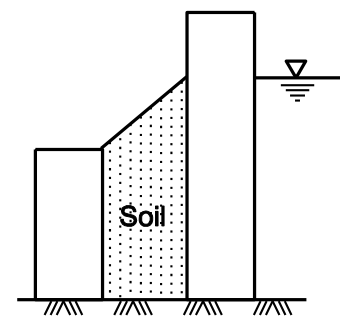
Plan



(i) Clover leaf



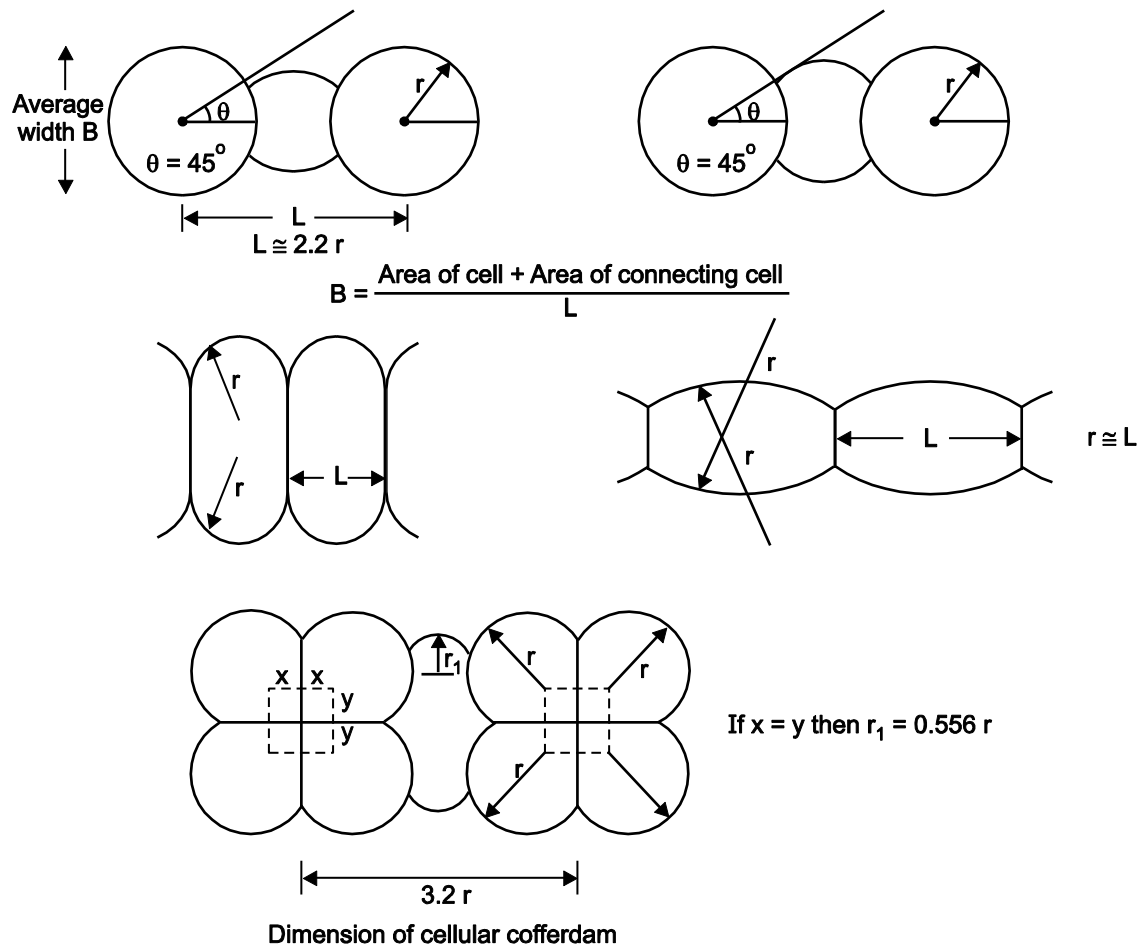
(ii) Circular with berm



(iii) Double circular cell

Elevation

(c) Cellular arrangement to resist high lateral loads



Merits and Demerits of Different Types of Cellular Cofferdams

	Merits	Demerits
Circular	1. Each cell is self supporting and independent of adjacent cells thus failure of any one cell does not affect the other cell.	1. Diameter of circular cofferdam is limited by interlocking tension. Thus, cannot be used for high heads.
	2. Can be used singularly or in a group or at end.	2. Great skill is required in setting and driving the pile.
	3. It can be filled with no regard to other cells.	
	4. Completed cells can be used as working platform for construction of adjacent cells.	

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	5. Requires less number of piles per unit length of cofferdam.	
	6. Only this cofferdam can be constructed without protection in rough base with flowing water ($v < 1.3 \text{ m/s}$)	
Diaphragm	1. Interlock tensions are smaller and uniform than those in circular type.	1. Cells are not independently stable.
	2. It can be widened easily by increasing length of diaphragm (without affecting interlock tension) if required for stability.	2. Fill elevation in adjacent cells must be reasonably uniform to avoid distortion of walls.
	3. They can be constructed cheaper and faster than circular cofferdam if done with floating equipment and progressive filling of cell.	3. Several templates are required during construction as each cell is not stable independently.
Cloverleaf	1. Cells are independently stable.	1. Require more number of sheet piles.
	2. Any size cell can be built to meet stability requirement but radii of arcs will be governed by interlock tension.	2. Fill elevation in adjacent compartments must be reasonably uniform to avoid distortion of walls of cofferdam.
	3. It can be very effectively used as an end cells or corner cells.	
For equal section modulus: (a) Circular cofferdam is more economical for $r \leq 6 \text{ m}$. (b) Diaphragm cofferdam is more economical for $r > 6 \text{ m}$.		
For equal cross-sectional area diaphragm type is more economical than circular type.		

Module 6: Slope stability (6 Lectures)

Different definitions of factor of safety, types of slope failures, stability of an infinite slope of cohesion less soils, stability of infinite slope in cohesive soils, stability of finite slopes – slip circle method, semi graphical and graphical methods, friction circle method, stability number concepts and its use

SLOPE STABILITY

Introduction: Slope is the surface of soil which is inclined at some angle with horizontal usually between $0 - 70^\circ$. (When angle is more than 70° then it is called wall). Slopes are required whenever soil is to be supported at two different levels without retaining wall in between them. Slopes are commonly required for following structures Highway, railway, canal, earth dam, Trench etc. Cost of the earthwork would be minimum if the slopes are made steepest. However steep slopes may not be stable. A compromise has to be made between economy and safety i.e. slopes provided are neither too steep nor too flat (steepest slope which are stable and safe should be provided)

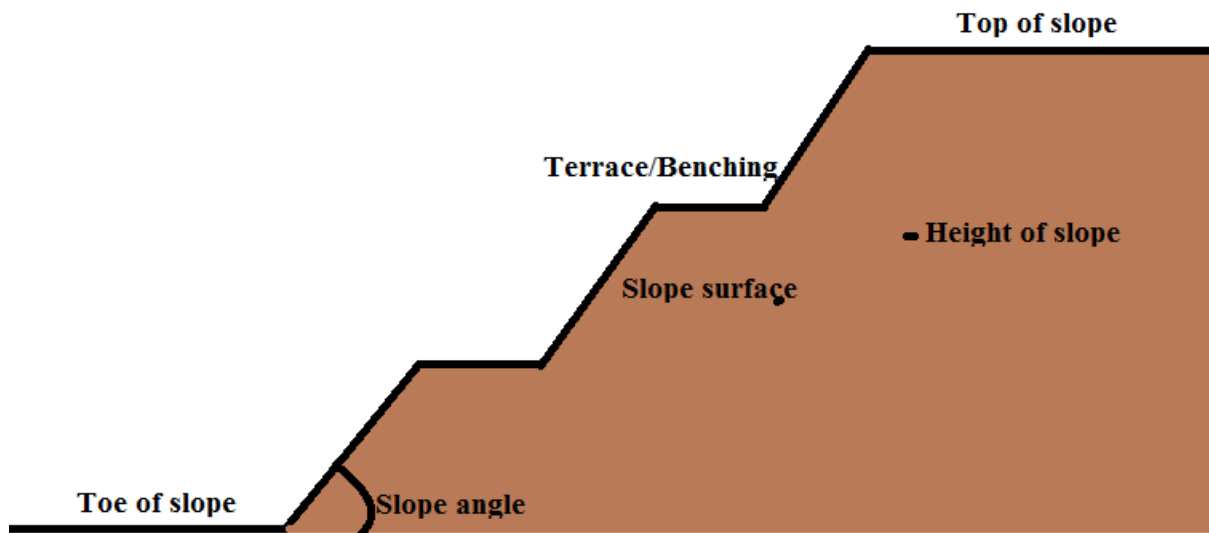


Fig: Slope nomenclature

Slope classification: Slopes are classified based on various criteria's which are as given below

- a) Based on agents of formation
 - i) Natural
 - ii) Artificial or man made
- b) Based on extent of slope
 - i) Infinite
 - ii) Finite
- c) Based on composition of slope
 - i) Homogeneous
 - ii) Heterogeneous or Non homogeneous

Concept of stability of slope: Whenever there is difference of level between two surfaces of soil gravitational force (weight) try to cause movement from the higher to lower level unless the shearing resistance within the material is sufficient to withstand these forces. The forces which tend to cause slippage are known as actuating forces (Gravitational, seepage, earthquake etc.) and the force which oppose slippage (shear strength) are called resisting forces.

The concept of stability analysis of slope is based on the fact that on every plane or curved surface across the slope shear resistance must be more than the shearing stress due to actuating forces; when this condition is not satisfied slope will fail. Slope stability is arguably the most complex and challenging of all the sub-disciplines of geotechnical engineering, and is often the least understood.

Generally, failure occurs due to natural or man-made causes. Natural failures primarily occur because of stresses imposed by weight of the soil mass itself and by changing soil properties. Man-made failures occur when the slope is physically altered. Irrespective of the mechanism causing failure, a slope fails when the imposed stresses exceed the shear strength of the soil along the failure.

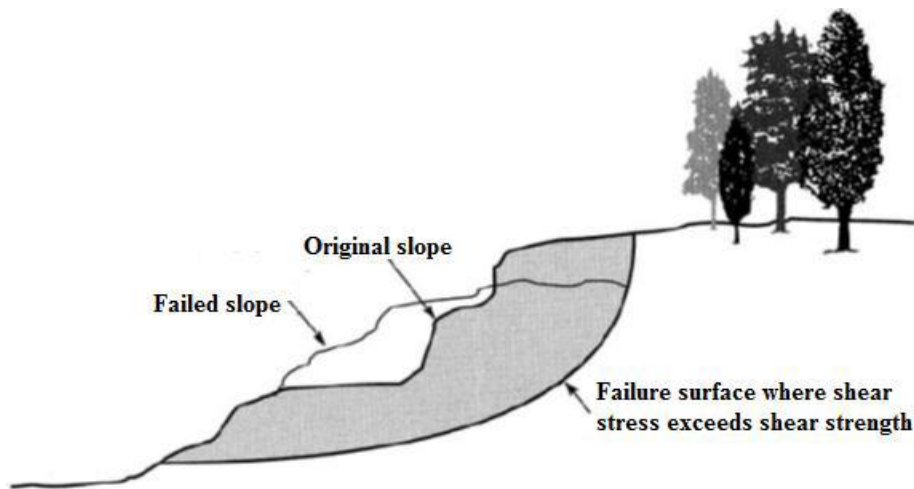


Fig Typical slope failure

Assumption in analysis of slope:

Actual analysis of slope failure is complex and indeterminate to simplify analysis following assumptions were made in slope analysis

- 1) Problem is 2 – D.
- 2) Shear strength of the soil is known and can be represented by Coulombs law.
- 3) Seepage condition and water level are known and the corresponding pore-pressure can be estimated.
- 4) Shear strain at all points along critical surface are large enough to mobilize all available shear strength.

Causes of slope failure: Generally, failure occurs due to natural or man-made causes. Natural failures primarily occur because of stresses imposed by weight of the soil mass itself and by changing soil properties. Man-made failures occur when the slope is physically altered. Irrespective of the mechanism causing failure, ***a slope fails when the imposed stresses exceed the shear strength of the soil along the failure.***

Failures of natural and man-made slopes are generally attributable to any activity that

results in either an increase in soil stress or a decrease in soil strength. The specific causes of slope instability are varied and depend on the nature of the soil, pore water pressure, climate, and stress within the soil mass (static and dynamic). Specific examples that cause a net increase in stresses include an increase in the unit weight of the soil through rainfall, loads imposed by fills or structures at the top of a slope or excavation at the toe of a slope, movement of water levels (such as rapid drawdown in a reservoir), earthquakes, and water pressure in cracks within the slope. These are shown in Figure

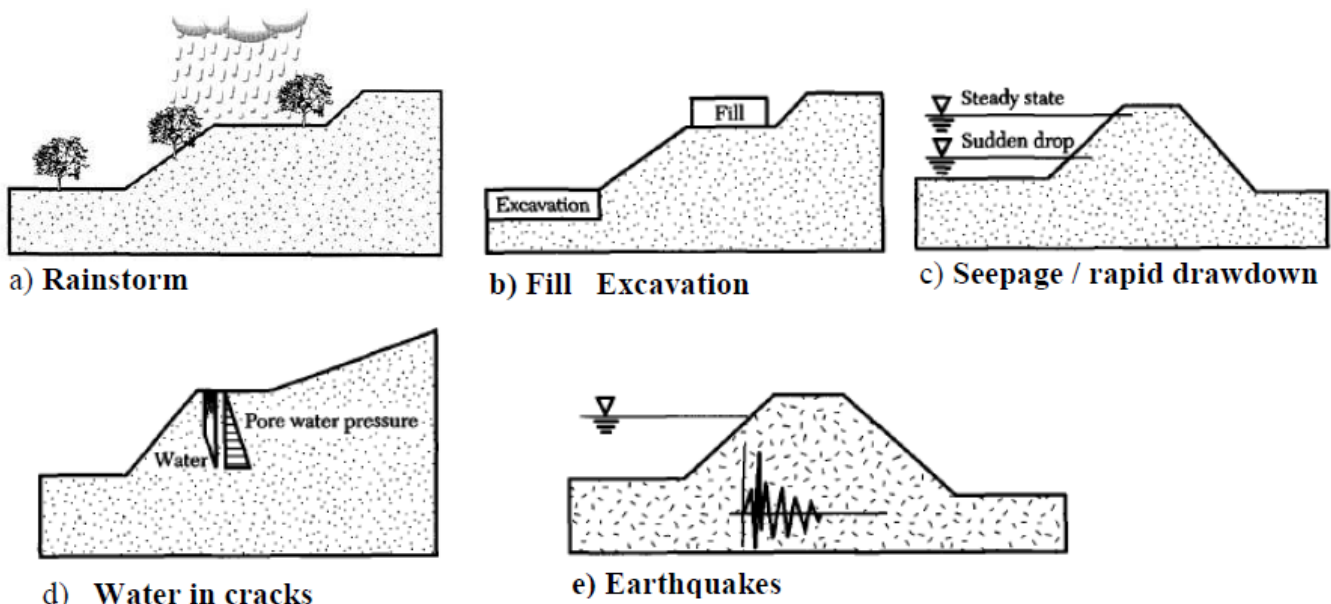


Fig: Various causes of slope failure

Water plays a role in many of the processes that reduce strength, and, as discussed in the following section, water is also involved in many types of loads on slopes that increase shear stresses. It is not surprising, therefore, that virtually every slope failure involves the destabilizing effects of water in some way, and often in more than one way.

Another factor involved in most slope failures is the presence of soils that contain clay minerals. The behavior of clayey soils is much more complicated than the behavior of gravels, sands, and non-plastic silts, which consist of chemically inert particles. The mechanical behavior of clays is affected by the physicochemical interaction between clay particles, the water that fills the voids between the particles, and the ions in the water.

When a slope fails, it is usually not possible to pinpoint a single cause that acted alone and resulted in instability. For example, water influences the stability of slopes in so many ways that it is frequently impossible to isolate one effect of water and identify it as the single cause of failure

Types of slope failure:

- 1) Translational failure
- 2) Rotational failure
- 3) Wedge failure
- 4) Compound failure
- 5) Miscellaneous failure

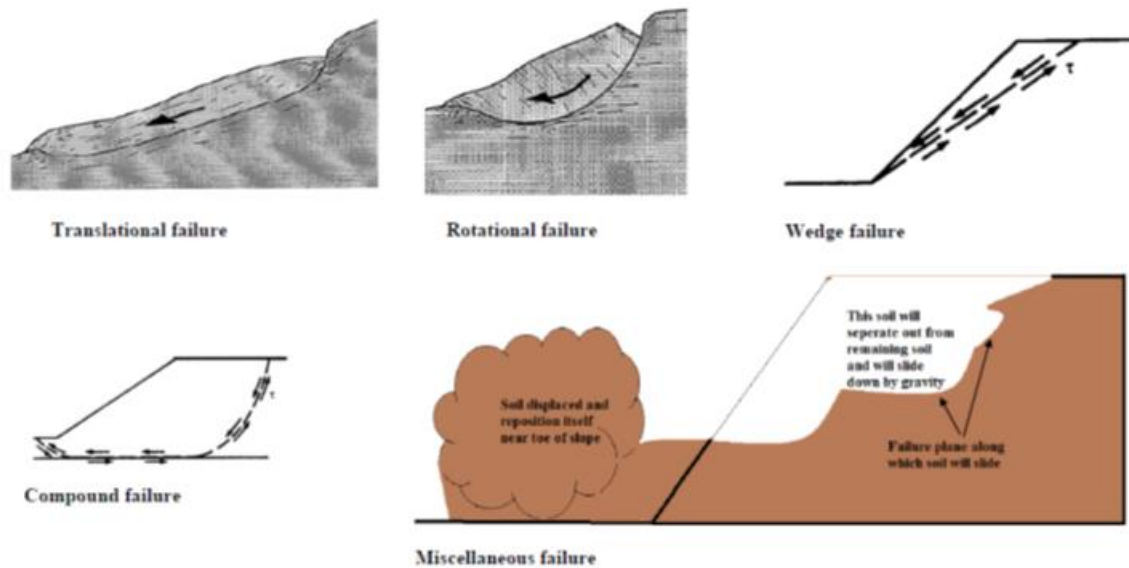


Fig: Types of slope failure



Fig: Types of rotational failure

If failure surface intersects the slope above toe then it is called face failure, if at the toe then it is called toe failure and if it intersects below toe then it is called base failure.

Factor of safety:

It is dimensionless number which will ensure the safety of a structure and it depends on

- The method of stability analysis used
- The method used to determine the shear strength.
- The degree of confidence in the reliability of subsurface data.
- The consequences of a failure.
- How critical the application is.

A minimum factor of safety as low as 1.25 is used for highway embankment side slopes. This value of the safety factor should be increased to a minimum of 1.30 to 1.50 for slopes whose failure would cause significant damage such as end slopes beneath bridge abutments, major retaining structures and major roadways such as regional routes, interstates, etc.

Types of factor of safety

- 1) F.S. with respect to shear strength
- 2) F.S. with respect to cohesion
- 3) F.S. with respect to friction
- 4) F.S. with respect to moment
- 5) F.S. with respect to height

These factor of safety are defined as below

- 1) F.S. with respect to shear strength

$$F_s = \frac{\text{shear strength of soil}}{\text{mobilised shear stress}}$$

- 2) F.S. with respect to cohesion

$$F_s = \frac{\text{cohesive strength of soil}}{\text{mobilised cohesion}}$$

- 3) F.S. with respect to friction

$$F_s = \frac{\tan \phi}{\tan \phi_m} \approx \frac{\phi}{\phi_m}$$

- 4) F.S. with respect to moment

$$F_s = \frac{\text{overturning moment}}{\text{resisting moment}}$$

- 5) F.S with respect to height

$$F_s = \frac{\text{critical height}}{\text{actual height}}$$

Analysis of slope: Slope stability analysis is performed to assess the potential for failure of the slope by rupture. The primary objective of a stability analysis is to determine the factor of safety (FS) of a particular slope, to predict when failure is imminent, and to assess remedial treatments when necessary. In many practical situations, an analytical assessment of stability can be made

Analysis of Infinite slope:

a) Dry soil:

Consider an infinite slope having slope angle β as shown in figure. Consider small element of this slope of width b ; various forces acting on this are as shown. For an infinite slope, the forces on the two ends of the block will be identical in magnitude, opposite in direction, and collinear. Thus, the forces on the ends of the block exactly balance each other and can be ignored in the equilibrium equations.

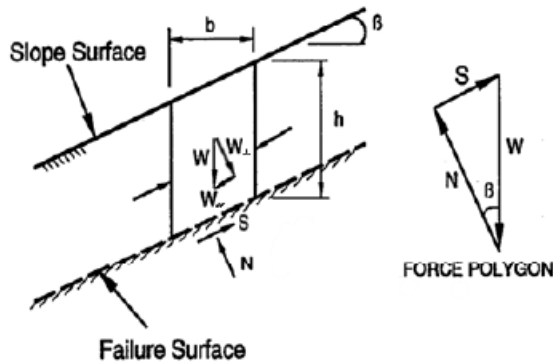


Fig Failure of infinite slope

Summing forces in directions perpendicular and parallel to the slip plane gives the following expressions for the shear force, S, and normal force, N, on the plane:

$$S = W \cos \beta \text{ and } N = W \sin \beta$$

But $W = \gamma b h = \text{weight of element of soil}$

$$\text{Normal stress} = \frac{N}{\text{Area parallel to failure surface}} = \frac{\gamma b h \cos \beta}{b / \cos \beta} = \gamma h \cos^2 \beta$$

$$\text{Shear stress} = \frac{S}{\text{Area parallel to failure surface}} = \frac{\gamma b h \sin \beta}{b / \cos \beta} = \gamma h \sin \beta \cos \beta$$

$$\text{Factor of safety} = \frac{\text{Shear strength}}{\text{Shear stress}} = \frac{c + \sigma \tan \phi}{\gamma h \cos^2 \beta} = \frac{C + \gamma h \cos^2 \beta \tan \phi}{\gamma h \sin \beta \cos \beta}$$

$$\text{When } c = 0 \text{ we get } FS = \frac{\tan \phi}{\tan \beta}$$

Height of the slope at which slope is on the verge of failure i.e. shear stress on failure surface is equal to shear strength of soil (FOS = 1)

Substituting FOS = 1 and $h = H_C$ in above equation we get

$$C + \gamma H_C \cos^2 \beta \tan \phi = \gamma H_C \sin \beta \cos \beta$$

$$\gamma H_c \sin \beta \cos \beta - \gamma H_c \cos^2 \beta \tan \phi = C$$

$$\gamma H_c \cos^2 \beta (\tan \beta - \tan \phi) = C$$

$$\frac{C}{\gamma H_c} = \cos^2 \beta (\tan \beta - \tan \phi) \quad \text{Term } \frac{C}{\gamma H_c} \text{ is called stability number}$$

$$H_c = \frac{C}{\gamma \cos^2 \beta (\tan \beta - \tan \phi)}$$

For submerged slope

$$H_c = \frac{C}{\gamma' \cos^2 \beta (\tan \beta - \tan \phi)}$$

Slope with seepage parallel to slope

$$H_c = \frac{C}{\cos^2 \beta (\gamma \tan \beta - \gamma' \tan \phi)}$$

When the seepage is parallel to slope

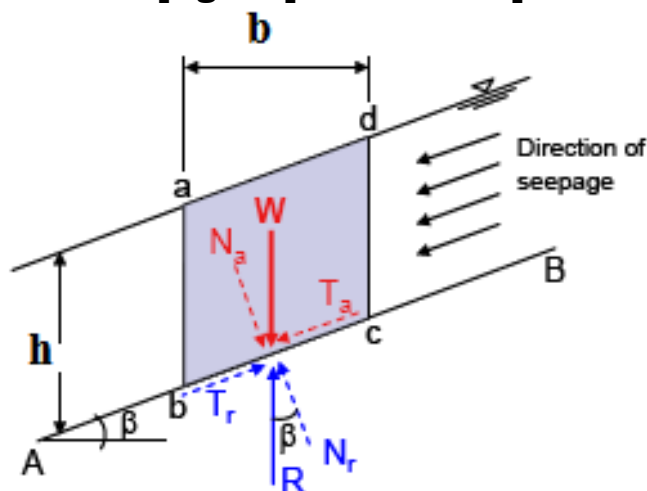


Fig Infinite slope with seepage parallel to slope

$$FOS = \frac{C}{\gamma_{sat} h \cos^2 \beta \tan \beta} + \frac{\gamma' \tan \phi}{\gamma_{sat} \tan \beta} \quad \text{for } c - \phi \text{ soil}$$

for granular soil $C = 0$ hence $FOS = \frac{\gamma' \tan \phi}{\gamma_{sat} \tan \beta}$

GENERAL CASE:

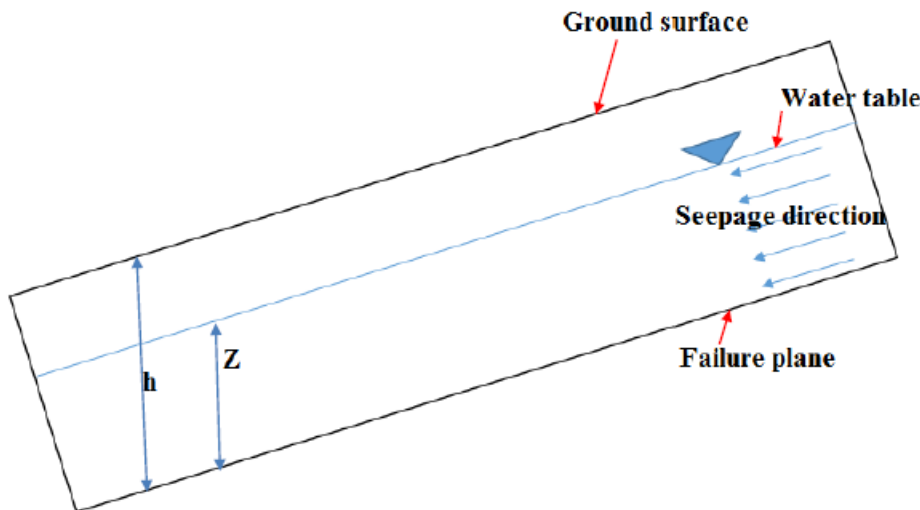


Fig 7.9 Infinite slope with seepage parallel to slope (partially submerged)

$$FOS = \frac{C + h \cos^2 \beta [(h-z)\gamma + z\gamma'] \tan \phi}{h \sin \beta \cos \beta [(h-z)\gamma + z\gamma_{sat}]}$$

$$FOS = \frac{C + h \cos^2 \beta [(1-m)\gamma + m\gamma'] \tan \phi}{h \sin \beta \cos \beta [(1-m)\gamma + m\gamma_{sat}]}$$

Where c and ϕ are shear parameters of soil

γ bulk unit weight of soil above water table

γ_{sat} , and γ' - Saturated and submerged bulk unit weight of soil

β - Slope angle

h - Height of slope

m - normalized height of water table above failure plane = z / h

Z - Height of water table above failure surface

h - Height of slope

Analysis of Finite slope: The finite slope can be analyzed by any of following method

- Swedish circle method or slip circle method or method of slice
- Friction circle method
- Taylors method

Swedish circle method:

Investigation carried out in Sweden on slope failures indicated that the failure surfaces resembles the arc of a circle. Fellenius developed a method for stability analysis of a slope assuming a circular failure surface, which is known as the Swedish circle method.

Case a) Stability analysis for cohesive soil ($\phi=0$ Analysis)

Fig. shows finite slope whose stability is to be determined. A trial slip surface of radius r is assumed and factor of safety of the slope is determined for the assumed trial surface. For analysis purpose unit length of slope is considered.

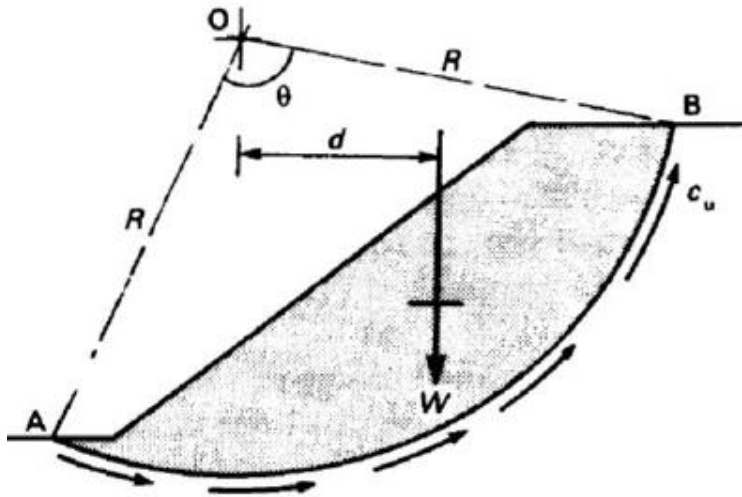


Fig: Finite slope in purely cohesive soil

Let R – radius of slip circle

c – cohesion of soil (undrained)

O – center of slip circle

W – weight of sliding soil (soil above failure surface)

d – Horizontal distance between center of circle and the point where weight acts (lever arm)

θ – central angle

$\hat{L}$ – Arc length

Moment causing sliding of soil mass downward = disturbing moment = $MD = W d$

Moment resisting sliding tendency of soil mass = resisting moment = $MR = c \hat{L} R$

Factor of safety = Resisting moment / disturbing moment

$$FS = \frac{c \hat{L} R}{W d} = \frac{c R^2 \theta}{W d}$$

Analysis is repeated for number of trial slip surfaces and the FOS is determined in each case. The slip circle corresponding to minimum FOS is the critical slip circle.

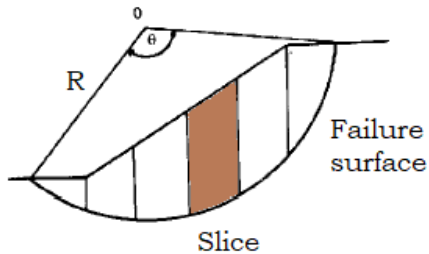
Note:

- Position of centroid of sliding wedge can be determined by dividing wedge of soil into small slices
- If soil is non-homogeneous or slope has benches, then it should be divided into small elemental areas and then moment of all such areas need to be taken about center of rotation

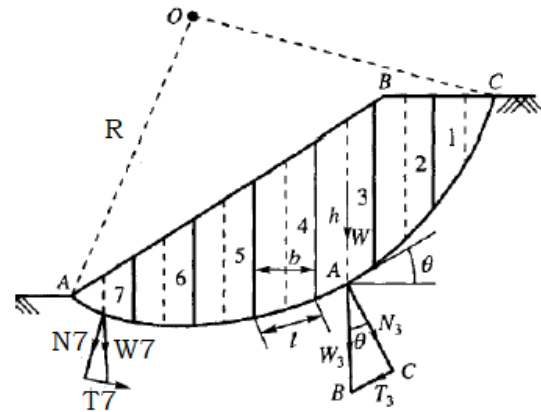
C – ϕ analysis (Method of slices): For soil which has both cohesion and friction component of shear strength, that is, $c - \phi$ soil, the shear strength along the slip surface is also contributed by the frictional component, which is a function of normal stress. The normal stress varies at every point on the slip surface both in magnitude

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and direction hence total sliding soil mass is divided into number of slices (vertical). The force between the slice is neglected and each soil is assumed to act independently as a column of soil. The weight of each slice is assumed to act at its Centre. Weight of each slice is resolved into normal (N) and tangential component (T). The normal component passes through the center of slip circle and does not cause any moment. However, the tangential component will cause a driving moment ($= T_i \cdot r$) where r is radius of slip circle. For some slices tangential component may cause resisting moment in which case it should be considered negative.



Sliding soil mass divided in to number of slices



Component of weight of slice in to normal and tangential component

$$FOS = \frac{c \bar{L} + \tan \phi \sum N}{\sum T}$$

When soil is submerged we need to consider pore-water pressure (U) in that case factor of safety is given by

$$FOS = \frac{c \bar{L} + \tan \phi \sum (N - U)}{\sum T}$$

Where $\sum N$ – algebraic sum of normal component of the weight of slice

$\sum T$ – algebraic sum of tangential component of the weight of slice

$\sum U$ – algebraic sum of pore pressure

c - cohesion of soil, ϕ - angle of internal friction

$\bar{L}$ - arc length of failure surface = $r \theta$

θ - central angle

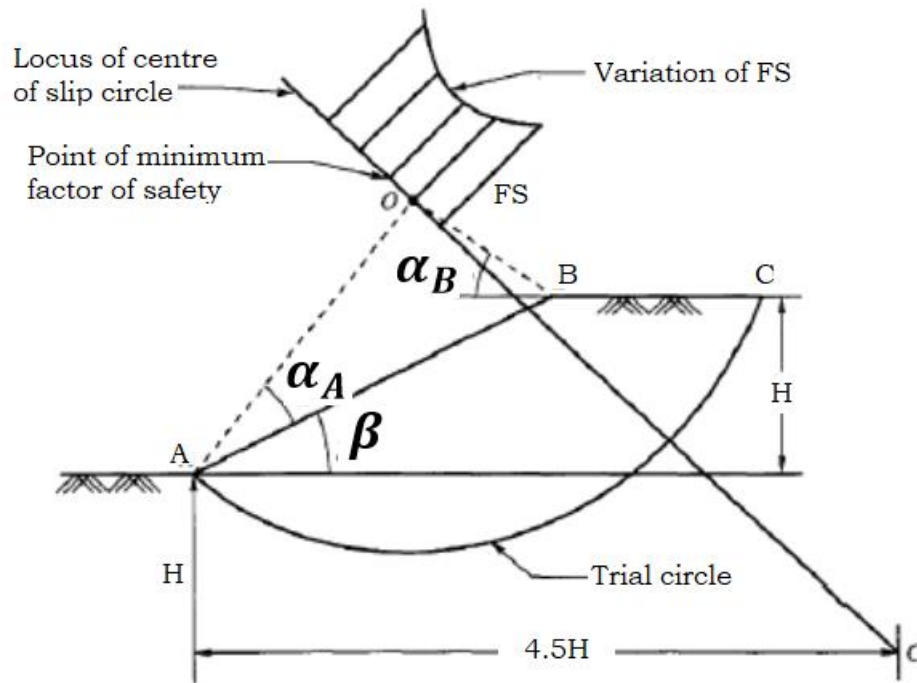
Calculations will be done in tabular column

Slice No.	Width of slice	Area of slice	Weight of slice	θ	Component of weight	
					Normal = $W \cos \theta$	Tangential = $W \sin \theta$

$$\sum N = \quad \quad \quad \sum T =$$

Note: If the calculation of factor of safety is done partially by graphical construction and partially by analytical method then it is called semi-analytical method, and if it is done entirely by graphical construction then it is called graphical method

Fellenious method of locating center of slip circle:



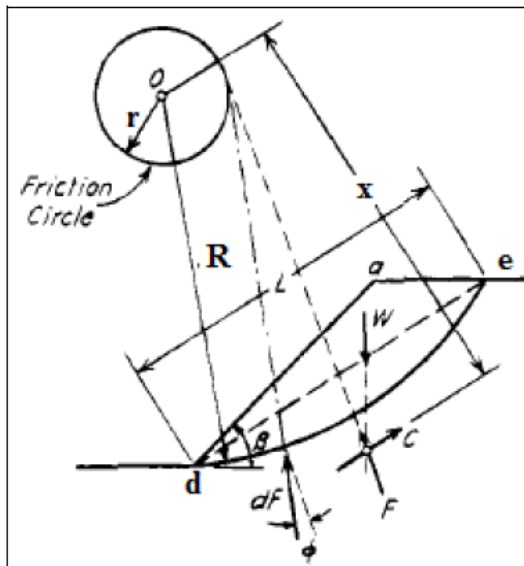
Slope	Slope angle	Direction angles	
		α_A	α_B
0.6 : 1	60	29	40
1 : 1	45	28	37
1.5 : 1	33.8	26	35
2 : 1	26.6	25	35
3 : 1	18.3	25	35
5 : 1	11.3	25	37

Procedure:

- 1) From given information i.e. slope angle and height Draw the slope with some suitable given scale

- 2) From toe of slip circle and the slope draw direction lines at angles α_A and α_B to intersect each other at O
- 3) With respect to A as origin Locate C (4.5H, H)
- 4) Join CO and extend it, which is locus of Centre of slip circle.
- 5) Choose any point on the locus. Draw trial slip circle passing through toe and find FOS for it
- 6) Repeat the process for five to six trial centers along the locus of center of slip circle
- 7) Plot all these FOS normal to locus line and join them by smooth curve. Find the lowest point on it which correspond to minimum factor of safety and its slip circle will be critical slip circle.

Friction circle method: The friction circle method, originally proposed by Taylor (1937), considers the stability of the entire sliding mass as a whole. The disadvantage of the friction circle method is that it can only be applied to a homogeneous slope with a given angle of internal friction.



The forces acting on the sliding mass are its weight W , the resultant cohesion C , and the resultant F of the normal and frictional forces acting along the surface of sliding.

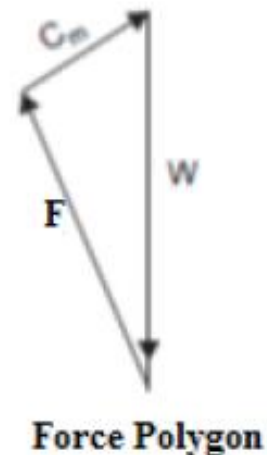
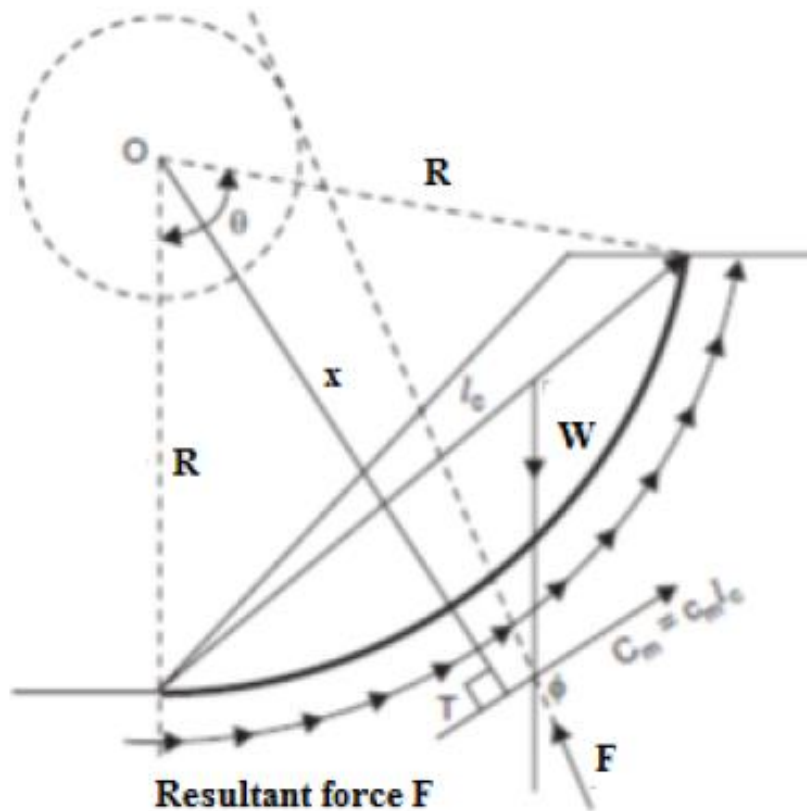
Weight W acts vertically downward through centroid of sliding mass

Resultant cohesion C acts in a direction parallel to the chord de and is equal to the unit cohesion c multiplied by length L of chord. Position of total cohesion C from center of rotation is determined by

$$C x = c (\text{arc length } de) R$$

$$c L x = c \hat{L} R$$

$$\text{i.e. } x = R \frac{\hat{L}}{L} = \text{radius} \times \frac{\text{Arc length}}{\text{Chord length}}$$



Sliding wedge is in equilibrium under the action of three forces so these forces should be concurrent i.e. W , C and F should pass through single point. Thus by constructing force polygon magnitude and direction of F can be determined.

If factor of safety against sliding is unity, then slope is on the verge of failure. Under this condition each of elementary reaction dF must be inclined at an angle ϕ to the normal to the slip circle. As a consequence, the line of action of each elementary

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reaction is tangential to the circle, known as friction circle whose radius is $R \sin \phi$ with center as center of slip circle

Factor of safety in case of friction circle method:

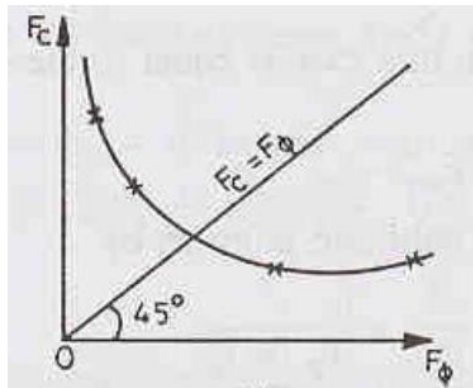
- 1) Assume trial value of factor of safety with respect to friction F_ϕ
- 2) Calculate mobilized friction angle ϕ_m by using $\tan \phi_m = \frac{\tan \phi}{F_\phi}$
- 3) Construct the given slope with suitable scale and trial slip circle of radius R
- 4) Construct the friction circle with same centre and radius $r = R \sin \phi_m$
- 5) Draw vertical line through the centroid of sliding mass to represent weight W
Whose magnitude will be $(= \text{unit weight} \times \text{c/s area of sliding wedge} \times 1)$
- 6) Draw a line parallel to chord and at a distance x $[x = R \frac{\hat{L}}{L}]$ from the centre of slip circle.
- 7) Draw resultant reaction passing through the point of intersection of weight and cohesion.
- 8) Construct force triangle. From force triangle measure the value of mobilized cohesion C_m

$$C_m = \frac{\text{Mobilised cohesion}}{\text{chord length}}$$

- 9) Calculate factor of safety with respect to cohesion F_c

$$F_c = \frac{c}{c_m}$$

- 10) Compare value of F_c and F_ϕ . If two values do not agree then procedure is repeated by choosing some other value of F_ϕ . Three trials will be sufficient to get $F_c = F_\phi$ or alternatively plot the graph between these two values of factor of safety and draw a line at an angle of 45° to the horizontal. Intersection of the line with the curve will be minimum FOS



Taylor's stability number and stability charts.

Taylor carried out stability analysis of large number of soil slopes with a wide range of properties such as slope angle (β), height (H), cohesion (c), friction angle (ϕ), and unit weight of soil (γ) using Friction circle method. Taylor presented the results of the stability analysis in the form of curves (stability chart) which gives the relationship

between stability number (S_n) and slope angle (β) for various values of friction angle (ϕ). Stability number is defined by the relation $S_n = c/(\gamma H c) = c/(\gamma F H)$

c – unit cohesion for the soil (kPa), γ – unit weight of soil (kN/m^3), H – height of slope

F – factor of safety and H_c – critical height of slope

The chart is divided into two zones, A and B. As shown in the inset for Zone A, the critical circle for steep slopes passes through the toe of the slope with the lowest point on the failure arc at the toe of the slope.

As shown in the inset for Zone B, for shallower slopes the lowest point of the critical circle is not at the toe, and three cases must be considered as follows:

Case 1: Where long dashed curves do not appear in the chart, the critical circle passes through the toe. This condition corresponds to Case 1. Stability numbers for Case 1 are given by the solid lines in the chart both when there is and when there is not a more dangerous circle that passes below the toe, i.e., the curves for Case 1 are an extension of the curves that correspond to a toe circle failure in Zone A.

In both Case 1 and Case 2 the failure circle passes through the soil below the toe of the slope. The depth ratio, D , which is a multiple of the slope height H , is used to define the depth (DH) from the top of the slope to an underlying strong material through which the failure circle does not pass.

Case 2: For shallow slope angles or small developed friction angles the critical circle may pass below the toe of the slope. This condition corresponds to Case 2 in the inset for Zone B. The values of N_s for this case are given in the chart by the long dashed curves.

Case 3: This case corresponds to the condition where there is an underlying strong layer at the elevation of the toe ($D=1$). This case is represented by short dashed lines in the chart.

Use of chart:

To find factor of safety - Given $\{\gamma, \beta, \phi, \text{ and } c\}$

1) Assume trial value of FOS F_ϕ and determine mobilized friction angle ϕ_m

By using $\tan \phi_m = \tan \phi / F_\phi$

2) By using stability chart find stability number from value of ϕ_m and β

3) Determine the factor of safety with respect to cohesion by using $F_c = \frac{c}{S_n \gamma H}$

4) If $F_c \neq F_\phi$ then procedure is repeated by choosing some other value of F_ϕ

To find maximum slope angle – Given $\{\gamma, F, \phi, \text{ and } c\}$

1) Determine mobilized friction angle ϕ_m by using $\tan \phi_m = \tan \phi / F_\phi$

2) Determine stability number by using $S_n = \frac{c}{F \gamma H}$

3) Use the stability chart determine slope angle β knowing S_n and ϕ_m

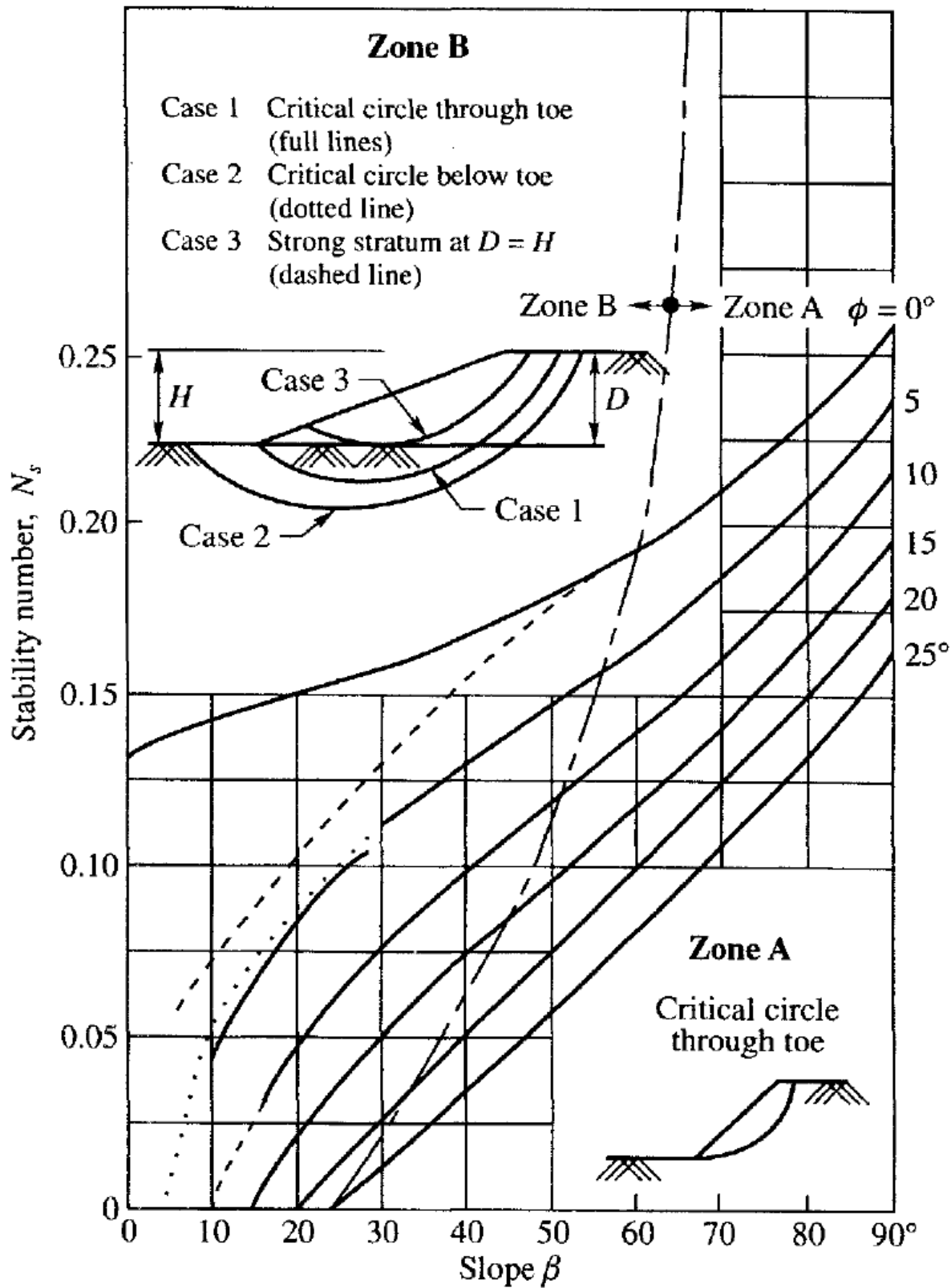


Fig Taylors stability chart

Stability Conditions for Analysis

Variations of the loads acting on slopes, and variations of shear strengths with time, result in changes in the factors of safety of slopes. As a consequence, it is often necessary to perform stability analyses corresponding to several different conditions reflecting different stages in the life of a slope. As conditions change, the factor of safety against slope instability may increase or decrease.

- When an embankment is constructed on a clay foundation, the embankment load causes the pore pressures in the foundation clay to increase. Over a period of time these excess pore pressures will dissipate, and the pore pressures will return eventually to values governed by the groundwater conditions. As the excess pore pressures dissipate, the effective stresses in the foundation clay increase, the strength of the clay will increase, and the factor of safety of the embankment will also increase the embankment height stays constant and there is no external loading, a critical condition occurs at the end of construction.
- When a slope in clay is created by excavation, the pore pressures in the clay decrease in response to the removal of the excavated material. Over time the reductions in pore pressures dissipate, and the pore pressures return eventually to values governed by the groundwater conditions. As the pore pressures increase, the effective stresses in the clay around the excavation decrease, and the factor of safety of the slope decreases with time. If the depth of excavation is constant and there are no external loads, the factor of safety decreases continually, and the minimum value is reached when the pore pressures reach equilibrium with the groundwater seepage conditions. In this case, therefore, the long-term condition is more critical than the end-of-construction condition
- In the case of a natural slope, not altered by either fill placement or excavation, there is no end-of-construction condition. The critical condition for a natural slope corresponds to whatever combination of seepage and external loading results in the lowest factor of safety. The higher the phreatic surface within the slope, and the more severe the external loading condition, the lower is the factor of safety

END-OF-CONSTRUCTION STABILITY

Slope stability during and at the end of construction is analyzed using either drained or un-drained strengths, depending on the permeability of the soil. Undrained strength (Total stress analysis) are used for soil with low permeability and drained strength (effective stress analysis) can be used for the soil which can drain the water easily. For many embankment slopes the most critical condition is the end of construction. In some cases, however, there may be intermediate conditions during construction that are more critical. In some fill placement operations, including some waste fills, the fill may be placed with a slope geometry such that the stability conditions during construction are more adverse than at the end of construction.

LONG-TERM STABILITY:

Over time after construction the soil in slopes may either swell (with increase in water content) or consolidate (with decrease in water content). Long-term stability analyses are performed to reflect the conditions after these changes have occurred. Shear strengths are expressed in terms of effective stresses, and the pore water pressures are estimated from the most adverse groundwater and seepage conditions anticipated during the life of the slope.

RAPID (SUDDEN) DRAWDOWN:

Rapid or sudden drawdown is caused by a lowering of the water level adjacent to a slope, at a rate so fast that the soil does not have sufficient time to drain significantly. Undrained shear strengths are assumed to apply for all but the coarsest free-draining materials ($k > 10^{-1}$ cm/sec).

- If drawdown occurs during or immediately after construction, the un-drained shear strength used in the drawdown analysis is the same as the un-drained shear strength that applies to the end-of-construction condition.
- If drawdown occurs after steady seepage conditions have developed, the un-drained strengths used in the drawdown analysis are different from those used in the end-of-construction analyses and are determined by the effective stresses during steady seepage.
- For soils that expand when wetted, the un-drained shear strength will be lower if drawdown occurs after a period of time following construction than if it occurs immediately after construction.

EARTHQUAKE:

Earthquakes affect the stability of slopes in two ways. First, the acceleration produced by the seismic ground motion during an earthquake subjects the soil to cyclically varying forces. Second, the cyclic strains induced by the earthquake loads may result in a decrease in the shear strength of the soil.

- If the strength of the soil is reduced less than 15 percent by cyclic loading, pseudo-static analyses of the earthquake loading can be used. In pseudo-static analyses, the effect of the earthquake is represented crudely by applying a static horizontal force to the potential sliding mass
- If the strength of the soil is reduced more than 15 percent as a result of cyclic loading, dynamic analyses are needed to estimate the deformations that would result from earthquakes.

Analysis case	Slope (critical)
1) End of construction (including staged construction)	u/s and d/s.
2) Long term (steady seepage, maximum storage pool, spillway crest or top of gates)	d/s
3) Maximum surcharge pool	d/s
4) Rapid drawdown	u/s
5) Earthquake loading	u/s and d/s